

UNIVERSIDAD POLITÉCNICA DE MADRID

**ESCUELA TÉCNICA SUPERIOR DE
INGENIEROS DE TELECOMUNICACIÓN**



**Thermal, stress, and traps effects in
AlGaIn/GaN HEMTs**

Tesis doctoral

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Doctoral dissertation

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List of abbreviations

2D	two dimension
2DEG	two dimensional electron gas
2DHG	two dimensional hole gas
3D	three dimension
AlGaN	aluminum gallium nitride
AlInN	aluminum indium nitride
CPS	converse piezoelectric stress
DH-HEMTs	double heterojunction high electron mobility transistors
EM	electro-mechanical
ET	electro-thermal
ETM	electro-thermo-mechanical
FEM	finite element method
FET	field effect transistor
GaAs	gallium arsenide
GaN	gallium nitride
GEDS	gate edge of the drain side
GLR	gate lag ratio
HEMTs	high electron mobility transistors
InP	indium phosphide
NDC	negative differential conductance
PAT	phonon-assisted tunneling
PF	Poole-Frenkel
TBR	thermal boundary resistance
TS	thermal stress
WKB	Wentzel-Kramers-Brillouin

List of notations

c_{ijkl}	elastic stiffness tensor
C_G	gate capacitance with $V_{DS}=0$
C_{GD}	gate-drain capacitance
C_{GS}	gate-source capacitance
C_{FP}	filed-plate capacitance
C_{FR}	fringing capacitance
D_i	electric displacement vector
e_n	electron emission rate
e_{ijk}	piezoelectric coefficient tensor
$\mathbf{E}$	electric field
E_0	electron vacuum energy
E_C	conduction band edge
E_g	bandgap
E_f	Fermi energy level
E_k	electric field vector
E_T	traps ionization energy
E_V	valance band edge
E_x	parallel electric field
E_y	vertical electric field
h	Planck constant
I_{DS}	drain current
$I_{DS,Lin}$	linear region of the drain current
$I_{DS,Sat}$	saturation region of the drain current
$I_{G,Rev}$	reverse gate current
$\mathbf{j}_n$	electron current density

k	thermal conductivity
k_B	Boltzmann constant
L_G	gate length
L_{GD}	distance between gate and drain
L_{SG}	distance between source and gate
m_e^*	electron effective mass
n	electron concentration
N_A	acceptor-like traps density
N_A^-	ionized acceptor-like traps density
N_C	state density of conduction band
P_D	power dissipation
P_i^s	spontaneous polarization vector
Q_G	gate charge density
Q_P	polarization charge density
s_{kl}	strain tensor
R_{TH}	thermal resistance
V_{BR}	breakdown voltage
V_{DS}	drain voltage
V_{GS}	gate voltage
$V_{DS,Q}$	quiescent drain voltage
$V_{GS,Q}$	quiescent gate voltage
V_{th}	threshold voltage
$I_{G,Rev}$	reverse gate current
V_T	electron thermal velocity
T_{amb}	ambient temperature
T_{ch}	channel temperature
T_L	lattice temperature
W_G	gate width

α	thermal expansion coefficient
τ_n	electron emission characteristic time
ϕ_n	quasi-Fermi potential
ψ	electric potential
μ_n	electron mobility
ρ	spatial charge density
χ	electron affinity energy
ε_{ij}	permittivity tensor
ε_o	vacuum permittivity
ε_r	relative permittivity
σ_n	electron capture cross-section
σ_{ij}	stress tensor
ΔE_C	discontinuity of conduction band
ΔE_V	discontinuity of valance band

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Abstract

GaN and AlN are group III-V piezoelectric semiconductor materials. The AlGaN/GaN heterojunction presents large piezoelectric and spontaneous polarization charge at the interface, leading to high 2DEG density close to the interface. A high power output would be obtained due to the high 2DEG density and mobility, which leads to elevated lattice temperature. The gate and drain biases induce converse piezoelectric stress that can influence the piezoelectric polarization charge and further influence the 2DEG density and output characteristics. Therefore, the device physics is relevant to all the electrical, thermal, and mechanical aspects. In this dissertation, by using the commercial finite-element-method (FEM) software COMSOL, we achieved the GaN HEMTs simulation with electro-thermal, electro-mechanical, and electro-thermo-mechanical full coupling. The coupling parts include the drift-diffusion model for the electron transport, the thermal conduction, and the piezoelectric effect. By simulations and some experimental characterizations, we have studied the device thermal, stress, and traps effects described in the following.

The device geometry impact on the self-heating was studied by electro-thermal simulations and electrical characterizations. Among the obtained interesting results, we found that, for same power output, the distance between the gate and drain contact can influence distribution of the heat generation in the channel and thus influence the channel temperature.

Diamond possesses high thermal conductivity. Integrated diamond with the device can spread the generated heat and thus potentially reduce the device self-heating effect. Electro-thermal simulations on this topic were performed. For the diamond integration on top of the device (top-side heat spreading), the determinant factors for the heat spreading ability are the diamond thickness, its thermal conductivity, and its distance to the heat source. The top-side heat

spreading can also reduce the impact of thermal boundary resistance between the buffer and the substrate on the device thermal behavior. The very low electrical conductivity of diamond allows that it can directly contact the gate metal (which is very close to the heat source), being quite convenient to reduce the self-heating for the device under pulsed bias. Also, the diamond coated in vias etched in the substrate as heat spreading path (back-side heat spreading) was simulated. A competing mechanism influences the heat spreading ability, i.e., the etched vias would increase the device temperature due to the reduced heat sink while the coated diamond would decrease the device temperature due to its higher thermal conductivity. Therefore, relative thick coated diamond is needed in order to reduce the self-heating effect.

The simulated local stress at the gate edge of the drain side for the device with standard and field plate gate structure were compared, which would be relevant to the device mechanical failure. Other stress simulations focused on the intrinsic stress in the diamond capping layer impact on the device electrical behaviors. The simulated stress and electrical output characteristics were compared to experimental data obtained by micro-Raman spectroscopy and electrical characterization, respectively. Results showed that the intrinsic stress in the capping layer caused the non-uniform distribution of 2DEG in the channel and the access region. Besides the enhancement of the device power output, intrinsic stress in the capping layer can potentially improve the device reliability by modulating the local stress at the gate edge of the drain side.

Finally, the surface, buffer, and barrier traps effects were simulated in this work. Pulsed measurements showed that long gates and distances between gate and drain contact can increase the gate lag ratio (decrease the current collapse). This was explained by simulations on the surface traps effect. The simulations on buffer traps effects focused on illustrating the dynamic trapping/detrapping in the buffer and the self-heating impact on the device transient drain current. A model was presented to describe the trapping and detrapping in the barrier. The

trapping was the electron direct tunneling from the gate metal while the detrapping was the electron emission into the conduction band described by phonon-assisted tunneling. The reverse gate current was simulated based on this model, whose mechanism can be attributed to the temperature and electric field dependent electron emission in the barrier. Furthermore, the mechanism of the device bias via the self-heating and electric field impact on the electron emission and the transient drain current were also illustrated.

Resumen

GaN y AlN son materiales semiconductores piezoelétricos del grupo III-V. La heterounión AlGaN/GaN presenta una elevada carga de polarización tanto piezoelétrica como espontánea en la interfase, lo que genera en su cercanía un 2DEG de grandes concentración y movilidad. Este 2DEG produce una muy alta potencia de salida, que a su vez genera una elevada temperatura de red. Las tensiones de puerta y drenador provocan un stress piezoeléctrico inverso, que puede afectar a la carga de polarización piezoeléctrica y así influir la densidad 2DEG y las características de salida. Por tanto, la física del dispositivo es relevante para todos sus aspectos eléctricos, térmicos y mecánicos. En esta tesis se utiliza el software comercial COMSOL, basado en el método de elementos finitos (FEM), para simular el comportamiento integral electro-térmico, electro-mecánico y electro-térmico-mecánico de los HEMTs de GaN. Las partes de acoplamiento incluyen el modelo de deriva y difusión para el transporte electrónico, la conducción térmica y el efecto piezoeléctrico. Mediante simulaciones y algunas caracterizaciones experimentales de los dispositivos, hemos analizado los efectos térmicos, de deformación y de trampas.

Se ha estudiado el impacto de la geometría del dispositivo en su auto-calentamiento mediante simulaciones electro-térmicas y algunas caracterizaciones eléctricas. Entre los resultados más sobresalientes, encontramos que para la misma potencia de salida la distancia entre los contactos de puerta y drenador influye en generación de calor en el canal, y así en su temperatura.

El diamante posee una elevada conductividad térmica. Integrando el diamante en el dispositivo se puede dispersar el calor producido y así reducir el auto-calentamiento, al respecto de lo cual se han realizado diversas simulaciones electro-térmicas. Si la integración del diamante es en la parte

superior del transistor, los factores determinantes para la capacidad disipadora son el espesor de la capa de diamante, su conductividad térmica y su distancia a la fuente de calor. Este procedimiento de disipación superior también puede reducir el impacto de la barrera térmica de intercara entre la capa adaptadora (buffer) y el sustrato. La muy reducida conductividad eléctrica del diamante permite que pueda contactar directamente el metal de puerta (muy cercano a la fuente de calor), lo que resulta muy conveniente para reducir el auto-calentamiento del dispositivo con polarización pulsada. Por otra parte se simuló el dispositivo con diamante depositado en surcos atacados sobre el sustrato como caminos de disipación de calor (disipador posterior). Aquí aparece una competencia de factores que influyen en la capacidad de disipación, a saber, el surco atacado contribuye a aumentar la temperatura del dispositivo debido al pequeño tamaño del disipador, mientras que el diamante disminuye esa temperatura gracias a su elevada conductividad térmica. Por tanto, se precisan capas de diamante relativamente gruesas para reducir el efecto de auto-calentamiento.

Se comparó la simulación de la deformación local en el borde de la puerta del lado cercano al drenador con estructuras de puerta estándar y con *field plate*, que podrán ser muy relevantes respecto a fallos mecánicos del dispositivo. Otras simulaciones se enfocaron al efecto de la deformación intrínseca de la capa de diamante en el comportamiento eléctrico del dispositivo. Se han comparado los resultados de las simulaciones de la deformación y las características eléctricas de salida con datos experimentales obtenidos por espectroscopía micro-Raman y medidas eléctricas, respectivamente. Los resultados muestran el *stress* intrínseco en la capa producido por la distribución no uniforme del 2DEG en el canal y la región de acceso. Además de aumentar la potencia de salida del dispositivo, la deformación intrínseca en la capa de diamante podrá mejorar la fiabilidad del dispositivo modulando la deformación local en el borde de la puerta del lado del drenador.

Finalmente, también se han simulado en este trabajo los efectos de trampas localizados en la superficie, el buffer y la barrera. Las medidas pulsadas muestran que tanto las puertas largas como las grandes separaciones entre los contactos de puerta y drenador aumentan el cociente entre la corriente pulsada frente a la corriente continua (*lag ratio*), es decir, disminuir el colapso de corriente (*current collapse*). Este efecto ha sido explicado mediante las simulaciones de los efectos de trampa de superficie. Por su parte, las referidas a trampas en el buffer se enfocaron en los efectos de atrapamiento dinámico, y su impacto en el auto-calentamiento del dispositivo. Se presenta también un modelo que describe el atrapamiento y liberación de trampas en la barrera: mientras que el atrapamiento se debe a un túnel directo del electrón desde el metal de puerta, el desatrapamiento consiste en la emisión del electrón en la banda de conducción mediante túnel asistido por fonones. El modelo también simula la corriente de puerta, debida a la emisión electrónica dependiente de la temperatura y el campo eléctrico. Además, también se ilustra la corriente de drenador dependiente de la temperatura y el campo eléctrico.

1 Introduction and research objectives

In this chapter, the basic operation principle of a high electron mobility transistor (HEMT) will be briefly described, as well as some advanced structures of the state of the art of GaN HEMTs. The advantages and disadvantages of these device structures will be summarized. Referring to the reported results, the existing issues that limit the GaN HEMTs commercial applications will be discussed. At last, based on these issues, the detailed research objectives of this dissertation will be presented.

1.1 State of the art of GaN HEMTs

The HEMT is a heterostructure field-effect transistor (FET). The contact between semiconductor materials with different bandgaps, usually grown by epitaxial techniques using different alloy fractions, can form a so called heterojunction. Because of the different bandgaps and their relative alignment to each other, conduction and valence band discontinuities occur at the interface between the two materials, and thus the carriers confined by a quantum well can form close to the heterojunction. These carriers have an apparent quantum effect in the growth direction and therefore have discrete energy levels, while showing a continuous energy distribution in the plane, usually called two dimensional electron gas (2DEG) or two dimensional hole gas (2DHG).

HEMTs based on the heterojunction formed by several III-V semiconductor families, mainly those based in GaAs and GaN with their compounds AlGaAs and AlGaN barriers, have been extensively studied. Comparing GaAs- with GaN-based HEMTs, there are two main differences. One is that the electron

mobility shows a much higher value (tends to $5000 \text{ cm}^2/\text{V.s}$) for GaAs HEMTs, compared to GaN HEMTs (around $1500 \text{ cm}^2/\text{V.s}$), due to the less Coulomb scattering in GaAs. Another one is about the source of electrons. The 2DEG in GaAs-based HEMTs mainly comes from the doping in the AlGaAs barrier layer. For GaN HEMTs, however, due to the relative large piezoelectric and spontaneous polarization coefficient of GaN and AlGaN, a high density of positive polarization charges can form at the AlGaN/GaN interface, leading in turn to a high density of 2DEG for compensation of the polarization charge. Therefore, the 2DEG density is mainly influenced by the interface polarization charge, but not by the doping in the barrier. On the other hand, in ideal conditions (no additional charges caused by defects, traps, etc.), the 2DEG should come from the valence band of AlGaN near the AlGaN surface when the barrier thickness exceed a critical value. Correspondingly, that will result in 2DHG near the AlGaN surface. However, it was reported that the electrons should come from the AlGaN surface states, according to an investigation of the modification of the 2DEG density with the AlGaN thickness [1].

Although the high electron mobility in GaAs-based HEMTs, the relative narrow bandgap of GaAs still limits these transistors applications, such as the applications in high temperature and high voltage. However, the wide bandgap of GaN makes the corresponding HEMTs possess a huge potential for those applications. In fact, it has been shown that this type of HEMTs shows a power density several times higher than commercially available devices based in Si, GaAs, or SiC, which makes the corresponding power amplifiers have good performance in the applications for wireless base-station, communication, and military radar systems. Properties of some competing materials used in transistors are listed in Table 1-1 [2].

	Si	GaAs	6H-SiC	GaN
Bandgap (eV)	1.1	1.4	2.9	3.4
Electron mobility (cm²V⁻¹s⁻¹)	1300	5000	260	1500
Electron saturation velocity (cm/s)	1×10^7	2×10^7	2×10^7	2.5×10^7
Breakdown field (MV/cm)	0.25	0.4	2.5	4
Thermal conductivity (Wm⁻¹K⁻¹)	150	50	350	170
Dielectric constant	11.8	12.8	10	9.5

Table 1-1: Comparison of the basic parameters for some competing semiconductor materials [2].

In order to improve the device performance and reliability, different heterojunction structures for GaN HEMTs have been studied. The basic properties of the most typical heterostructures are summarized in Table 1-2.

1) AlInN barrier

The use of AlInN as the barrier layer to form heterojunction of AlInN/GaN can lead to much higher 2DEG density ($\sim 1.9 \times 10^{13} \text{ cm}^{-2}$) compared to the traditional barrier layer AlGaN ($\sim 1 \times 10^{13} \text{ cm}^{-2}$), due to the much larger spontaneous polarization of AlInN than that of AlGaN [3]-[5]. On the other hand, the AlInN layer with appropriate Al mole fraction (~ 0.82) can have same lattice constant as the GaN buffer, i.e., the AlInN layer is not strained, which might improve the device reliability. However, there is a disadvantage, namely, the large alloy scattering at the AlInN/GaN interface, which may play an important role to

reduce the electron mobility.

2) AlN barrier

AlN can be directly used as the barrier layer to form the AlN/GaN heterojunction [6]. This heterostructure shows similar behavior to the AlInN/GaN heterojunction, which means higher 2DEG density ($\sim 2.1 \times 10^{13} \text{ cm}^{-2}$) induced by both large spontaneous and piezoelectric polarization and lower electron mobility. Due to the large lattice mismatch between AlN and GaN, the AlN barrier layer cannot be grown very thick (generally 3-6 nm AlN [6] compared to traditional 15-25 nm AlGaIn). This thin AlN layer could lead to elevated gate leakage current by direct electron tunneling from the gate metal.

3) AlN spacer

In AlGaAs/GaAs HEMTs, a thin undoped spacer was typically introduced between the doped barrier and the undoped channel to reduce scattering. Similarly, a thin 1 nm AlN layer can be inserted into the traditional heterostructure to form AlGaIn/AlN/GaN or AlInN/AlN/GaN. These structures can increase the electron mobility due to the reduced alloy scattering [7][8].

4) Double heterojunction HEMTs (DH-HEMTs)

AlGaIn/GaN/InGaIn/GaN DH-HEMTs [9] have been demonstrated to increase the confinement of 2DEG and reduce the buffer leakage current.

5) Cap layers

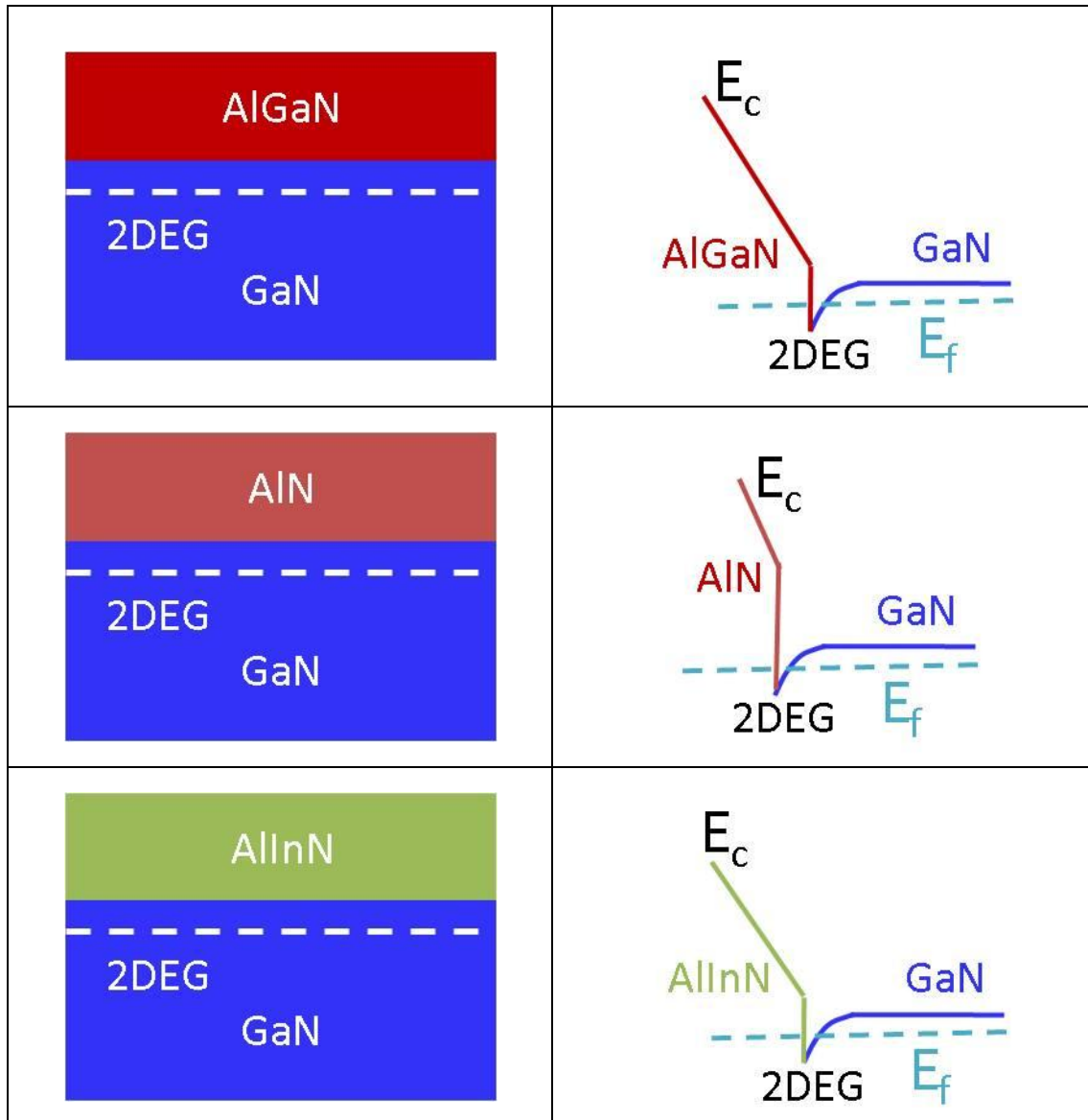
With an additional GaN cap layer, the GaN/AlGaIn/GaN heterojunction has been shown to improve the device electrical performance [10] and also to improve the device reliability from the mechanical failure point of view due to the reduced total strain in the device [11].

An InGaIn cap layer in InGaIn/AlGaIn/GaN heterojunctions can introduce a polarization field, and thus raise the conduction band at the AlGaIn/GaN interface, which makes the threshold voltage positive shift and thus convert the

device from D-mode to E-mode [12][13]. Another approach of E-mode HEMTs fabrication is the gate recess in the barrier layer [14].

6) Gate dielectric

For most of the heterostructures, it is possible to deposit an additional dielectric layer, such as the usually used SiN or SiO₂, to form MOS-HEMTs. This dielectric layer can effectively reduce the gate leakage current.



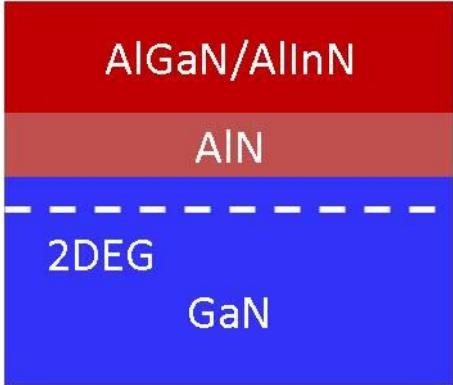
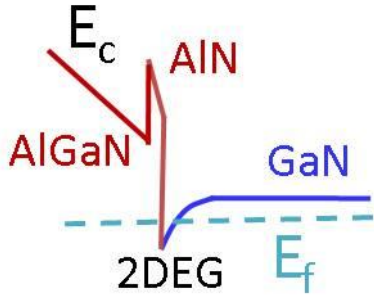
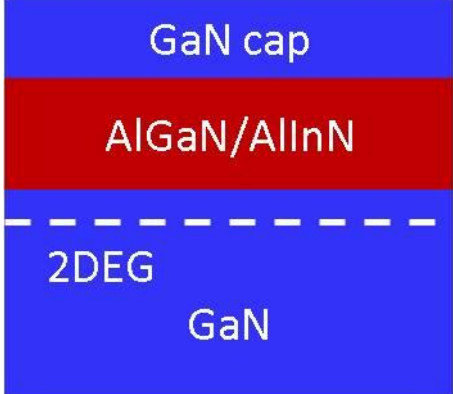
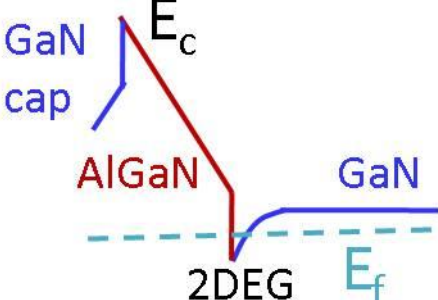
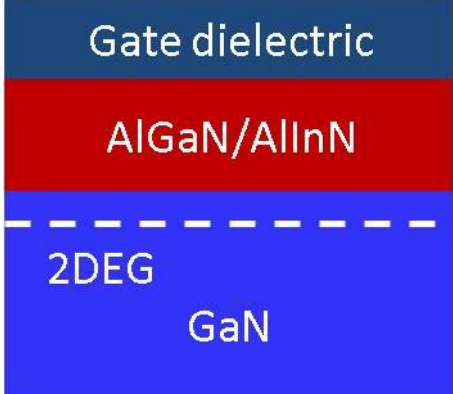
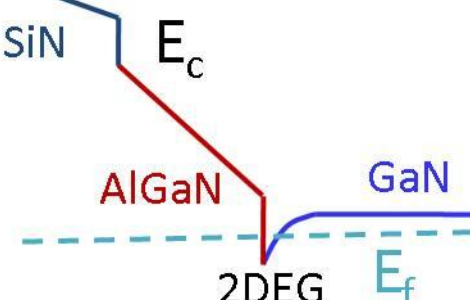
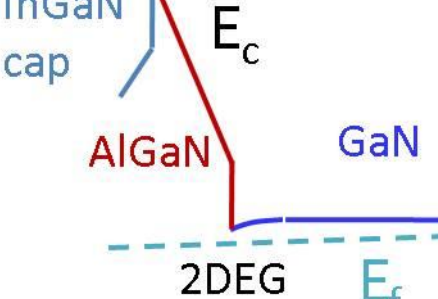
	
	
	
	

Table 1-2: Different structures for GaN HEMTs

1.2 Current issues in GaN HEMTs

Although the material properties and device design have become matured with intensive studies in recent years, there are still some issues that limit the device practical applications. They are described in the following.

1.2.1 Reliability/degradation

A. Electrical stress

The electrical degradation of GaN HEMTs was mainly manifested as the gate current increase and drain current drop [15][16].

The AlGaIn layer epitaxially grown on GaN has built-in tensile stress due to its smaller lattice constant compared to that of the GaN layer. This tensile stress limits the thickness of the grown AlGaIn layer, because relative thick AlGaIn layer might relax, presenting considerable defect or crack density which would limit the device performance.

GaN and its alloy AlGaIn are piezoelectric materials. They suffer from the converse piezoelectric stress when an external biases (gate and drain bias) are applied on them. It has been observed that cracks/pits could be formed at the gate edge of the drain side when the converse piezoelectric stress exceeds a critical value [17] (see Figure 1-1). These cracks/pits are responsible for the drain current drop and gate current increase, as shown in Figure 1-2.

In order to investigate the electrical degradation, the device can be biased in four different situations [11]:

- i) high power: the device is biased at high drain current and high drain voltage;
- ii) on-state: the device is biased in the saturation region of the drain current with relative low drain voltage;

iii) off-state: the device has very low drain current but biased at high drain voltage; and

iv) $V_{DS}=0$ state: the device is biased at large negative gate voltage while the source and drain terminals are shorted.

The biases were varied in a step-stress manner. For the latter two cases, the self-heating can be neglected due to the low drain current. In all of these cases, the device was observed to have electrical degradation. Especially, a critical

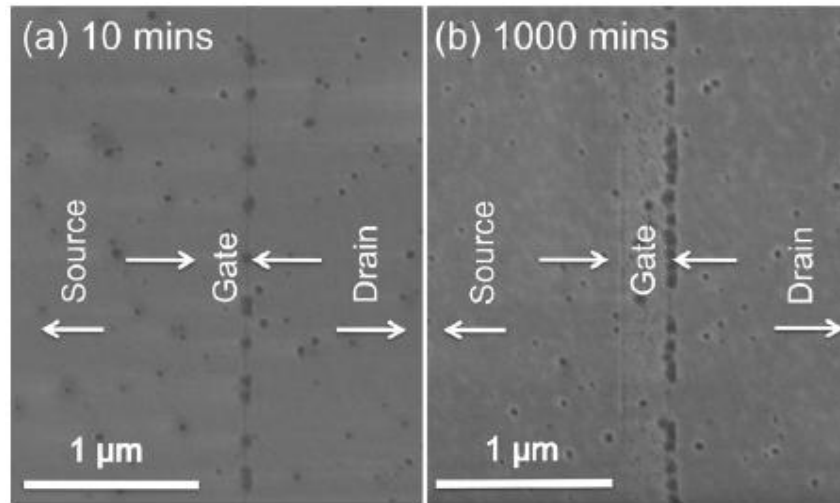


Figure 1-1: SEM images of pits formation at the gate edge of the drain side after electrical stress during 10 minutes (a) and 1000 minutes (b) [17].

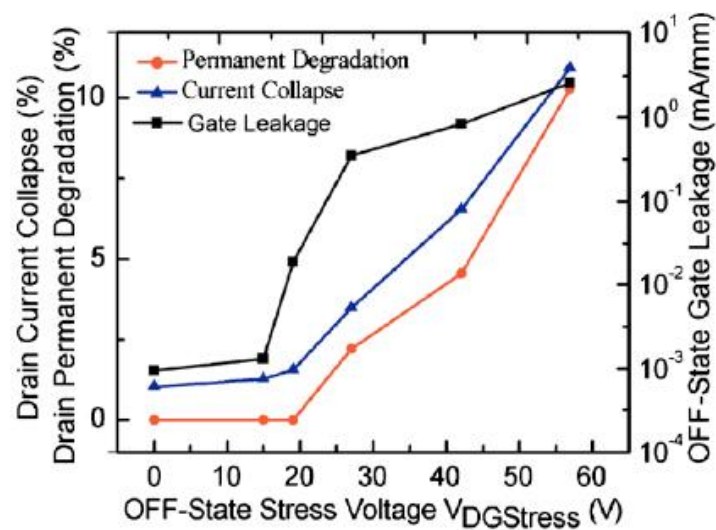


Figure 1-2: Electrical behavior degradation after electrical stress [17].

bias can be identified from the sudden increase of the gate current, as illustrated in Figure 1-3.

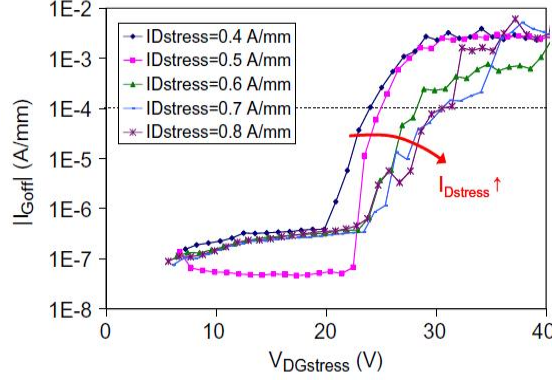


Figure 1-3: Gate current for the device in off-state as a function of the voltage between the gate and drain contact. The device is biased in a step-stress manner [11].

B. Thermal stress

GaN HEMTs have potential applications at high temperature due to the wide bandgap of GaN. However, the degradation under thermal stress described in the following is a significant limitation for this application. Also, high temperature characterization constitutes an adequate procedure for aging and reliability tests.

An increase of the Schottky barrier height was observed for Ni/Au Schottky contacts, due to an additional interfacial layer formed between the gate and the AlGaN surface induced by the elevated junction temperature when the device in high DC stress [18][19]. Long time thermal storage tests showed an increase resistance of Ti/Al/Ni/Au ohmic contact together with an increase of surface roughness. This degradation was attributed to Au inter-diffusion within the metal layers, and Ga out-diffusion from the semiconductor into the metallic compounds [19][20]. The changes of Schottky barrier height and ohmic contact resistance induced by thermal stress are always manifested as the threshold voltage shift and drain current drop.

In addition to the degradation/reliability issues described above, the traps and the self-heating effects are the other two main factors limiting the device performance. The traps effects are mainly determined by the material properties of GaN as well as the device fabrication process. The self-heating effect is determined by the fact of the device in high power operation conditions as well as the technique of the thermal management. These two effects are illustrated in the following.

1.2.2 Traps effect

It is well known that GaN HEMTs have transconductance frequency dispersion, gate/drain-lag transient [21], and limited microwave power output. In DC conditions, the electron emission and capture process by the trap centers has an equilibrium state. This yields to the drain current drop and threshold voltage shift compared to the case of absence of trap centers in the device. The transient gate- and drain-lag or DC-RF dispersion means that the change of drain current cannot follow the applied bias signal, leading to the current or power output collapse, due to the large characteristic time for the detrapping process. It was demonstrated that probably the drain-lag was related to the buffer and substrate traps, whereas the gate-lag was related to the surface traps [22].

Many techniques have been developed to characterize the traps effect including electrical approaches (frequency-dependent capacitance and conductance characterization [23], or pulse/transient measurements) and optical approaches (DLTS, DC electroluminescence, and photo-ionization experiments) [24][25]. Figure 1-4 shows a typical pulse (a) and transient (b) measurement for the gate-lag characterization [26]. In the measurements, the gate terminal was pulsed from off- to on-state while the drain voltage was fixed. As shown in

Figure 1-4(a), there occurs a current collapse compared to DC measurements, especially for the narrower pulse width. Figure 1-4(b) shows the transient drain current measured at different ambient temperature. The shift of this transient current reflects the variation of electron emission rate (e_n) with temperature. By performing a stretched exponential fitting for the transient current, the electron emission characteristic time ($\tau_n = 1/e_n$) and further the Arrhenius plot can be obtained, as shown in the inset of Figure 1-4(b). Finally, with a linear fitting, the traps ionization energy can be extracted.

Although the origin of the traps in the device is still under debate, it seems to be mostly generated by nitrogen vacancies caused by plasma damage in the processing, the dislocations at the heterojunction interface, and the unintentional Fe or C doping in the buffer layer. In [27], an electrical approach was described to characterize the traps location in the device. It was concluded that traps in the barrier led to the threshold voltage shift while the traps on the AlGaN surface between the gate and drain contact led to the reduction of the peak transconductance. DLTS measurements have shown that both deep-level acceptor and donor-like traps exist in the GaN buffer, whereas capacitance and conductance characterization indicated that traps exist at the AlGaN/GaN

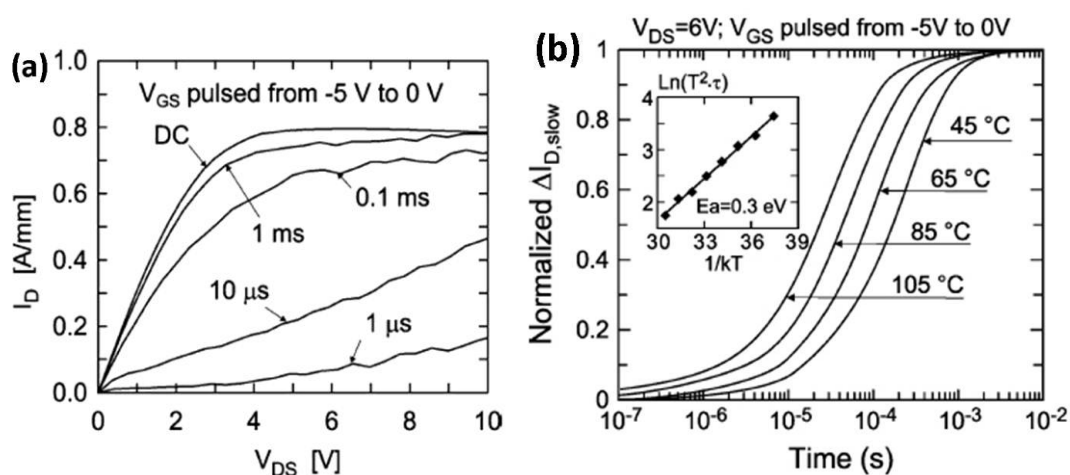


Figure 1-4: Gate-lag characterization (gate terminal was pulsed from off- to on-state while drain voltage was fixed): pulse measurement (a) and transient measurement (b) [26].

interface and under the gate.

Regarding to mechanisms of current collapse caused by the presence of traps in the device, a model based on the large surface states of AlGaN has been typically used [28]. It was illustrated that the surface states of AlGaN near the gate edge of the drain side can capture the electrons tunneling from the gate metal or the channel. The surface area with the trapped electrons could reduce the 2DEG density in the correspondent position of the drain-side access region (see Figure 1-5), i.e., the channel resistance was increased and thus the drain current was decreased. This surface area acts like the increase of the gate length and therefore was called “virtual gate”. The traps effects could be reduced by adding a passivation layer on top of the device or using field-plate gate structure. This will be described below.

The use of some materials for the passivation on top of the device, such as SiN, SiO₂, AlN, Al₂O₃, HfO₂, etc, were found to effectively reduce the current collapse. The mechanism commonly explained for this reduction is that the passivation materials reduce the surface states of AlGaN and thus the resulting traps effects. Intensive researches have been devoted to the characterization and

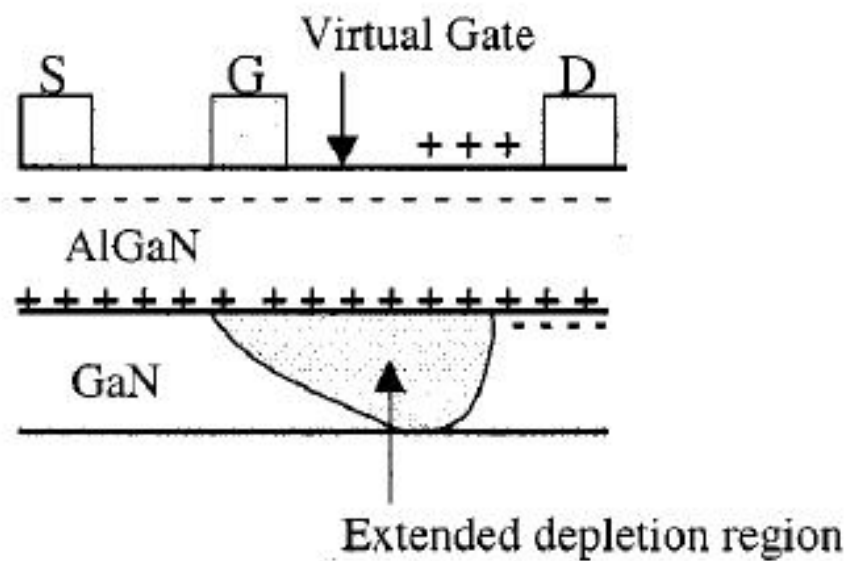


Figure 1-5: Schematic of the virtual gate effect [28].

processing of the passivation.

Using a gate field-plate for spreading the electric field near the gate edge of the drain side is a common approach to enhance the device breakdown voltage. In addition to the breakdown voltage enhancement, the field-plate was also found to reduce the current collapse from the point of view of traps effects. A transient drain current simulation including the traps effect for comparison of the current collapse for the device with and without field-plate is showed in Figure 1-6. We can see that the drain current collapse was suppressed for the device with gate field-plate [29][30]. This can be explained because the reduction of the electric field by the field-plate suppresses the electron tunneling from the gate metal to the surface traps or capping layer traps and therefore reduce the traps occupation probability. Such type of simulations was also performed for the traps in the GaN buffer [31].

Figure 1-7(b) shows that, due to the reduced electric field, fewer electrons were pulled into the GaN buffer and therefore less electrons were trapped there for the device with a gate field-plate, compared to the device with a standard gate (Figure 1-7(a)). As a result, I_{DS} collapse was also reduced [31].

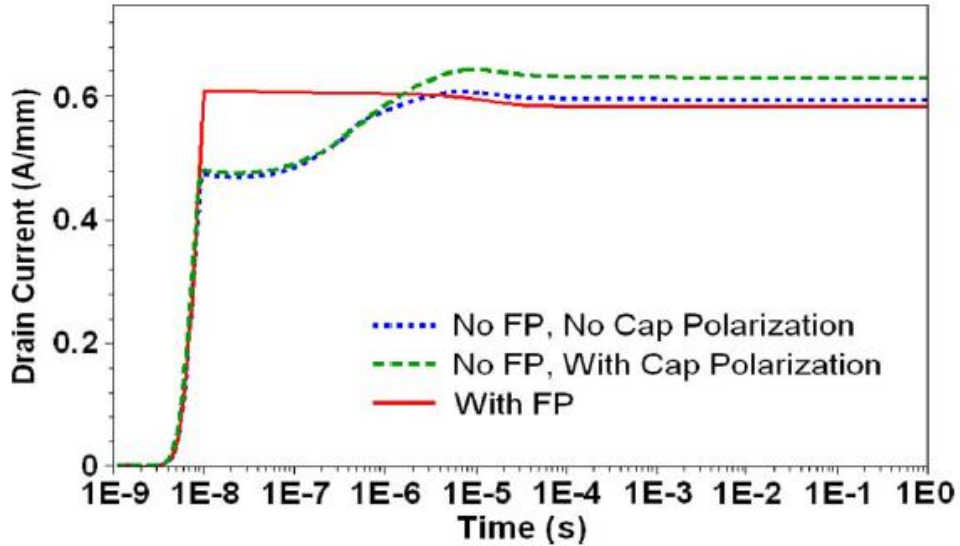


Figure 1-6: Simulation of the transient I_{DS} including the traps effect for the device under pulsed gate bias. The device with and without gate field plate were compared [29].

We note that there is a disadvantage, namely, that the field plate can increase the gate parasitic capacitance, and therefore limit the device RF performance. More details will be illustrated in Section 3.1.3.

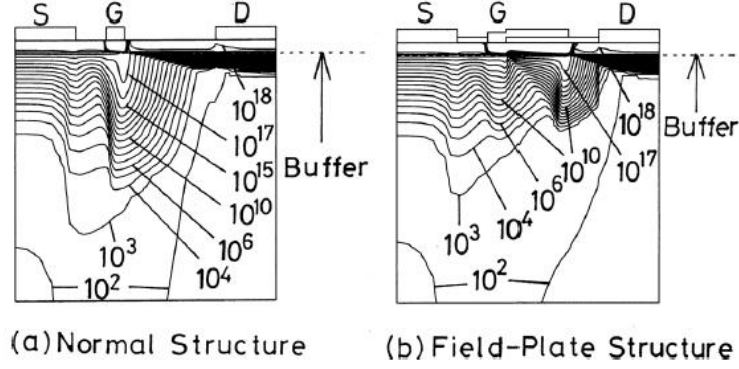


Figure 1-7: Simulation of electron concentration in the GaN buffer for the device with standard gate (a) and field-plate gate (b) [31].

1.2.3 Self-heating effect

Due to the high electron density and mobility in the channel, AlGaN/GaN HEMTs show a huge potential for high-power applications. However, the device in high power operation may output decreased drain current, which is mainly attributed to the decreased electron mobility caused by the elevated lattice temperature, named self-heating effect [32]-[35]. This effect is even severer for the device with multi-fingers due to the thermal crosstalk between the fingers [36][37]. Therefore, good thermal managements would be demanded for reduction of the self-heating effect.

1.2.4 Hot electron effect

When the device working in the on-state with high power level, i.e., there is a large electron current in the channel. The electrons are accelerated by high electric field to own large kinetic energy (hot electrons), so that they may have enough energy to induce traps either in the channel, in the barrier or at the

interface. In addition to that, the electrons may directly tunnel from the channel to the gate or to the AlGaN surface and be trapped there. The induced traps can also partially deplete the 2DEG in the channel and lead to the drain current collapse. The hot electrons can be characterized by electroluminescence (EL) measurement. Figure 1-8 shows reported equivalent hot electrons temperature characterized by this method for the device in a typical on-state bias [24].

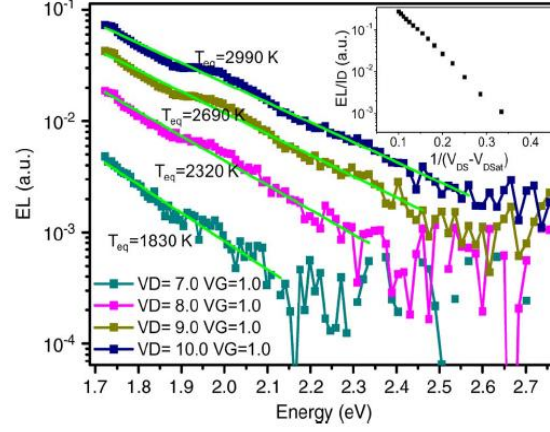


Figure 1-8: Equivalent electron temperature T_{eq} characterized by EL measurement [24].

1.3 Motivations and research objectives

Up to now, the GaN HEMTs have been investigated in a wide range of topics, such as materials growth techniques, device fabrication techniques, characterizations for thermal effects, traps effects, and the device performance degradations caused by thermal or electrical stress, device passivation for reduction of traps effects/improvement of device performance, thermal management for reduction of self-heating effect, etc. It is well known that the device degradation, the self-heating effect, and the traps effects are important issues that limit the practical applications of GaN HEMTs. Intensive studies are now being devoted to understand their mechanisms and eliminate their deleterious influences, which are also our research objectives in this

dissertation. In particular, they include:

- 1) Develop model and correspondent simulation method that can comprehensively study the device electrical, thermal, and stress behavior (Chapter 2).
- 2) Investigate the substrate and the device geometry impact on the self-heating effect for optimization of the device design by simulations and electrical characterization method. The device stress behavior will also be studied for standard and field plate gate structure for investigation of mechanical failure mechanisms (Chapter 3).
- 3) Simulate the device thermal management by integrating a high thermal conductivity diamond layer on top of the device (top-side heat spreading) or in vias etched in the substrate (back-side heat spreading) for potential reduction of the device self-heating effect (Chapter 4).
- 4) Investigate the intrinsic stress in the diamond layer impact on the device electrical behavior by both simulations and experiments for potential improvement of the device performance by stress engineering, as well as reduction of the mechanical failure (Chapter 5).
- 5) Develop model and simulation method of traps effects that can include both the temperature and electric field impact on the electron emission rate, and furthermore comprehensively investigate mechanisms of the reverse gate current and the device bias via self-heating and electric field impact on the device transient drain current (Chapter 6).

2 Simulation and characterization method

In this chapter, details of the simulation method including electro-thermal, electro-mechanical, and electro-thermo-mechanical coupling will be described. The applied boundary conditions for the relevant equations (ohmic and Schottky contact for the electrical active regions of the device, and the thermal as well as stress boundary conditions for the substrate) will be described. A brief introduction of the commercial FEM simulation tool COMSOL, the electrical method for channel temperature (T_{ch}) characterization, and the micro-Raman spectroscopy technique for stress characterization, will also be presented.

2.1 Simulation method

2.1.1 Coupling equations

The semiconductor device simulation method described in the following is applied to GaN HEMTs. The relevant equations taken into account in our simulations are Poisson, electron current continuity, thermal conduction, and elasticity equations. Due to the general large gate width/length ratio, the simulations will be restricted to two-dimensions (2D) for simplicity, unless other conditions are specially stated.

A. Poisson equation

The Poisson equation is expressed as

$$\nabla^2 \psi = -\frac{\rho}{\varepsilon_0 \varepsilon_r}, \quad (2.1)$$

where ψ is the electrical potential, ρ is the space charge density, ε_0 is the

vacuum permittivity, and ε_r is the relative permittivity. ρ can be constituted by electrons, holes, doping, defects, traps, etc. Some of them can be neglected depending on the specific model and simulations. In the ideal situation, holes would exist near the AlGaIn surface as the source of the 2DEG in the channel. However, it was concluded that the 2DEG should come from the surface states of the AlGaIn layer [1], leading to negligible hole concentration in the device. So holes were not included in our simulations.

B. Electron current continuity equation

As an approximation of Boltzmann equation, the electron current continuity equation can be expressed as

$$\frac{\partial n}{\partial t} = \frac{1}{q} \nabla \cdot \mathbf{j}_n, \quad \mathbf{j}_n = -qn\mu_n \nabla \phi_n, \quad (2.2)$$

where n is the electron concentration, t is time, q is the basic charge unit, $\mathbf{j}_n$ is the electron current density, μ_n is the electron mobility, and ϕ_n is the quasi-Fermi level. n is dependent on ψ and ϕ_n (Eq. 2.3), which is calculated by Fermi-Dirac statistics with the Aymerich-Humet approximation using an analytical expression [38]. Eq. 2.2 is used as a transient simulation for the device in pulse operation conditions. The time dependent term $\frac{\partial n}{\partial t}$ is replaced by zero for the steady-state simulation. Thus,

$$n = \frac{N_c}{\left[\exp(-\xi) + 3 \sqrt{\frac{\pi}{2}} (\xi + 2.13 + (|\xi - 2.13|^{2.4} + 9.6)^{\frac{5}{12}})^{1.5} \right]}, \quad (2.3)$$

$$\xi = \frac{[q(\psi - \phi_n) + \chi]}{k_B T}, \quad (2.4)$$

where k_B is the Boltzmann constant, T is temperature, and χ is the electron affinity energy. N_c is the conduction state density, expressed as

$$N_c = \frac{2(2\pi m_e^* kT)^{\frac{3}{2}}}{h^3}, \quad (2.5)$$

where m_e^* is the electron effective mass and h is the Planck constant. We note

that if n is approximated by Boltzmann distribution, the drift-diffusion model for electron transport can be obtained, with n as a dependent variable. In all of our simulations, we chose ϕ_n instead of n as the dependent variable for electron transport (Eq. 2.2) in order to avoid the non-physical phenomenon of negative carrier concentration. Another consideration is that the Fermi-level (E_f) is generally higher than the conduction band edge (E_c) for the GaN area close to the heterojunction, where the Boltzmann distribution is not valid.

Some of the mentioned physical quantities (E_c , χ , etc.) described above are showed schematically in Figure 2-1 and specific used values are listed in Table 2-1. As AlGaN has a larger bandgap (E_g) than GaN (Figure 2-1(a)), after their contact there are conduction (ΔE_C) and valance (ΔE_V) band discontinuities at the AlGaN/GaN interface and the 2DEG is formed near this interface in the GaN buffer (Figure 2-1(b)).

To model the electron mobility μ_n , we will express it as dependent on the temperature and parallel electric field (E_x),

$$\mu_n = \mu_{n0} \left(\frac{300}{T} \right)^\beta \left(1 + \frac{\mu_{n0}|E_x|}{v_{sat}} \right)^{-1}, \quad (2.6)$$

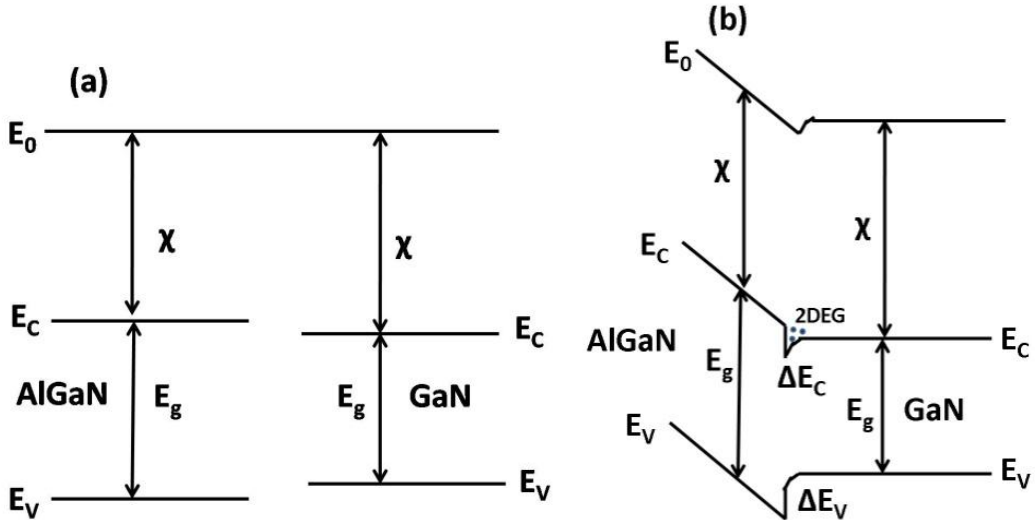


Figure 2-1: Schematic of physical quantities of AlGaN and GaN before contact (a) and after contact (b). Note that the specific values of E_c (conduction band edge), χ (electron affinity), etc., labeled in the graphs are listed in Table 2-1.

where μ_{n0} is the electron mobility at room temperature 300 K, β is a coefficient describing μ_n variation with T , and v_{sat} is the electron saturation velocity. It is clear that μ_n decreases when T increases, representing an increased lattice scattering. According to Eq. 2.6, for increasing E_x , the electron velocity will tend to its saturation value v_{sat} .

	Al_{0.25}Ga_{0.75}N	GaN
χ	3.8 eV	4.1 eV
E_g	3.9 eV	3.4 eV
ΔE_C	0.3 eV	
ΔE_V	0.1 eV	
m_e^*	0.23 m_0	0.2 m_0

Table 2-1: Basic parameters of AlGa_{0.25}N and GaN at room temperature (similar to the ones used in [39]) used in the simulations. Other parameters, such as thermal conductivity, electron mobility, Schottky barrier, etc., were presented in specific simulations.

C. Thermal conduction equation

The thermal conduction equation used to describe the device thermal behavior can be expressed as

$$\rho c \frac{\partial T}{\partial t} = \nabla \cdot k \nabla T + \mathbf{j}_n \cdot \mathbf{E}, \quad (2.7)$$

where ρ is the mass density, c is the heat capacitance, k is the thermal conductivity, $\mathbf{E}$ is the electric field, and $\mathbf{j}_n \cdot \mathbf{E}$ is the Joule heat generated by the electron current. Again, the time-dependent term $\frac{\partial T}{\partial t}$ is replaced by zero in the thermal steady-state simulation.

C. Equation of motion governing linear elasticity

The equation of motion governing linear elasticity can be expressed as

$$\rho \frac{\partial^2 u}{\partial t^2} = \nabla \cdot \sigma, \quad (2.8)$$

where u is the displacement field, ρ is the mass density, and σ is the mechanical stress.

In order to analyze the device stress behavior (thermal stress, converse

piezoelectric stress, and intrinsic stress or built-in stress), the following constitutive equations will be used,

$$\begin{aligned}\sigma_{ij} &= c_{ijkl} [s_{kl} - \alpha(T - T_{ref})] - e_{kij} E_k \\ D_i &= e_{ijk} s_{jk} - \varepsilon_{ij} E_j + P_i^s,\end{aligned}\tag{2.9}$$

where σ_{ij} is the stress tensor, c_{ijkl} is the elastic stiffness tensor, s_{kl} is the strain tensor, e_{ijk} is the piezoelectric coefficient tensor, ε_{ij} is the permittivity tensor, E_k is the electric field vector, D_i is the electric displacement vector, and P_i^s is the spontaneous polarization vector.

We note that, if the piezoelectric effect is included in the simulations, the AlGaIn surface and the AlGaIn/GaN interface polarization charge can be determined automatically. However, if the piezoelectric stress is not included in the simulations, due to lack of information on the surface states or defects in the device, these polarization charges are usually set manually to make the 2DEG density similar to the result from Hall measurements.

Figure 2-2 shows a schematic of the simulation process with a full electro-thermo-mechanical coupling. The built-in stress in the AlGaIn and GaN epitaxial layers, intrinsic stress in the passivation, converse piezoelectric stress induced by the device bias, and thermal stress induced by the elevated device temperature influence the piezoelectric polarization charge, the 2DEG density, and further the device output characteristics. At the same time, conversely, these output characteristics influence the device piezoelectric behavior, for example, the output power determines the device temperature and thus the thermal stress, presenting a fully coupling process as shown in the graph of Figure 2-2.

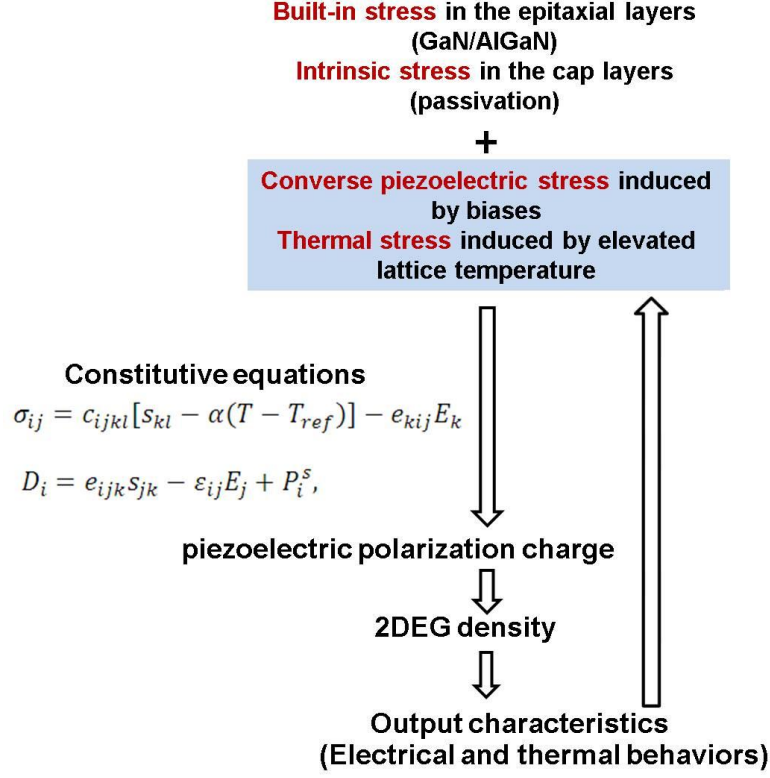


Figure 2-2: Schematic of coupling process for a full electro-thermo-mechanical simulation.

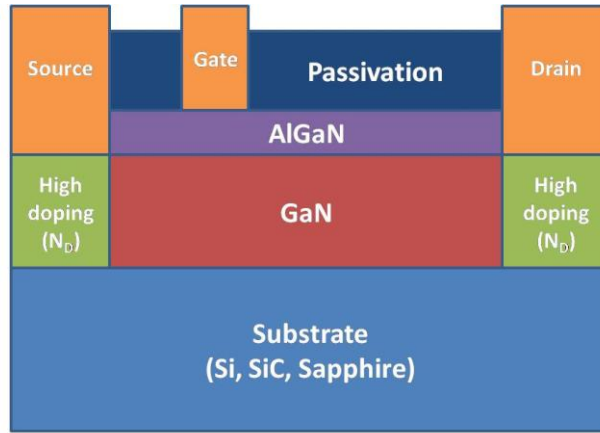
In principle, all the equations should be solved and using a full coupling manner for optimum simulations. However, the complexity, large amount of computation, and numerical problems (non-convergent results) present difficulties for these simulations. Therefore, depending on specific topics, only some of the equations will be solved and correspondent coupling manner will also be adjusted (one-way coupling or full coupling). See a summary of the different simulation methods and the correspondent solved equations in Table 2-2. More details on the procedures will be described in the following chapters.

Simulation methods	Coupling equations
Equilibrium state ($V_{DS}=0$)	• Poisson equation
Electrical simulation	• Poisson equation • Current continuity equation
Electro-thermal simulation	• Poisson equation • Current continuity equation • Thermal conduction equation
Electro-mechanical simulation	• Poisson equation • Current continuity equation • Motion equation governing linear elasticity
Electro-thermo-mechanical simulation	• Poisson equation • Current continuity equation • Thermal conduction equation • Motion equation governing linear elasticity

Table 2-2: Summary of the different simulation methods and correspondent solved equations.

We note that different equations are applicable to different parts of the device (see Figure 2-3). The Poisson equation (Eq. 2.1) and piezoelectric coupling (Eq. 2.9) are solved in the AlGa_N and Ga_N layer. Due to the low electron density in the AlGa_N barrier, the electron continuity equation (Eq. 2.2) is solved only in the Ga_N buffer. Simulations of the thermal (Eq. 2.7) and mechanical stress (Eq. 2.8) behavior are applied in all the parts of the device: AlGa_N, Ga_N, substrate, metal contact (ohmic and Schottky), and passivation.

Finally, note that for all simulations presented in this dissertation, the quantum effect that needs to be coupled solving Schrodinger equation was not taken into account. How the results are influenced by the quantum effect is not addressed. Generally, a significant difference is that the maximum electron density appears close to the interface (not at the interface) with including the quantum effect.



Eq. 2.1 (Poisson equation) and **Eq. 2.9** (piezoelectric coupling) are used in GaN and AlGaIn layer



Eq. 2.2 (Electron current continuity equation) is used in GaN layer



Eq. 2.7 (Thermal conduction equation) and **Eq. 2.8** (motion equation governing linear elasticity) are applied in GaN , AlGaIn layer, substrate, contact metal, and passivation



Figure 2-3: Schematic of a basic structure of AlGaIn/GaN HEMT applied in the simulations. Note that different equations are applicable to different parts of the device.

2.1.2 Boundary conditions

A schematic of the applied boundary conditions for GaN HEMT simulation is displayed in Figure 2-4 and details are described in the following.

A. Ohmic contact

Ohmic contact is implemented as Dirichlet boundary condition, meaning ψ and ϕ_n are fixed. ϕ_n is set as zero and V_{DS} for the source and drain contact, respectively. In the simulations, the source and drain contact domain are manually set to have high density of donor doping and assume them completely ionized (N_D^+). The total charge density should be zero (neutrality). This means that

$$N_D^+ - n = 0 \quad (2.10)$$

Inserting Eq. 2.3 into Eq. 2.10 and performing numerical solution for Eq. 2.10, ψ at the source and drain Ohmic contact can be calculated.

B. Schottky contact

Schottky contact on the AlGa_N surface is also implemented as Dirichlet boundary condition, where ψ is fixed and calculated as the following (ϕ_n is not solved in the AlGa_N layer, see details in the following)

$$\psi = V_{GS} + \psi_{in}, \quad (2.11)$$

where ψ_{in} is the built-in potential determined by the surface potential of the AlGa_N/Ga_N heterojunction and the Fermi-level difference between the gate metal and the AlGa_N/Ga_N heterojunction.

C. Thermal and stress boundary conditions

Isothermal boundary condition is applied at the bottom of the substrate, representing a perfect heat sink to room temperature. Because the generated heat is highly concentrated at the gate edge of the drain side (GEDS), the area for heat exchange with the air is very small, leading to a negligible heat dissipation from the top-side of the device. Indeed, our numerical experiments show that the temperature distribution in the device is unchanged when trying different thermal exchange coefficients ranging from 10 to 100 W/m²K. Thus, adiabatic boundary conditions may be applied on the surface, as well as the

sidewalls of the device.

For the stress boundary condition, fixed constraint and mechanical free are applied at the bottom of the substrate and the sidewalls of the device, respectively.

See the appendix that describes details about how to use the coupled equations and boundary conditions described before in COMSOL.

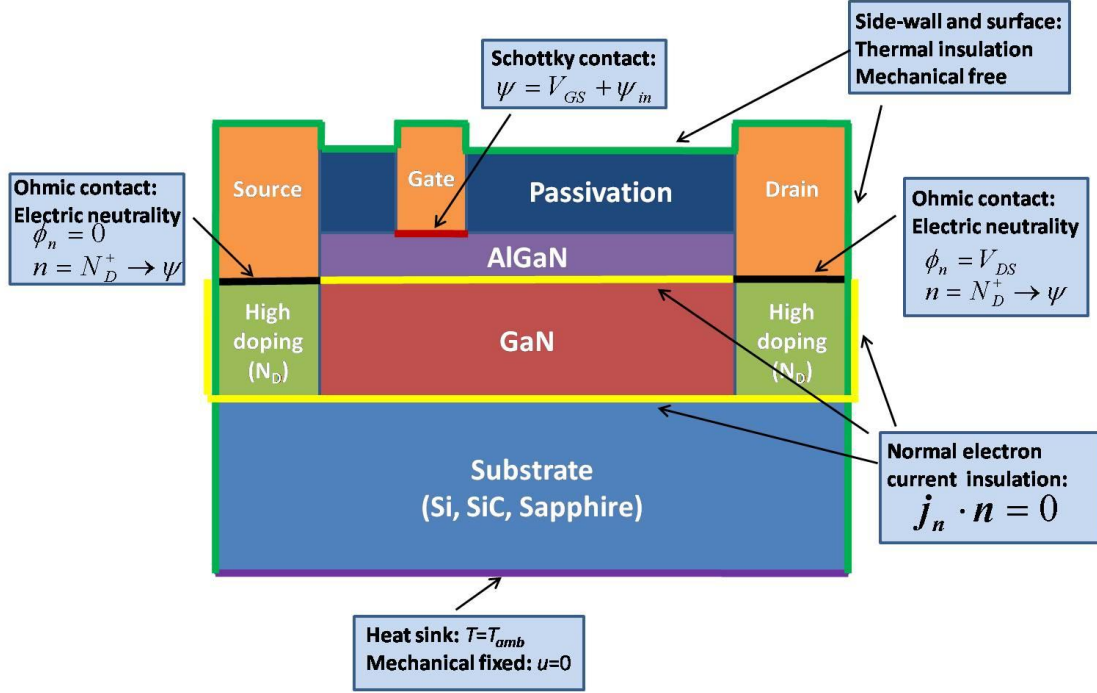


Figure 2-4: Schematic of the applied boundary conditions for GaN HEMT simulation.

2.2 Simulation tool: COMSOL

The commercial FEM software COMSOL [40] is a powerful tool for modeling and simulating many kinds of scientific and engineering problems. It includes many modules for specific physical field simulation, such as microfluidics, thermal conduction simulation, electromagnetic field simulation, electrical current simulation, mechanical stress simulation, etc., and also mathematics-based simulation in which you can input some coefficients to

form a set of equations for the description of almost any physical problems. For the specific physical interface, some physical quantities can be directly introduced or obtained, such as fluxes, sources, constraints, loads, etc. By coupling different physical interfaces in COMSOL, a complex multi-physical process can be simulated. COMSOL provides steady-state, transient state, frequency domain, engine frequency studies, etc.

In the area of semiconductor device simulation, electrical, thermal, and mechanical physics are involved. COMSOL does not provide a professional module for the simulation of GaN-based HEMTs, instead we have to manually couple the electrostatics interface, thermal conduction interface, mathematics-based partial differential equation (PDE) interface, and piezoelectric device interface. Steady and transient studies were applied in our simulations. The appendix section describes some the details of how to build models in COMSOL.

One important issue in modeling is non-convergence, which we have also faced in this work. Generally, determined by the properties of electron current in GaN-based HEMT, the drift current is much higher than the diffusion current. Thus, that may lead some numerical problems manifesting as the oscillation of the results and non-convergent results for the device in the sub-threshold state. These numerical problems can be overcome by an alternate numerical discretization method named box-integration implemented in commercial TCAD software.

2.3 Channel temperature characterization

Electrical methods [41][42], micro-Raman spectroscopy [43], and IR thermography [44] are the most used tools for the characterization of the channel temperature (T_{ch}). Electrical methods are simple and fast but cannot

determine the channel temperature distribution. Since the phonon vibration (phonon energy) is related to temperature, using appropriate material-dependent coefficients, T_{ch} can be determined by the GaN phonon frequency difference when the device is biased and unbiased [45]. Micro-Raman spectroscopy has high spatial resolution, reaching to submicron scale. Measurement of temperature using IR thermography is based on acquiring the intensity of the thermal radiation. This method allows large-area overviews of temperature distribution with relative low spatial resolution of 5-10 μm compared to 0.5-1 μm for micro-Raman spectroscopy [45]. These mentioned optical characterization methods need direct access to the device, limiting their applicability regarding the device geometry and design.

The simulated T_{ch} in this dissertation was compared with experimental data obtained by a simple and accurate electrical characterization method developed in our group [46]. This method for T_{ch} extraction is based on two assumptions. Firstly, T_{ch} can be changed by either self-heating or external heating. Secondly, self-heating is negligible for narrow pulses (≤ 350 ns) with very short duty cycles, since $I_{DS,sat}$ keeps constant for these measurement conditions (not the same behavior as $I_{DS,sat}$ decreases with V_{DS} increasing, due to the self-heating). This method consists of a calibration and a measurement step. In the calibration step, I_{DS} variation (ΔI_{DS}) is measured under pulsed bias as a function of ambient temperature (T_{amb}). Note that in this measurement the self-heating is negligible (i.e., $T_{ch} = T_{amb}$) because of the narrow pulse width (350 ns) and very low duty cycle (0.01%). Therefore, ΔI_{DS} is determined by only the external heating. For a fixed bias, relationship between I_{DS} and T_{amb} has a linear behavior (see Figure 2-5(a)). Therefore, relationship of ΔI_{DS} (ΔI_D shown in Figure 2-5(b)) and variation of ambient temperature ΔT_{amb} can be expressed as

$$\Delta I_{DS} = \Delta T_{amb} \theta, \quad (2.12)$$

where θ is calculated as the linear fit slope of I_{DS} versus T_{amb} .

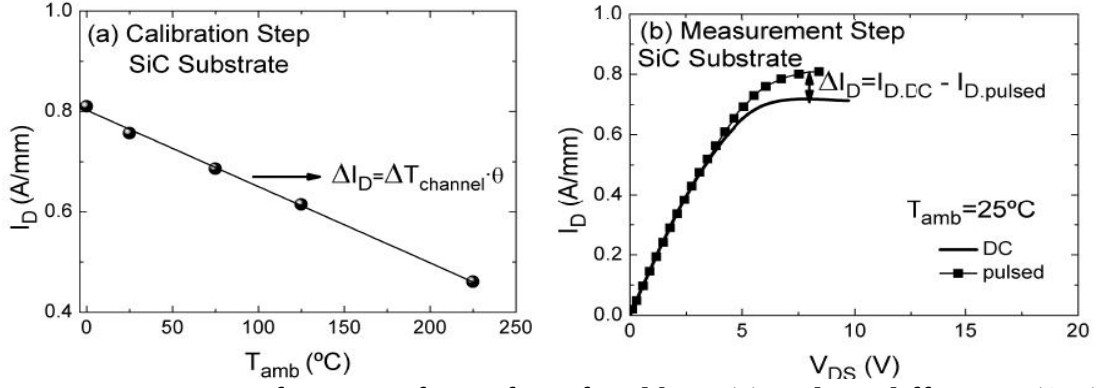


Figure 2-5: I_{DS} as a function of T_{amb} for a fixed bias (a) and I_{DS} difference (ΔI_D) between DC and pulsed measurement (b) [46].

Assuming I_{DS} reduction caused by the ambient temperature increase is same as that caused by the device self-heating, so we have

$$\Delta T_{ch} = \Delta T_{amb} \quad (2.13)$$

Inserting Eq. 2.13 into Eq. 2.12, we have

$$\Delta I_{DS} = \Delta T_{ch} \cdot \theta, \quad (2.14)$$

or

$$\Delta T_{ch} = \Delta I_{DS} / \theta, \quad (2.15)$$

Next, in the measurement step, DC measurements of I_{DS} are performed under the same bias conditions for a given T_{amb} . In this case, ΔT_{ch} is due to only the self-heating, so,

$$T_{ch} = T_{amb} + \Delta T_{ch} \quad (2.16)$$

Inserting Eq. 2.15 into Eq. 2.16, then

$$T_{ch} = T_{amb} + \Delta I_{DS} / \theta \quad (2.17)$$

In this step, ΔI_{DS} is defined as the I_{DS} difference between DC and pulsed measurement (see Figure 2-5(b)). It is caused by the device self-heating and can be expressed as

$$\Delta I_{DS} = I_{DS,DC} - I_{DS,Pulsed} \quad (2.18)$$

Finally, T_{ch} can be determined as

$$T_{ch} = T_{amb} + (I_{DS,DC} - I_{DS,Pulsed}) / \theta \quad (2.19)$$

We note that the accuracy of this method depends on the assumption that the

difference of $I_{DS,DC}$ and $I_{DS,Pulsed}$ is solely caused by the device self-heating. However, it is well known that AlGaIn/GaN HEMTs have amounts of trapping centers. In order to suppress the current collapse induced by the electron trapping, the device is biased in a double pulsed manner (a schematic of double-pulsed measurement is shown in Figure 2-6) and the quiescent gate ($V_{GS,Q}$) and drain voltage ($V_{DS,Q}$) are zero volt [47][48]. This is different from the case that the device is biased from off- ($V_{GS,Q}$ lower than V_{th}) to on-state, in which the electron trapping and detrapping process are involved.

In this dissertation, DC output characteristics were measured by HP/Agilent 4156C semiconductor parameter analyzer and a Janis probe station, which allows measurements under low/high temperature. The double-pulsed measurements were performed by a Yokogawa DLM2000 digital oscilloscope and an Agilent 81150A pulse function arbitrary noise generator remotely controlled by a computer.

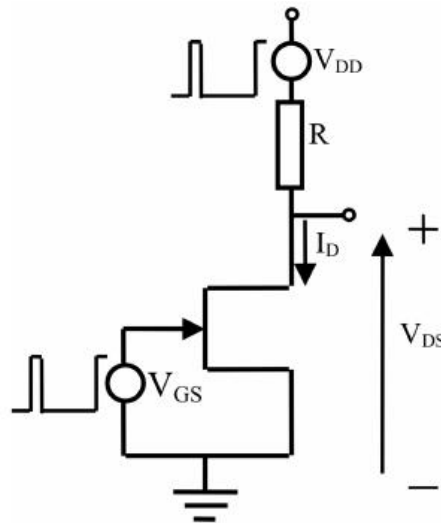


Figure 2-6: Schematic of double-pulsed measurement. Both of the gate and drain terminals are under pulsed biases.

2.4 Stress characterization

The built-in stress induced by epitaxial grown and converse piezoelectric stress induced by the gate and drain bias in AlGaIn/GaN HEMTs were

considered to be related to the device reliability issues. Therefore, this stress characterization is also important, and intensive researches have been performed on this aspect [49]-[52]. Generally, the phonon vibration is not only relevant to the device temperature, but also to the stress via the parameter of deformation potential. Based on this fact, micro-Raman spectroscopy can be used to determine the converse piezoelectric stress distribution [53], in which the device is biased in off-state to avoid the self-heating, so that the difference of the phonon frequency is only determined by the converse piezoelectric stress without the temperature superposition effect. Figure 2-7 shows the strain and stress distribution characterized by micro-Raman spectroscopy and corresponding modeling results for the device in off-state [53]. We can see that the maximum converse piezoelectric stress appears at the gate edge of the drain side (GEDS) because of the maximum electric field there. Also the strain is proportional to the applied voltage, as shown in the inset of Figure 2-7.

Thermal stress is induced in GaN HEMT under high power operation due to the different thermal expansion coefficients for different materials in the device as well as the large temperature gradient. By considering two modes of

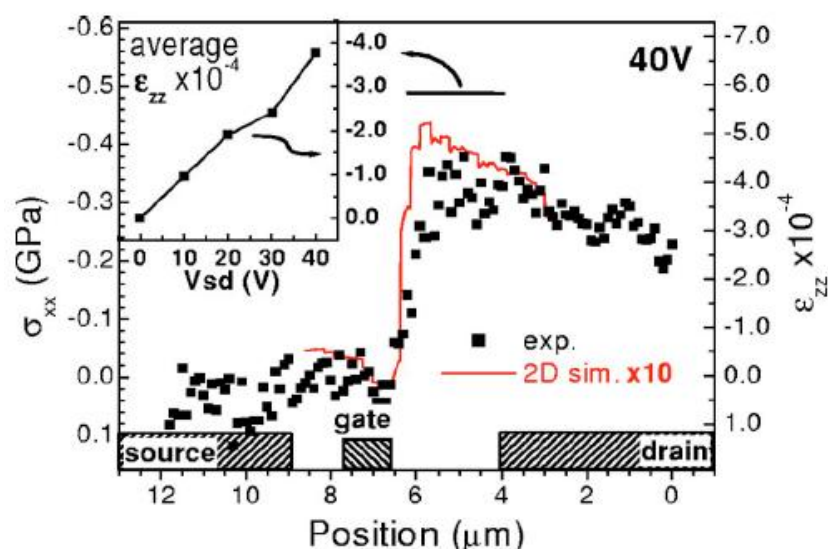


Figure 2-7: Strain and stress characterized by Micro-Raman spectroscopy and correspondent modeling results. The device is biased in off-state [53].

phonon vibration in GaN, the temperature and thermal stress can be characterized simultaneously by Micro-Raman spectroscopy [54]. Figure 2-8 shows the thermal stress and temperature distribution characterized by this method for the device in on-state [54]. We can see that the maximum thermal stress appears at the GEDS (note that this stress is compressive so the value is negative) because of the maximum temperature there. Overall, mechanical failure might happen at the GEDS because of the maximum converse piezoelectric and thermal stress (maximum elastic energy), further leading to defects/traps.

Our simulation results with electro-thermo-mechanical coupling will address the additional stress in the channel induced by the intrinsic stress in diamond layer. The characterization of this additional stress by micro-Raman spectroscopy is based on the difference of phonon frequency in GaN for the device with and without integrated diamond layer. See more details in Chapter

5

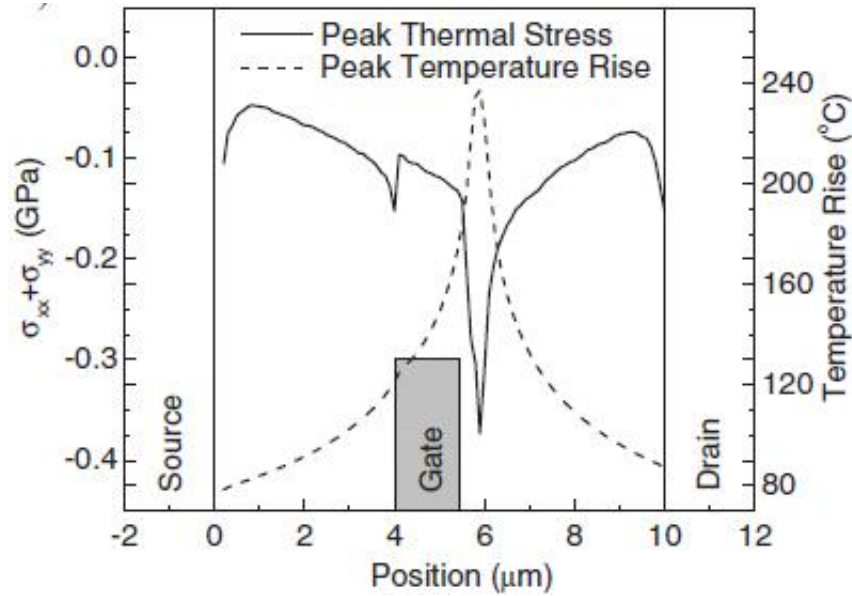


Figure 2-8: Stress and temperature distribution characterized by Micro-Raman spectroscopy and correspondent modeling results. The device is biased in on-state [54].

3 Basic electrical, thermal, and stress behavior of GaN HEMTs

In this chapter, we will describe how the distribution of electrical potential, electric field, electron density, electron conduction band edge, and quasi-Fermi level change with V_{DS} and V_{GS} for better understanding the device operation principle. The fringing and field plate capacitance will be taken into accounts, as they are important for the device in high-frequency applications. We will simulate the substrate and device geometry impact on the device thermal behavior, and results will be compared with experimental results from electrical characterizations. Finally, simulations on the device stress behavior for standard and filed-plate gate structure will be compared.

3.1 Electrical behavior

3.1.1 Description of the electrical behavior

Figure 3-1 shows simulations on the distribution of the electrical potential (a), electric field (b), electron density (c), electron velocity (d), and electron mobility (e) in the channel influenced by V_{DS} with fixed V_{GS} (the device is in $I_{DS, Sat}$ region). With V_{DS} increasing, the electrical potential drop in the access region between the gate and source/drain contact is not changed and it mainly appears at the GEDS (Figure 3-1(a)), because the device is biased in $I_{DS, Sat}$ region. As a result, the increased electric field appears at the GEDS (Figure 3-1(b)). Figure 3-1(c) shows that the electron density in the channel under the gate decreases with V_{DS} increasing. In particular, the electron density near the GEDS will be several orders lower than that in the access region, forming the pinch-off area. Also, we note that the electron density in the access region does not change with V_{DS} . Figure 3-1(d) shows that the electron velocity tends to

saturate due to the relative high electric field there, conforming to the electric field-dependent mobility model we have described (Eq. 2.6). Correspondingly, the electron mobility near the GEDS is lower than that in the access region (Figure 3-1(e)).

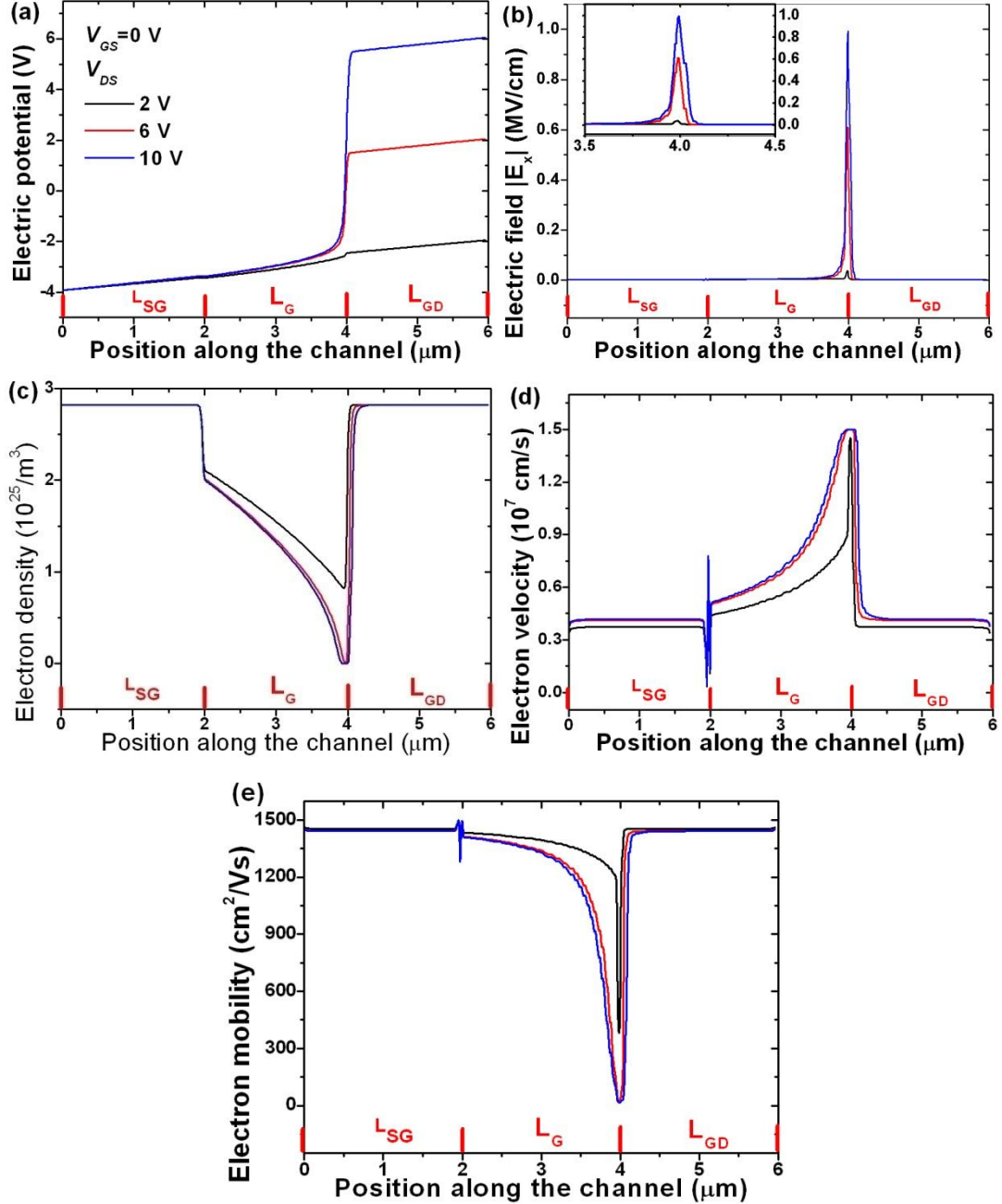


Figure 3-1: Simulations on distribution of the electrical potential (a), horizontal electric field $|E_x|$ ((b), the inset is the enlarged graph for $|E_x|$ near the GEDS), electron density (c), electron velocity (d), and electron mobility (e) in the channel influenced by V_{DS} with fixed V_{GS} (the device is biased in the $I_{DS,Sat}$ region).

Figure 3-2 shows similar simulation results to those in Figure 3-1, but for variable V_{GS} ranging from -2 to 0 V with fixed $V_{DS}=5$ V. Figure 3-2(a) shows that for the device in off-state ($V_{GS}=-2$ V with I_{DS} tending to zero) the electrical potential drop mainly happens at the GEDS while its drop in the access region is negligible. When the device in on-state ($V_{GS}=0$ V), partial electrical potential drop happens in the access region due to the relative high I_{DS} . Also, increase of

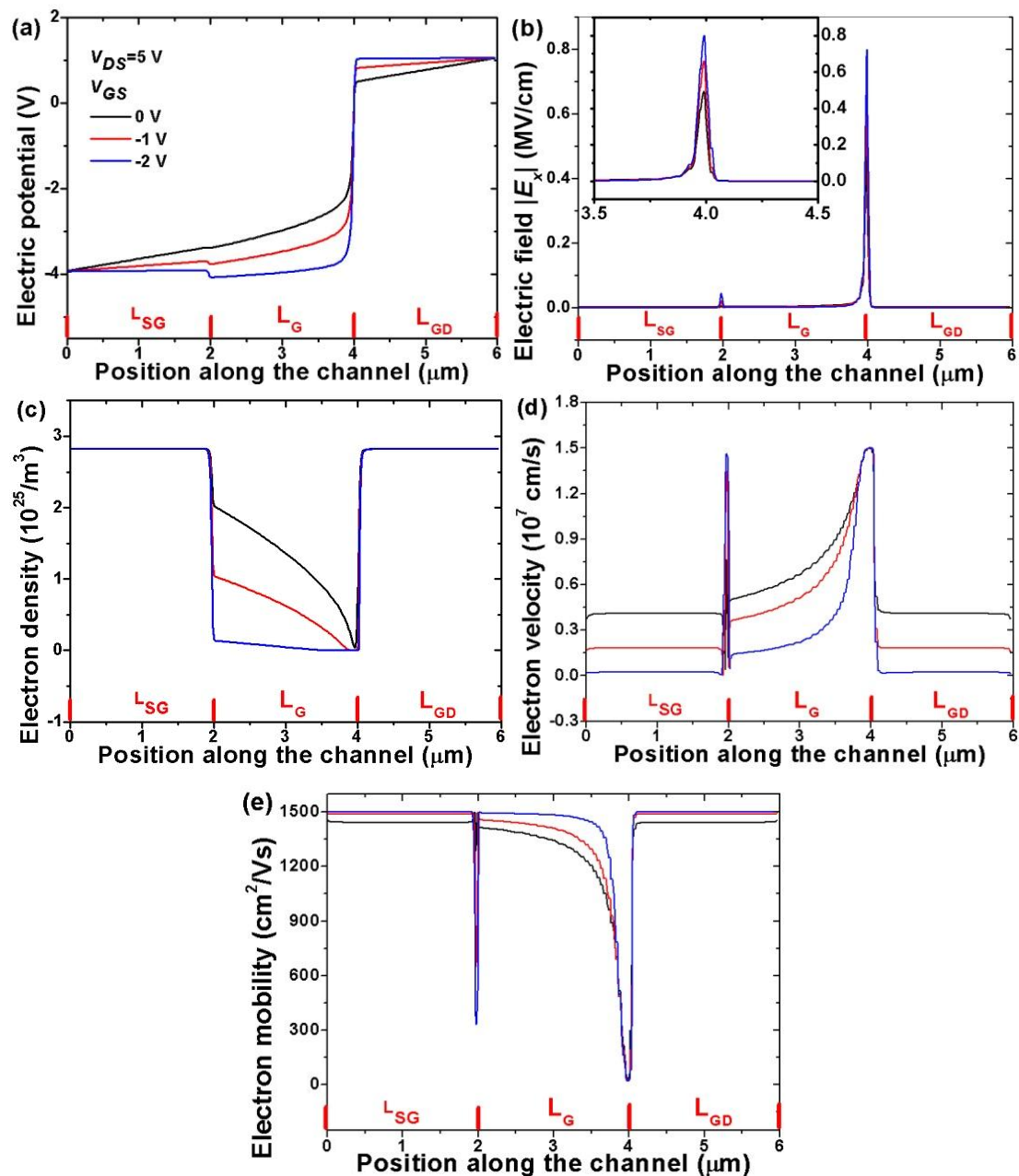


Figure 3-2: Similar simulations to that shown in Figure 3-1, but for variable V_{GS} ranging from -2 to 0 V with fixed $V_{DS}=5$ V.

electric field mainly happens at the GEDS with V_{GS} decreasing (Figure 3-2(b)). Figure 3-2(c) shows that the electron density in the channel region tends to be depleted with V_{GS} decreasing while it is not changed in the access region. Again, Figure 3-2(d) shows the electron velocity saturates near the GEDS due to the relative high electric field there. Electron mobility is varied in the access region due to the variation of electric field there, conforming to the electric field-dependent mobility model we used (Figure 3-2(e)).

Figure 3-3 shows distribution of both the electron conduction band and quasi-Fermi level from the source to drain contact for the device in steady state. It is clear that the electron has a higher energy at the source contact than that at the drain contact and the energy barrier mainly concentrates at the GEDS. Note that in the access region the quasi-Fermi energy is higher than the conduction band energy, leading to the high density of 2DEG close to the AlGaIn/GaN interface. Conversely, the quasi-Fermi energy is lower than the conduction band energy in the channel under the gate due to the reduced electron density modulated by V_{GS} .

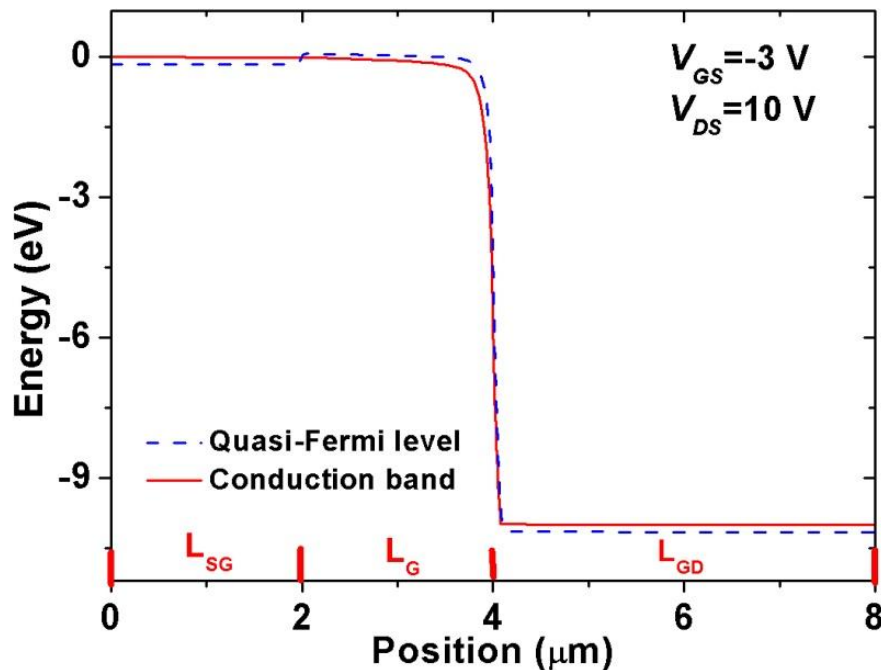


Figure 3-3: Distribution of electron conduction band and quasi-Fermi energy level from the source to the drain contact for the device in steady state.

We have illustrated how the distribution of electrical quantities along the horizontal direction change with the device biases. In the following, we will show their distribution in the vertical direction. Figure 3-4 shows vertical distribution of electron conduction band energy under the GEDS through the AlGaIn and GaN layer as a function of the biases. Figure 3-4(a) shows that there is a quantum well for $V_{DS}=2$ V. With V_{DS} increasing, the 2DEG under the GEDS is depleted, forming the pinch-off area (see for example $V_{DS}=10$ and 18 V). The electron conduction band energy near the heterojunction is higher than that in the deep GaN buffer ($V_{DS}=18$ V), meaning the electrons are pulled into the GaN buffer.

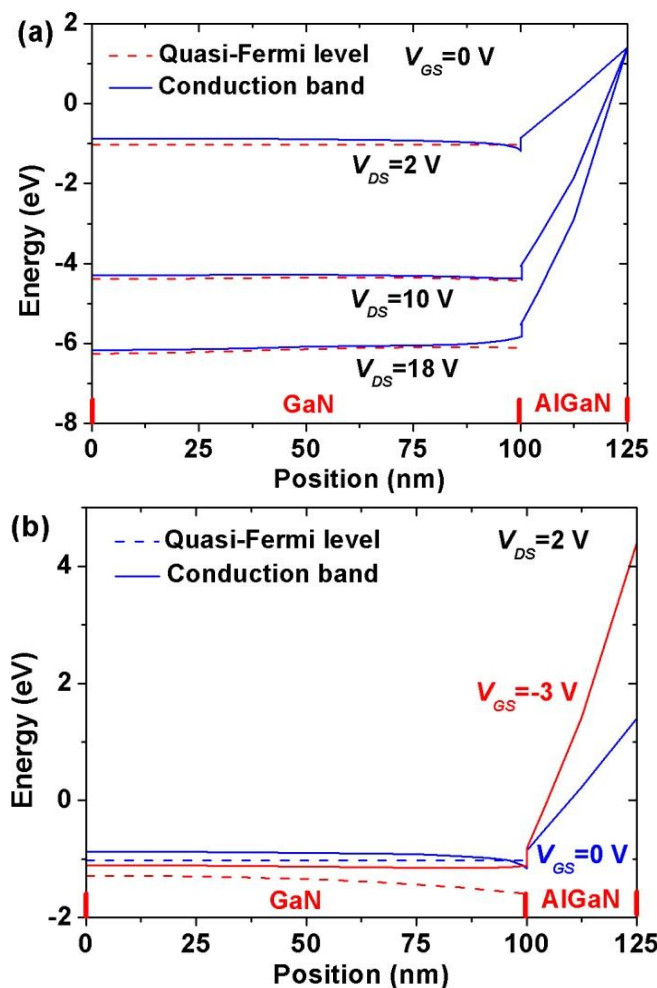


Figure 3-4: Vertical distribution of electron conduction band energy under the GEDS through the AlGaIn and GaN layer (100 nm thickness is displayed in the graphs) as a function of V_{DS} (a) and V_{GS} (b).

3.1.2 Breakdown behavior influenced by gate structures

Device with over-hanging and slant field-plate gate have been intensively investigated for potential enhancement of breakdown voltage [55]-[57]. In this section, we will simulate the device breakdown behavior for standard gate and these two field-plate gate structures.

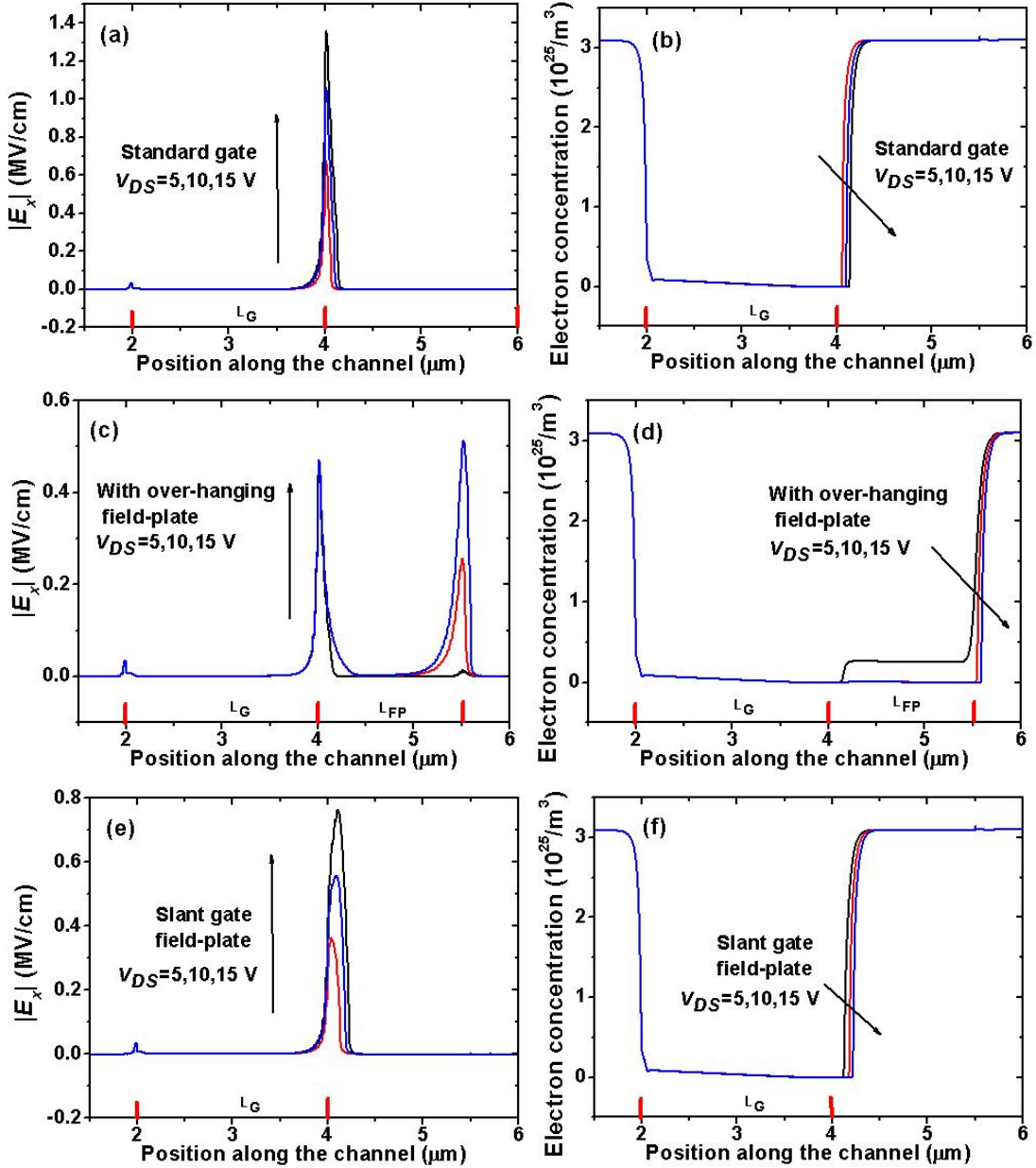


Figure 3-5: Simulation results of horizontal electric field (E_x) and electron concentration distribution in the GaN channel for standard gate (a, b), over-hanging field-plate gate (c, d), and slant field-plate gate (e, f). The dielectric under the field plate is 50 nm SiN. The device is biased with fixed $V_{GS} = -3 \text{ V}$.

Figure 3-5 shows simulation results of horizontal electric field and electron concentration distribution in the GaN channel for the device with standard gate, over-hanging field plate gate, and slant field-plate gate. As the graphs show, the devices with over-hanging and slant field plate have reduced maximum electric field compared to the device with standard gate, meaning that the device breakdown voltage can be enhanced by these two field-plate structures. The 2DEG under the over-hanging field plate will be depleted as V_{DS} increasing (Figure 3-5(d)). Before this depletion, the increased electric field is spread at both the GEDS and field-plate edge of the drain side. After this depletion, the increased electric field appears only at the field-plate edge of the drain side while it keeps unchanged at the GEDS (Figure 3-5 (c)).

3.1.3 Capacitance

Capacitance measurement for GaN-based diodes or HEMTs is important for interfacial traps characterization [58]-[61] and for the extraction of the 2DEG density and barrier height [62].

For the capacitance simulation in this dissertation, the device was assumed in steady-state. This means that change of the 2DEG density can follow the bias signal and the traps effects on the transient change of the 2DEG was neglected.

For the Schottky diode, we did not consider the current and only 1D Poisson equation was solved.

For the transistors, both the Poisson equation and electron current continuity equation were solved in 2D to obtain the gate capacitance (C_G), input capacitance (C_{iss}), and reverse transfer capacitance (C_{rss}). Because the simulations were performed in 2D, they can include the gate fringing capacitance that will be illustrated in the following. These capacitances are defined as

$$C_G = \left(\frac{dQ_G}{dV_{GS}} \right)_{V_{DS}=0} \quad (3.1)$$

$$C_{iss} = \left(\frac{dQ_G}{dV_{GS}} \right)_{V_{DS}=constant} \quad (3.2)$$

$$C_{rss} = \left(\frac{dQ_G}{dV_{DS}} \right)_{V_{GS}=constant} \quad (3.3)$$

In these formulas, Q_G is the gate charge density.

A. Temperature-dependent capacitance of Schottky diodes

Figure 3-6 shows the simulated capacitance of circular Schottky diodes (circular with 100 μm diameter) as a function of temperature from 150 K to 550 K. The simulation results at 300 K were compared to the experimental data and good agreement was obtained. The device has the following structure: Si substrate, 2 μm GaN/25 nm $\text{Al}_{0.25}\text{Ga}_{0.75}\text{N}$ epitaxial layers, and 3 nm GaN cap (Figure 3-6 (a)). As Figure 3-6(b) shown, the temperature affects the capacitance in the threshold region due to the thermal broadening of the 2DEG Fermi-distribution [39], while it does not influence the capacitance in the flat region. As mentioned above, the traps effect was neglected in these simulations. If the traps effect is considered, the capacitance would be influenced by the temperature because the electron emission rate is dependent on the temperature. However, capacitance measured under relative high frequency (such as 1 MHz)

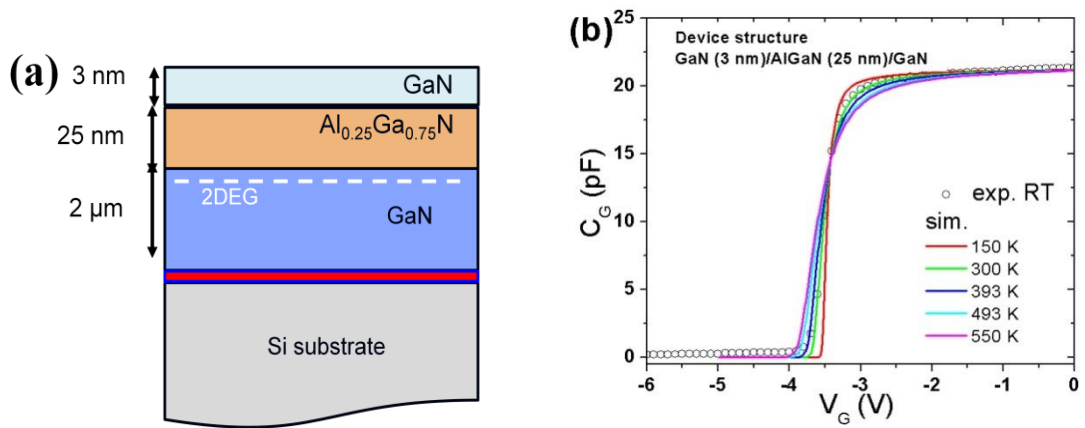


Figure 3-6: Schematic of the device structure (a) and simulated capacitance C_G of Schottky diodes as a function of temperature (b). Simulated C_G at 300 K (RT) is compared to the experimental one.

can reduce the deep-level traps influence due to the long characteristic time of electron emission from the traps.

B. Fringing (C_{FR}) and field-plate (C_{FP}) capacitance

As we mentioned before, the field-plate can enhance the device breakdown voltage. However, the field-plate also increases the parasitic capacitance that could restrict the device applications in high-frequency. C_{FR} has been illustrated to be significant for the highly scaled device and it is dependent on the passivation [63]. In the following, we will show the mechanisms of C_{FR} and C_{FP} .

The 2DEG distribution as a function of V_{DS} (fixed V_{GS}) for the device with field plate was simulated (see the schematic in Figure 3-7), as shown in Figure 3-8. The device is in off-state as we can see that the 2DEG under the gate is completely depleted. The 2DEG under the field plate is not completely depleted for relative low $V_{DS}=10$ V and it will also be depleted with V_{DS} increasing (see for example $V_{DS}=30$ V). Before this depletion, there has capacitance caused by the field-plate. Also we note that with V_{DS} increasing the 2DEG in the access region near the field-plate edge will also be depleted by the fringing electric field, which is the origin of C_{FR} as indicated in the graph.

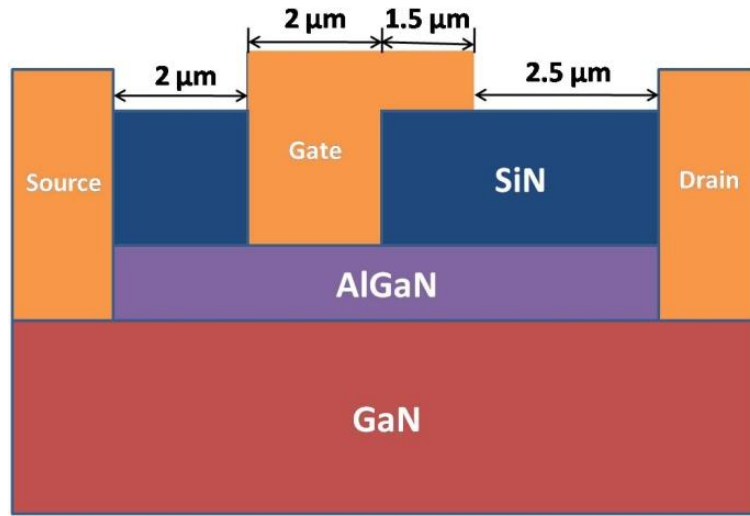


Figure 3-7: Schematic of the simulated device structure with field-plate.

Figure 3-9 shows the simulated C_{rss} and C_{iss} as a function of V_{DS} for the device with field-plate. Compared to the experimental results obtained from a similar device structure [64][65], they show similar behavior. For $V_{DS} < 20$ V, C_{FP} has the main contribution for C_{rss} and C_{iss} . For $V_{DS} > 22$ V, these two capacitances do not tend to zero and this is due to the main contribution from C_{FR} .

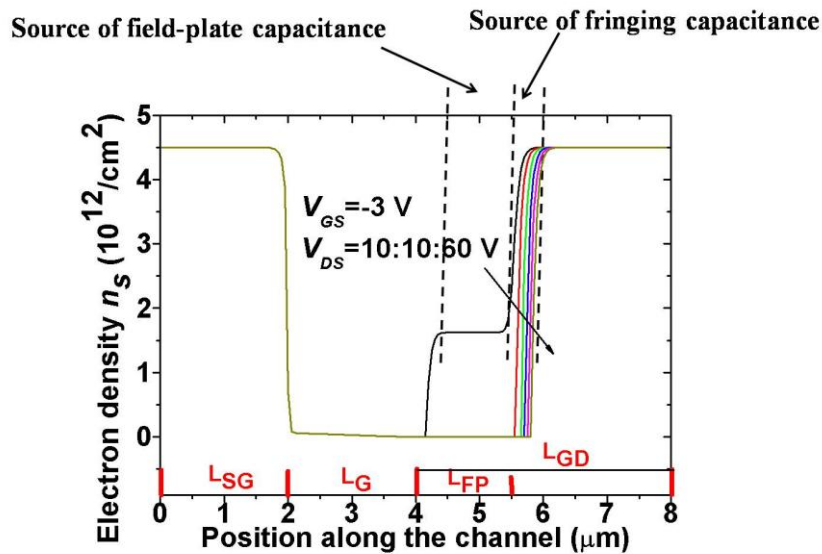


Figure 3-8: Simulation of the 2DEG distribution as a function of V_{DS} (fixed V_{GS}) for the device with field plate. Source of C_{FP} and C_{FR} are indicated in the graph. The device structure consists of 20 nm $\text{Al}_{0.25}\text{Ga}_{0.75}\text{N}$ barrier and 80 nm SiN passivation. The gate width is 120 nm.

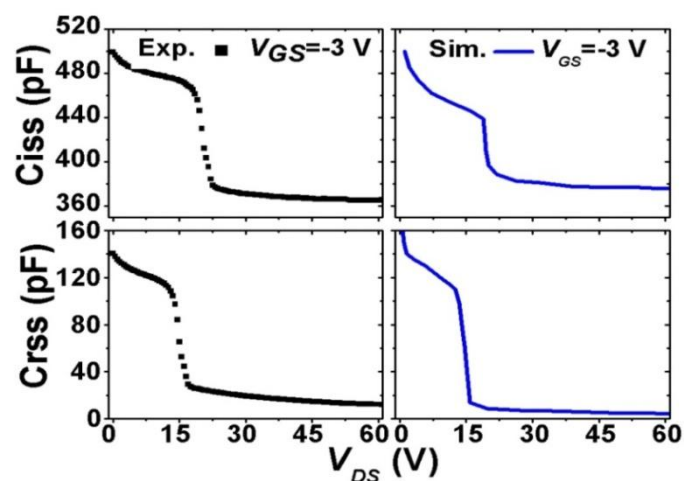


Figure 3-9: Experimental and simulated C_{iss} and C_{rss} as a function of V_{DS} (fixed $V_{GS} = -3$ V) for the device with field-plate. The experimental data were taken from [64][65]

C. Impact of dielectric thickness on C_{FP} and C_{FR} .

In Chapter 4 we will simulate the heat spreading by diamond integrated with the device to illustrate the potential reduction of the self-heating effect. Here we will first simulate the parasitic capacitance caused by the diamond on top of the device and the results will be compared to that of the SiN passivation. The device structure used in the simulation is same as that showed in Figure 4-1.

C_{FP} was calculated by considering the charge density only in the field-plate. C_{FR} was extracted by extrapolation of the gate capacitance with the gate length tending to zero (see more details in [63]). Figure 3-10 shows the simulated capacitance (total capacitance is the summation of C_{FP} and C_{FR}) as a function of the dielectric thickness (diamond or SiN). As the graph shows, C_{FP} decreases and C_{FR} increases when increasing the dielectric thickness. In addition to the dielectric thickness, the permittivity also plays an important role in the parasitic capacitance. Device with SiN passivation has a higher parasitic capacitance due to its higher permittivity ($\epsilon_r=7.2$), compared to device with diamond heat spreading layer ($\epsilon_r=5.5$). Here we only compared SiN and diamond. For other types of dielectric, such as aluminum nitride, silicon oxide, etc., the conclusions are straightforward depending on their permittivity.

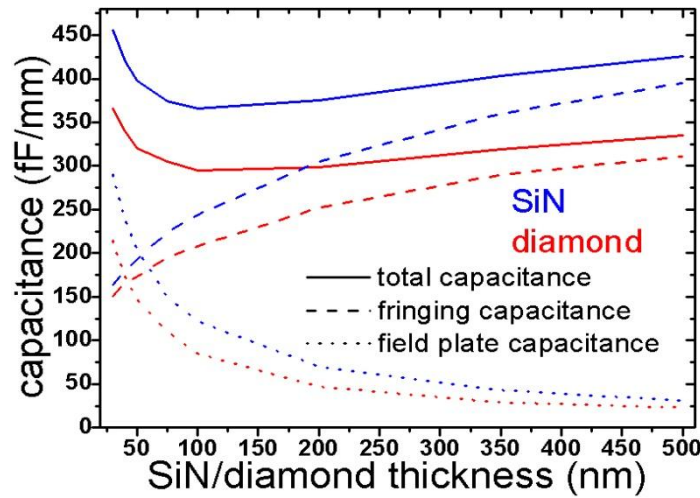


Figure 3-10: Simulation of the total capacitance, C_{FR} , and C_{FP} as a function of the dielectric (diamond or SiN) thickness. The device is in off-state ($V_{GS}=-3$ V and $V_{DS}=0$ V). See the device structure showed in Figure 4-1.

Finally, a summary about the advantages and disadvantages of using the field plate and passivation would be meaningful. The field plate can increase the device breakdown voltage and reduce traps effects due to the reduced electric field. The passivation can also reduce the traps effects due to the reduced traps density. They were both found to increase the device parasitic capacitance. Therefore, an optimized device design should address these aspects.

3.2 Thermal behavior

3.2.1 Channel temperature influenced by substrate

AlGaIn/GaN heterostructures are typically grown on Si, SiC, and sapphire substrates. In addition to the self-heating, the substrate can also influence the device breakdown voltage (V_{BR}) [66][67] and the electron density as well as mobility because of the intrinsic stress in the substrate. Si substrate have large size wafer and low cost, and allows the potential integration of GaN power electronics on an advanced and mature Si technology [68]. High thermal conductivity of SiC substrate allows to achieve a high power output, due to the reduced self-heating, but the relative high cost and small size wafer are important issues. Sapphire substrate has a low cost, but its low thermal conductivity limits the device power output. Diamond substrate has the highest thermal conductivity, which can potentially minimize the self-heating.

In this section, we will illustrate how the substrate impact on the device channel temperature (T_{ch}) and thermal resistance (R_{TH}) by means of electro-thermal coupling simulations and the electrical characterization described in Section 2.3. Electro-thermal simulation means that only Eq. 2.1 and Eq. 2.2 are solved simultaneously.

The device structures were grown on sapphire, Si, and 4H-SiC substrates by metal–organic chemical vapor deposition, and the AlGaN barrier thickness is between 21 and 25 nm and Al content is between 27% and 30%. The detailed structures are the following:

- 1) 25-nm Al_{0.28}Ga_{0.72}N/1.25- μ m GaN/330- μ m sapphire;
- 2) 2-nm GaN/22.2-nm Al_{0.27}Ga_{0.73}N/1.1- μ m GaN/400- μ m Si (111);
- 3) 1-nm GaN/21.2-nm Al_{0.30}Ga_{0.70}N/1.7- μ m GaN/400- μ m 4H-SiC.

A pulse width 350 ns and period 3.5 ms was used in the double-pulsed measurements.

The simulated I_{DS} was first calibrated to the experimental one, meaning the same power output between them. Temperature-dependent thermal conductivity was considered. Both the maximum and the average T_{ch} have been simulated. The average T_{ch} was obtained considering the area under the gate of the device.

Figure 3-11 shows the simulated and experimental maximum and average T_{ch} under different ambient temperature (T_{amb}) as a function of power dissipation (P_D) for the device on SiC (a), Si (b), and sapphire (c) substrate. P_D was calculated as $P_D = V_{DS} \times I_{DS}$. There is a good agreement between the simulation and experimental results. It is clear that for same P_D the substrate significantly influences T_{ch} , and the device on SiC (sapphire) substrate has lowest (highest) T_{ch} due to its highest (lowest) thermal conductivity. The simulated average T_{ch} was closer to the experimental results, probably meaning T_{ch} from electrical characterization is an average value in the channel. Narrowing the pulse width in the measurements to further eliminate the self-heating can improve agreement between the experimental and simulation results.

Table 3-1 shows the simulated and experimental device thermal resistance (R_{TH}) extracted as a linear fitting of T_{ch} vs. P_D . Also, we can see that the device on SiC (sapphire) substrate has lowest (highest) R_{TH} .

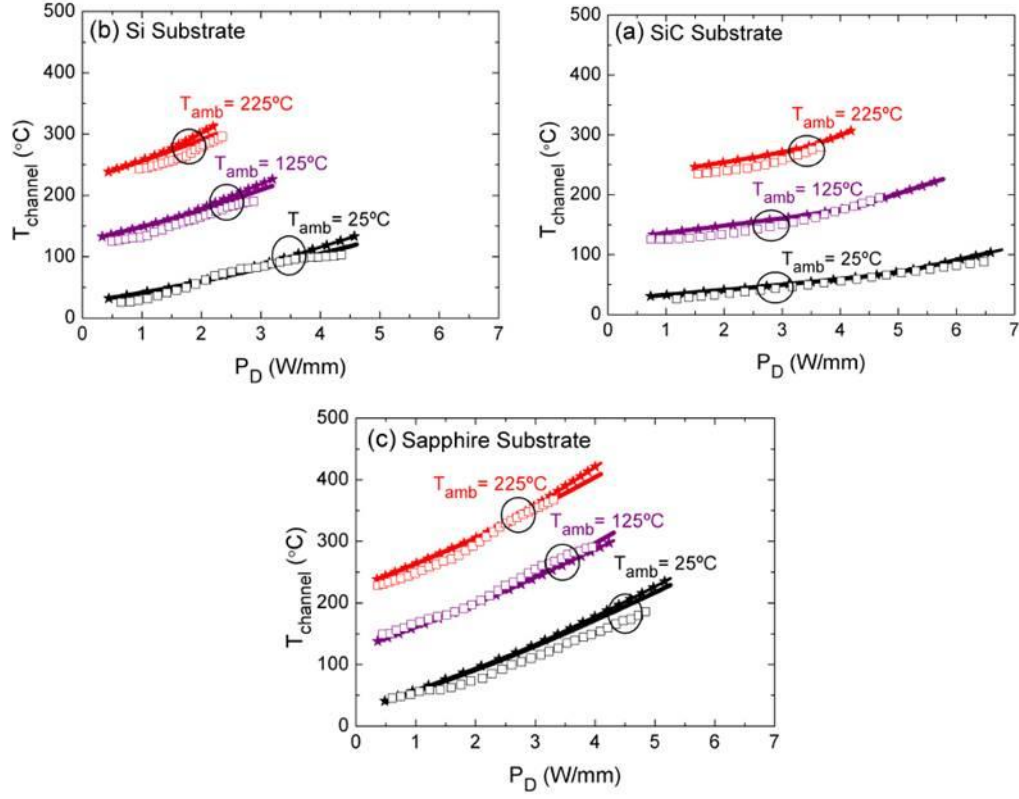


Figure 3-11: Simulated and experimental T_{ch} under different ambient temperature (T_{amb}) as a function of power dissipation (P_D) for the device on SiC (a), Si (b), and sapphire (c) substrate. The star-line, solid line, and empty square represent the simulated maximum T_{ch} , simulated average T_{ch} , and experimental results, respectively.

T_{amb} (°C)	Experimental R_{TH} (°C mm/W)			Simulated R_{TH} (°C mm/W)		
	SiC	Si	Sapphire	SiC	Si	Sapphire
25	11.5	23.0	34.3	12.1	25.1	41.6
125	17.7	29.9	43.1	17.2	33.1	44.3
225	21.2	39.8	50.1	21.3	42.4	50.3

Table 3-1: Simulated and experimental R_{TH} as a function of T_{amb} for the device on SiC, Si, and sapphire substrate.

Due to the elevated T_{ch} , I_{DS} decreases with V_{DS} increasing, termed negative differential conductance (NDC) that can be characterized as dI_D/dV_{DS} in $I_{DS,Sat}$ region, as shown in the inset of Figure 3-12 [46]. We can see that the device on

SiC (sapphire) substrate has highest (lowest) NDC. NDC increased with T_{amb} increasing, which can be explained that the device under higher T_{amb} has lower P_D and thus lower T_{ch} due to the reduced electron mobility (Figure 3-12).

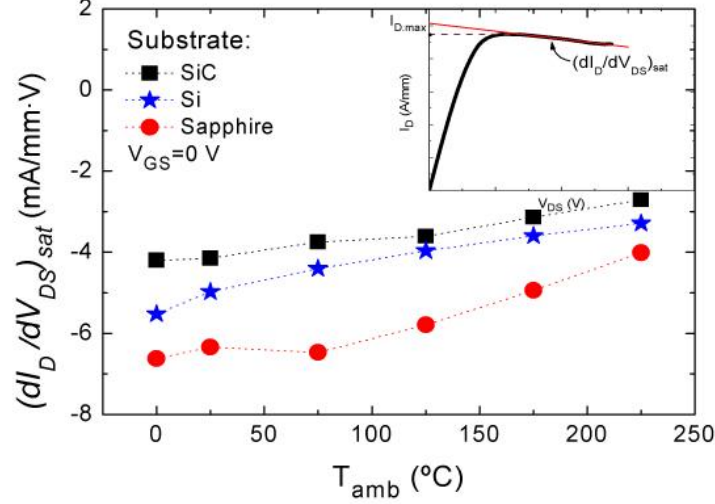


Figure 3-12: Self-heating effect characterized by negative differential conductance (NDC) expressed as dI_D/dV_{DS} in the $I_{DS,Sat}$ region. The device on different substrates and under varied T_{amb} , were compared [46].

3.2.2 Channel temperature influenced by device geometry

The device can be designed with different geometries for specific purposes of application. According to the expression of cut-off frequency ($f_T = \frac{v}{2\pi L_G}$) [69], device with shorter gate length (L_G) can be applied at higher frequency. However, highly scaled L_G will lead to short channel effect [70][71], mainly manifested by a not well pinch-off behavior. Because the 2DEG between the gate and drain contact (L_{GD}) can be depleted by the gate fringing electric field and partial V_{DS} drop appears in the drain access region, large L_{GD} can increase the device V_{BR} . A complete simulation of the device geometry impact on I_{DS} showed that scaling of L_G or distance between gate and source (L_{SG}) can increase I_{DS} while scaling of L_{GD} does not significantly influence I_{DS} [72]. Wide gate (W_G) can reduce the device on-resistance and increase the output power, which is an optimized design for application in power switch. In addition to the

electrical behaviors mentioned above, the device geometry can also influence its thermal behavior.

Bertoluzza *et al.* have analyzed impact of the finger width, length and pitch spacing on the thermal behavior of multi-fingers AlGaN/GaN HEMTs, using a 3D thermal simulation [73]. In this section, we will present a complete study of the impact of main geometrical parameters (L_G , L_{GD} , and W_G) on the thermal behavior for the device under different T_{amb} , by electro-thermal coupling simulations and electrical characterization described in Section 2.3.

For this study, the device heterostructures were grown on sapphire and 4H-SiC substrates by metal-organic chemical vapor deposition. The detailed structure of each sample is the following:

- 1) 25 nm $\text{Al}_{0.28}\text{Ga}_{0.72}\text{N}/1.25\text{ }\mu\text{m GaN}/330\text{ }\mu\text{m}$ sapphire substrate;
- 2) 3 nm GaN/10 nm $\text{In}_{0.17}\text{Al}_{0.83}\text{N}/1\text{ nm AlN}/2\text{ }\mu\text{m GaN}/300\text{ }\mu\text{m}$ 4H-SiC substrate.

The device is one-finger layout as shown in Figure 3-13. The measurement conditions for the electrical characterization are same as that described in Section 2.3.

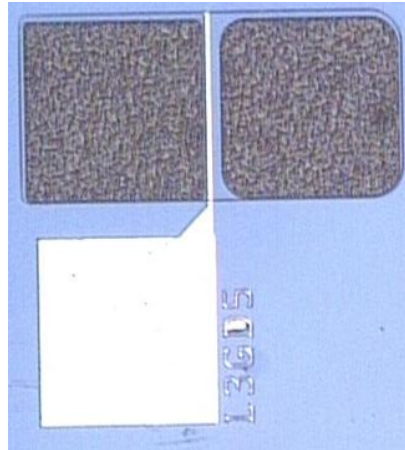


Figure 3-13: Picture of the studied one-finger device

A. W_G influence

Figure 3-14 shows T_{ch} as a function of P_D at different T_{amb} for device on sapphire substrate with different W_G . T_{ch} increases almost linearly with P_D

regardless of the device geometry. This increment is more significant for higher T_{amb} . T_{ch} is higher for larger W_G , e.g., for $P_D=1$ W/mm at $T_{amb}=25$ °C, T_{ch} is 46 °C, 55 °C and 57.5 °C for W_G of 100, 200 and 300 μm , respectively, as shown in Figure 3-15(a). This increase of T_{ch} with W_G is due to the worse heat dissipation in the third dimension of the device with larger W_G , which was also showed elsewhere [74]. Consequently, higher R_{TH} was observed for the device with larger W_G , as shown in Figure 3-15(b). Moreover, an increase in R_{TH} value with T_{amb} was observed regardless of the W_G , being more significant for the larger W_G . Note that R_{TH} increases with T_{amb} , as shown in Figure 3-15(b), which can be attributed to the decreased thermal conductivity of the substrate and epitaxial layers with T_{amb} increase.

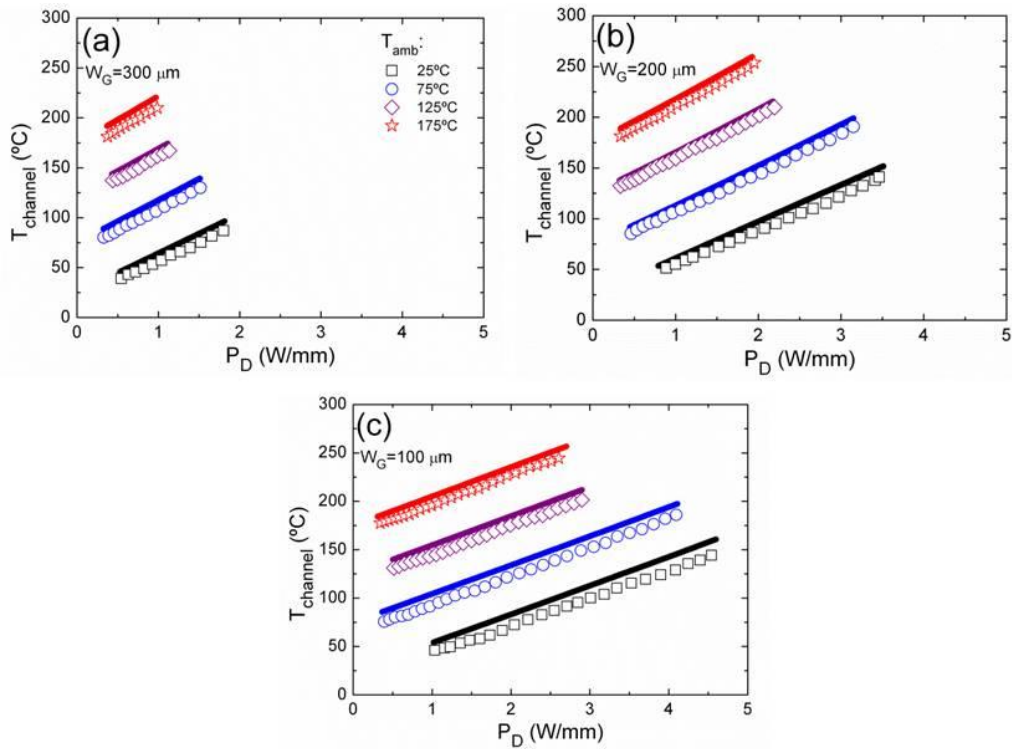


Figure 3-14: T_{ch} as a function of P_D at different T_{amb} for devices on sapphire substrate with different geometries: $W_G=300$ μm (a), $W_G=200$ μm (b), and $W_G=100$ μm (c). The empty symbols and solid lines correspond to experimental and simulation results, respectively.

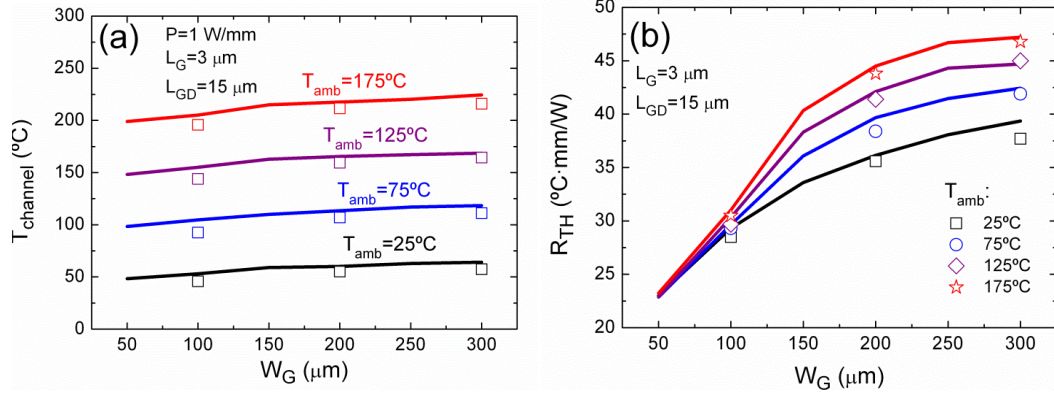


Figure 3-15: T_{ch} (a) and R_{TH} (b) as a function of W_G with fixed $P_D = 1$ W/mm. The devices are on sapphire substrate with varied W_G . The empty symbols and solid lines correspond to experimental and simulation results, respectively.

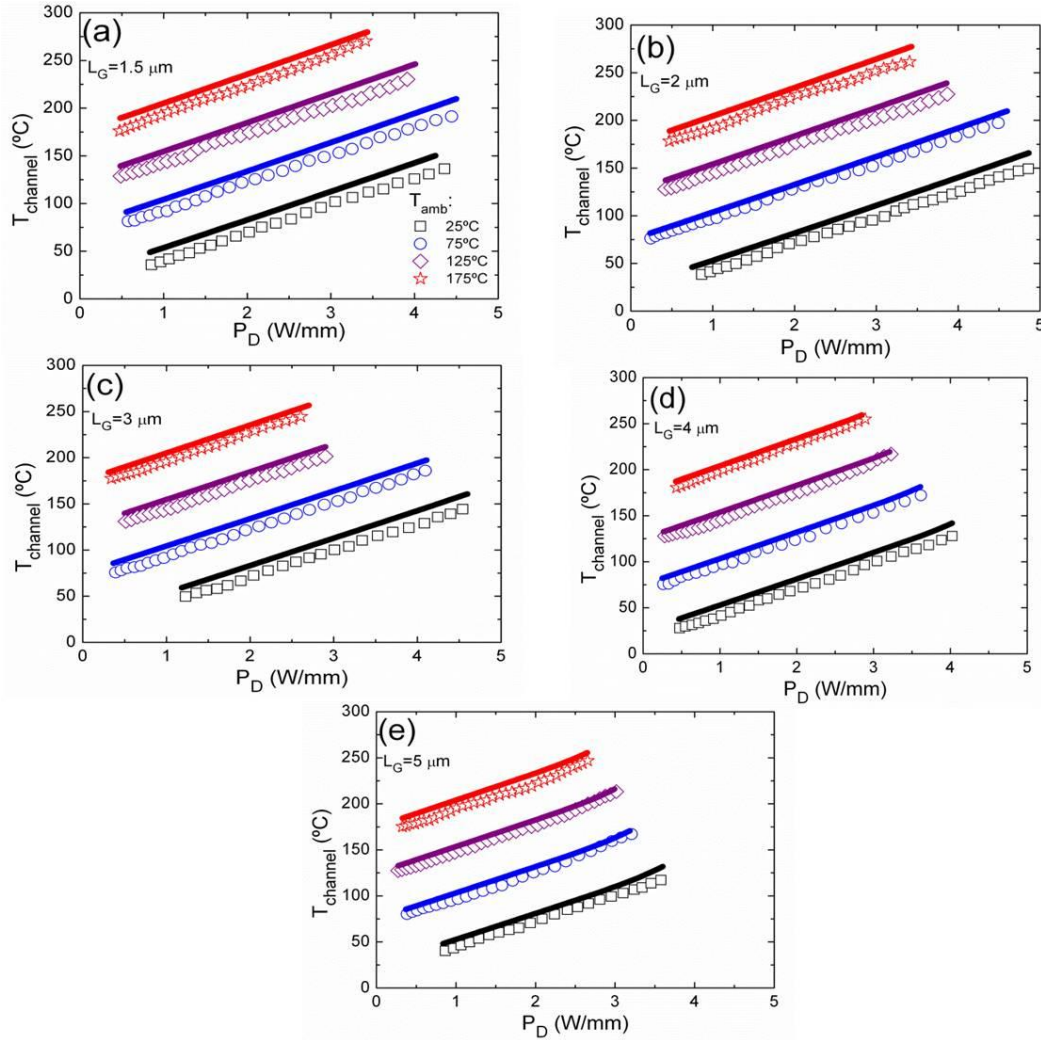


Figure 3-16: T_{ch} as a function of P_D at different T_{amb} for devices on sapphire substrate with different geometries: $L_G = 1.5$ μm (a), $L_G = 2$ μm (b), $L_G = 3$ μm (c), $L_G = 4$ μm (d), and $L_G = 5$ μm (e). The empty symbols and solid lines correspond to experimental and simulation results, respectively.

B. L_G influence

Figure 3-16 shows T_{ch} as a function of P_D at different T_{amb} for devices on sapphire substrate with different L_G . A slight reduction of T_{ch} was observed with the L_G increasing for fixed P_D , as shown in Figure 3-17(a). Accordingly, R_{TH} has a slight decrease with L_G increasing, as shown in Figure 3-17(b).

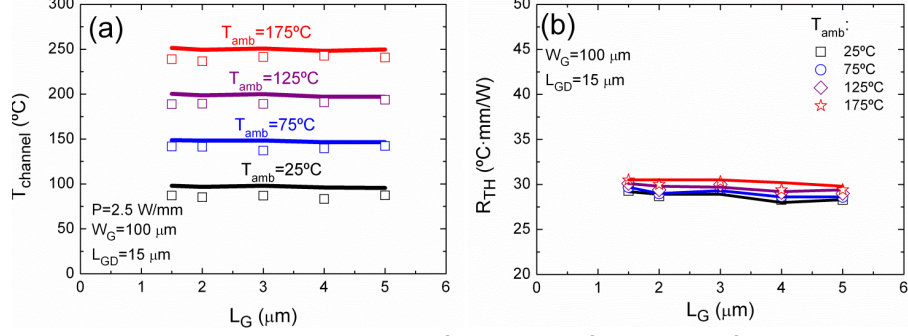


Figure 3-17: T_{ch} (a) and R_{TH} (b) as a function of L_G with fixed $P_D = 2.5$ W/mm (the devices are on sapphire substrate). The empty symbols and solid lines correspond to experimental and simulation results, respectively.

C. L_{GD} influence

T_{ch} at different T_{amb} as a function of L_{GD} for devices on sapphire substrate is showed in Figure 3-18.

Figure 3-19(a) shows that T_{ch} increases with L_{GD} decreasing and its increase is more significant for shorter L_{GD} , e.g., T_{ch} increases around 13 °C for L_{GD} variation from 2.5 μm to 5 μm while it only increases 1.5 °C for L_{GD} variation from 15 μm to 20 μm. Figure 3-19(b) shows that shorter L_{GD} leads to higher R_{TH} . We note that this R_{TH} dependence on L_{GD} is more relevant at higher T_{amb} .

In order to understand physical mechanisms of L_{GD} impact on T_{ch} , electro-thermal simulations were carried out to investigate the distribution of electric field magnitude (E_x), electron density (N_s), power dissipation density P_D , and T_{ch} . Figure 3-20 shows the simulation results for two different devices with $L_{GD} = 2.5$ μm and 20 μm (fixed $L_G = 3$ μm and $W_G = 100$ μm) under same power dissipation $P_D = 4.5$ W/mm. The device with shorter L_{GD} has higher E_x (Figure 3-20(a)) and lower N_s (Figure 3-20(b)) near the gate edge of the drain side. This leads to a concentration of the generated heat at the gate edge

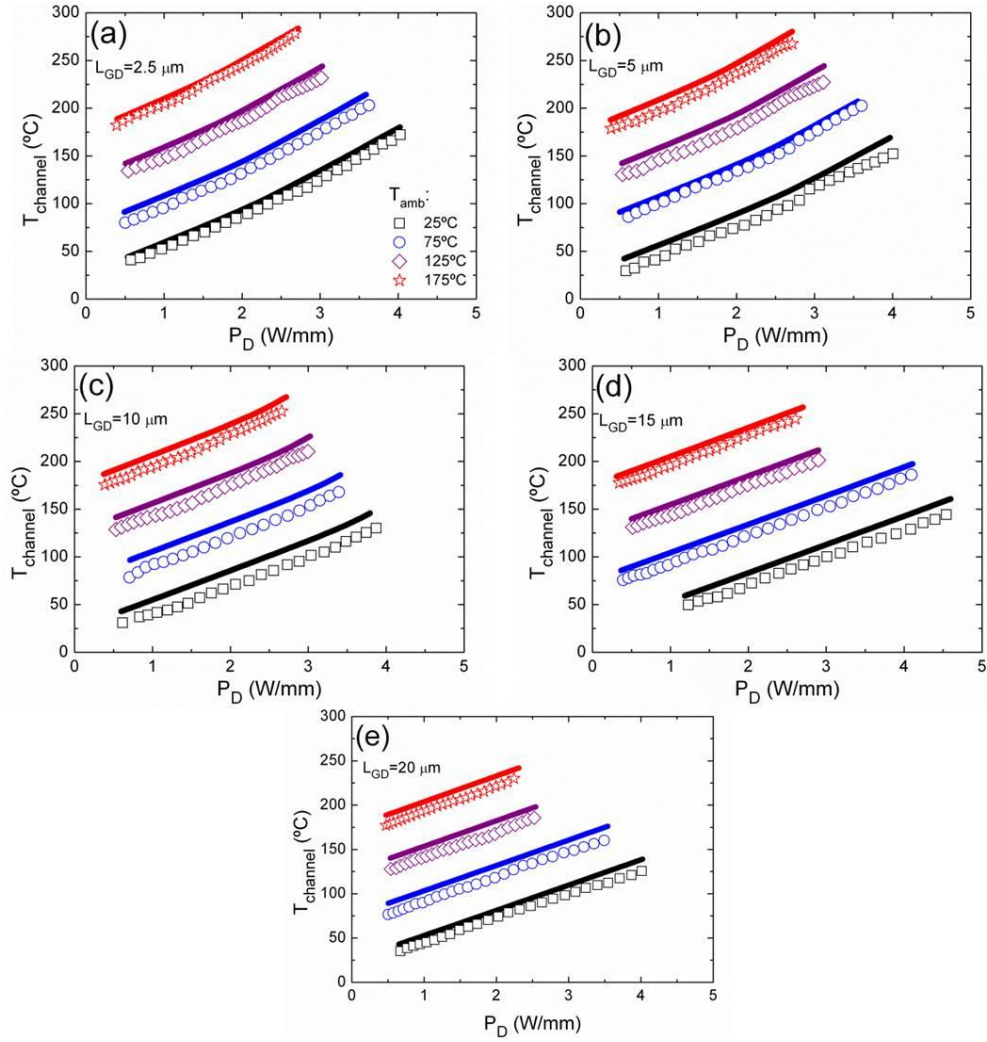


Figure 3-18: T_{ch} as a function of P_D at different T_{amb} for devices on sapphire substrate with different geometries: $L_{GD}=2.5 \mu\text{m}$ (a), $L_{GD}=5 \mu\text{m}$ (b), $L_{GD}=10 \mu\text{m}$ (c), $L_{GD}=15 \mu\text{m}$ (d), and $L_{GD}=20 \mu\text{m}$ (e). The empty symbols and solid lines correspond to experimental and simulation results, respectively.

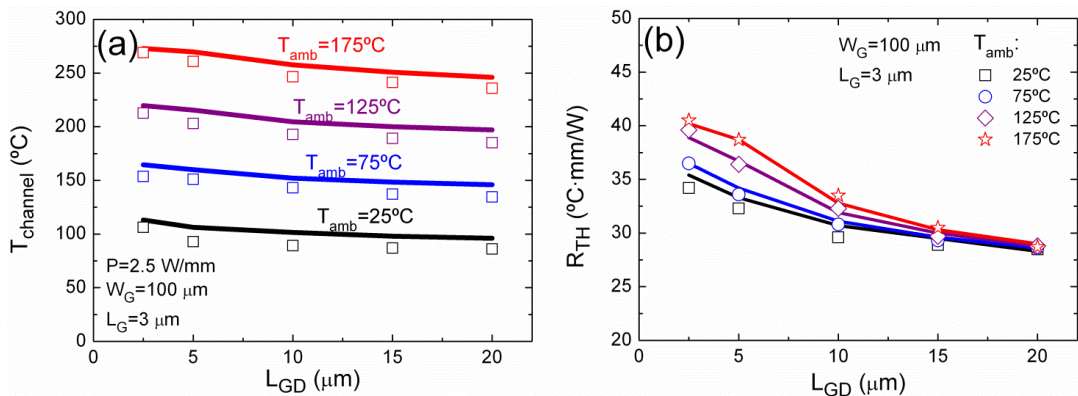


Figure 3-19: T_{ch} (a) and R_{TH} (b) as a function of L_{GD} with fixed $P_D=2.5 \text{ W/mm}$. (the devices are on sapphire substrate). The empty symbols and solid lines correspond to experimental and simulation results, respectively.

(Figure 3-20(c)), and thus higher T_{ch} in this region (Figure 3-20(d)). Thus, the influence of W_G and L_{GD} on T_{ch} is related to different issues. On one hand, the influence of W_G is linked to the heat dissipation in the third dimension, determined by the thermal conductivity of the substrate and epitaxial layers. On the other hand, the influence of L_{GD} is related to the distribution of the generated heat in the channel, determined by the electrical behavior of the device itself.

The impact of the heat distribution on T_{ch} has been verified by Micro-Raman measurements in [75], where Choi *et al.* demonstrated that, for the same power operation, the device has higher T_{ch} when biased with lower V_{GS} and higher V_{DS} (more heat concentration at the gate edge), than when biased with higher V_{GS} and lower V_{DS} . Our studies can present a guiding about the thermal simulation of AlGaIn/GaN HEMTs. A simple method has been presented to simulate T_{ch} , in which the heat source power is calculated as $V_{DS} \times I_{DS}$ and it is manually put near the GEDS. This method can be easily performed in 3D so that the heat dissipation in the third dimension that is essential for the simulation of the device with small gate width/length ratio can be included. However, disadvantages for this method are that the simulation result is sensitive to the size and position of the heat source manually set although the heat dissipation mainly concentrates at the GEDS and influence of L_{GD} on T_{ch} cannot be properly evaluated. The electro-thermal coupling method is more accurate, but it is very difficult to carry out the simulation in 3D, because the numerical calculation of the electron current needs a large density of spatial discretization points. Therefore, an approach to consider both the simulation simplicity and accuracy is to perform thermal simulation including proper modelling the relationship between the heat source size and the dissipated power (not manually set it as mentioned before).

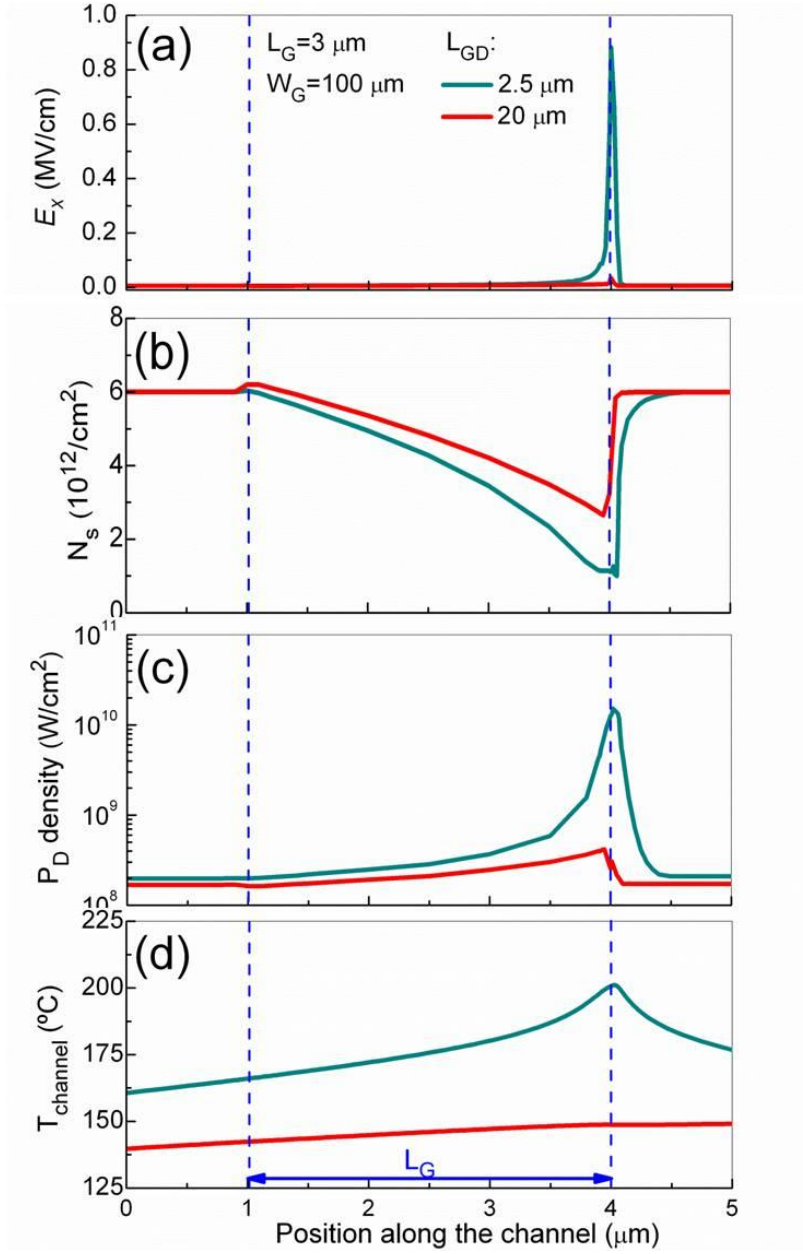


Figure 3-20: Simulations of (a) E_x magnitude, (b) N_s , (c) P_D density, and (d) T_{ch} in the channel region under the gate and 1 μm near the gate edge. Two devices with $L_{GD}=2.5\text{ }\mu\text{m}$ and 20 μm (fixed $L_G=3\text{ }\mu\text{m}$ and $W_G=100\text{ }\mu\text{m}$) for the same power dissipation $P_D=4.5\text{ W/mm}$ are compared.

3.2.3 Heat dissipation in the third dimension

In most of situations, the simulations were performed in 2D for simplicity, in which the heat dissipation in the third dimension cannot be included. In this section, we will compare the simulations with 2D electro-thermal coupling, 2D

electrical and 3D thermal coupling (see details in the following), and 3D electro-thermal coupling to analyze impact of the heat dissipation in the third dimension on simulation results.

Figure 3-21 shows schematic for 2D electrical and 3D thermal coupling process. Only half of the gate width was included in the simulations due to the device symmetry. For 2D electro-thermal simulation, Poisson (Eq. 2.1), electron current continuity equation (Eq. 2.2), and thermal conduction equation (Eq. 2.7) were solved in the front of the schematic. For 3D electro-thermal simulation, the three equations were solved in the whole device structure showed in the schematic. However, complete 3D electro-thermal simulation needs expensive computations and also there are some numerical problems. In order to reduce the computations and also include the heat dissipation in the third dimension, we can perform 2D electrical and 3D thermal full coupling simulation. This means that the Poisson and electron current continuity equation were solved in 2D (in front of the schematic) and the thermal conduction equation was solved in 3D for the whole device structure as shown in the schematic.

Figure 3-22 shows temperature distribution along the gate width (a)

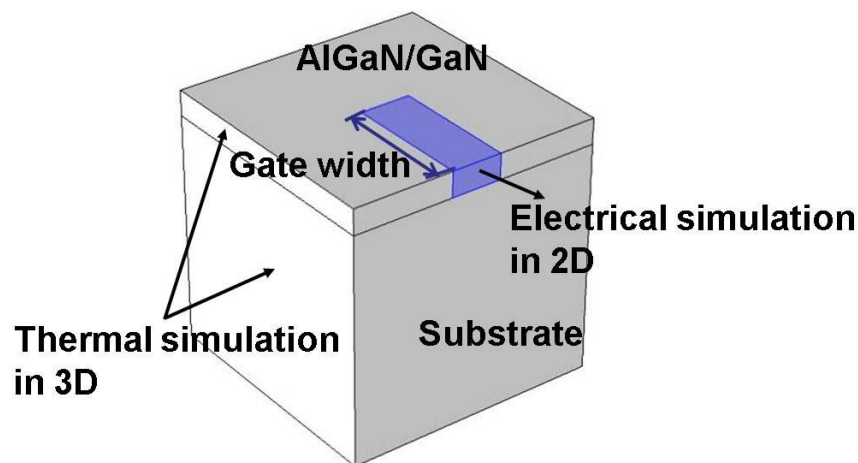


Figure 3-21: Schematic for the 2D electrical and 3D thermal coupling process.

and I_{DS} (b) simulated by different coupling method. As the graph shown, because the heat dissipation in the third dimension was included, the electro-thermal simulation of T_{ch} in 3D is lower than that in 2D (Figure 3-22(a)). Correspondingly, I_{DS} has a higher value for the simulations carried out in 3D (Figure 3-22(b)). Interestingly, the simulated results with 2D electrical and 3D thermal coupling are very close to that with complete 3D coupling, while this simulation needs much less computations and memories. For a general computer: 4 G memory and 2.4 GHz processor, the complete 3D

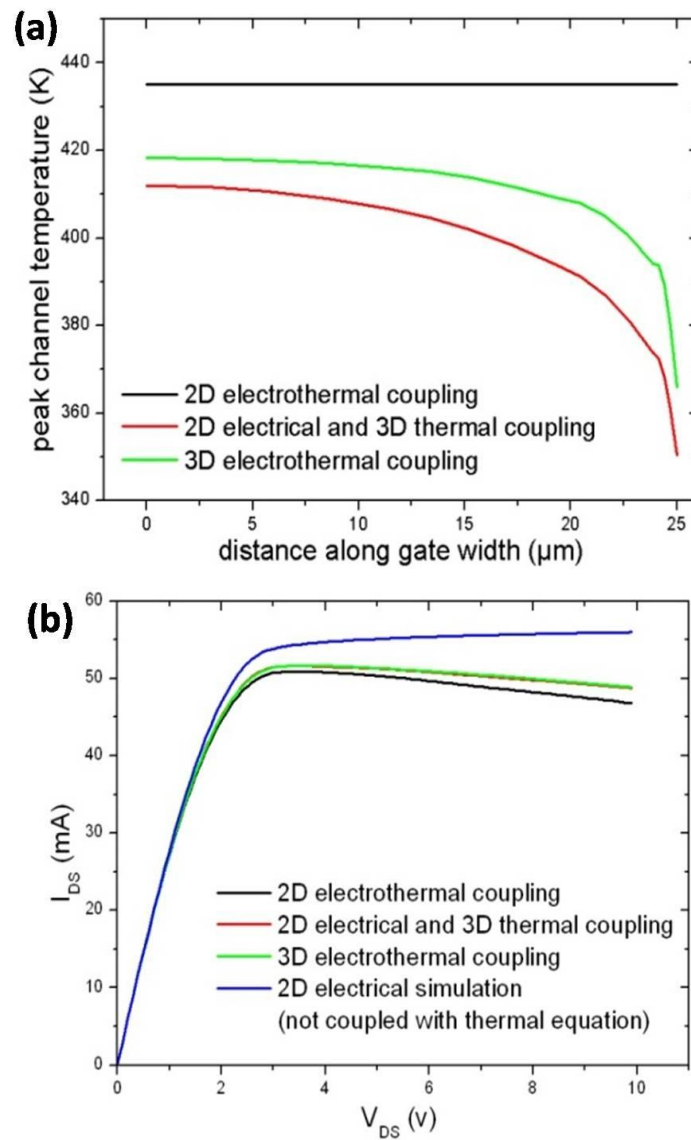


Figure 3-22: Temperature distribution along the gate width (a) and I_{DS} (b) simulated by different coupling methods. The device is on Si substrate and has geometry: $L_G=0.6 \mu\text{m}$, $L_{SG}=0.9 \mu\text{m}$, $L_{GD}=1.5 \mu\text{m}$, and $W_G=50 \mu\text{m}$.

coupling simulation is difficult to be achieved. We note that in the simulations a relative narrow gate (50 μm) was used, therefore, there has a significant T_{ch} difference for the simulations in 2D and 3D. It is straightforward that this difference will become negligible as gate width increase, i.e., for a relative wide gate the heat dissipation in the third dimension has a negligible impact on the device temperature.

3.3 Stress behavior

As mentioned in Chapter 1, the device stress would be related to the device degradation/reliability [76]-[79]. Therefore, it is valuable to do correspondent simulations to better know the device degradation mechanisms

In this section, we will compare the device converse piezoelectric stress (CPS) and thermal stress (TS) for standard and field plate gate by electro-mechanical (EM), electro-thermal (ET), and full electro-thermo-mechanical (ETM) coupling simulations. EM coupling means that only CPS was included and device temperature induced by the self-heating impact on the electron mobility and TS were neglected. ET coupling means that TS and device temperature impact on the electron mobility were included, while CPS was neglected. At last, ETM means that the TS, CPS, the device temperature, and the device electrical behavior were fully coupled.

The simulated device structure includes a 100 μm Si substrate, 1 μm GaN buffer, 20 nm AlGaN barrier with 25% Al mole fraction, 50 nm SiN passivation, and 100 nm thick gate metal. The device geometry is as the following: $L_{SG}=0.5$ μm , $L_G=0.5$ μm , and $L_{GD}=1$ μm for the device with standard gate, and additional over-hanging field-plate $L_{PF}=0.5$ μm for the device with field-plate gate.

Figure 3-23 shows a comparison of the simulated I_{DS} for a device with standard gate by using the EM and ETM coupling method. Note that the big

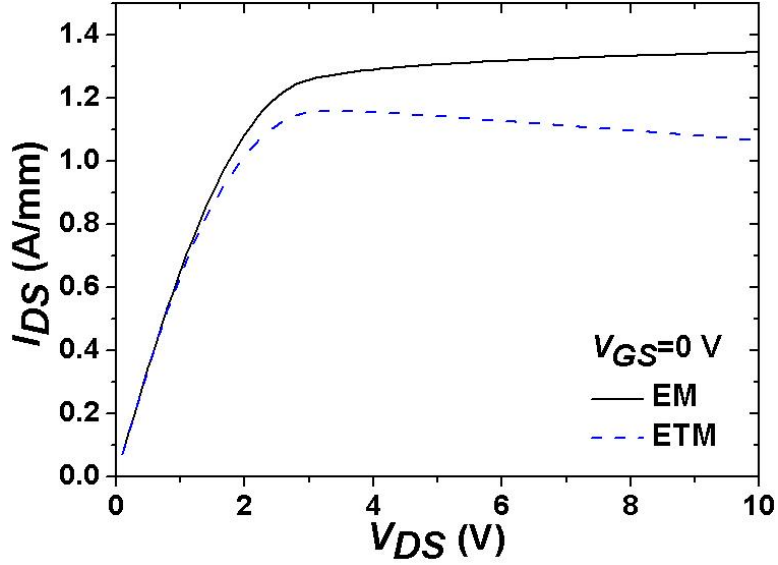


Figure 3-23: Comparison of the simulated I_{DS} for the standard gate by using EM and ETM coupling method.

difference of I_{DS} is due to the reduced electron mobility caused by the device self-heating.

Figure 3-24 shows simulated 2D distribution of electric field magnitude ($|E|$) and principal stress for standard gate (a) and field-plate gate (b) by using EM coupling method. Note that the first principal stress showed in the graph is a total value of built-in stress caused by lattice mismatch in the epitaxial layers and CPS. This built-in stress was calculated using the method described in [80][81] and it was introduced as the initial stress for the simulations in the software COMSOL. For a standard gate structure, both the maximum $|E|$ and stress appear at the GEDS, tending to 8.3 MV/cm and 3.7 GPa, respectively. As expected, for a field-plate gate, the maximum $|E|$ and stress are reduced to 4 MV/cm and 3.4 GPa, respectively, because the electric field is spread by the field-plate.

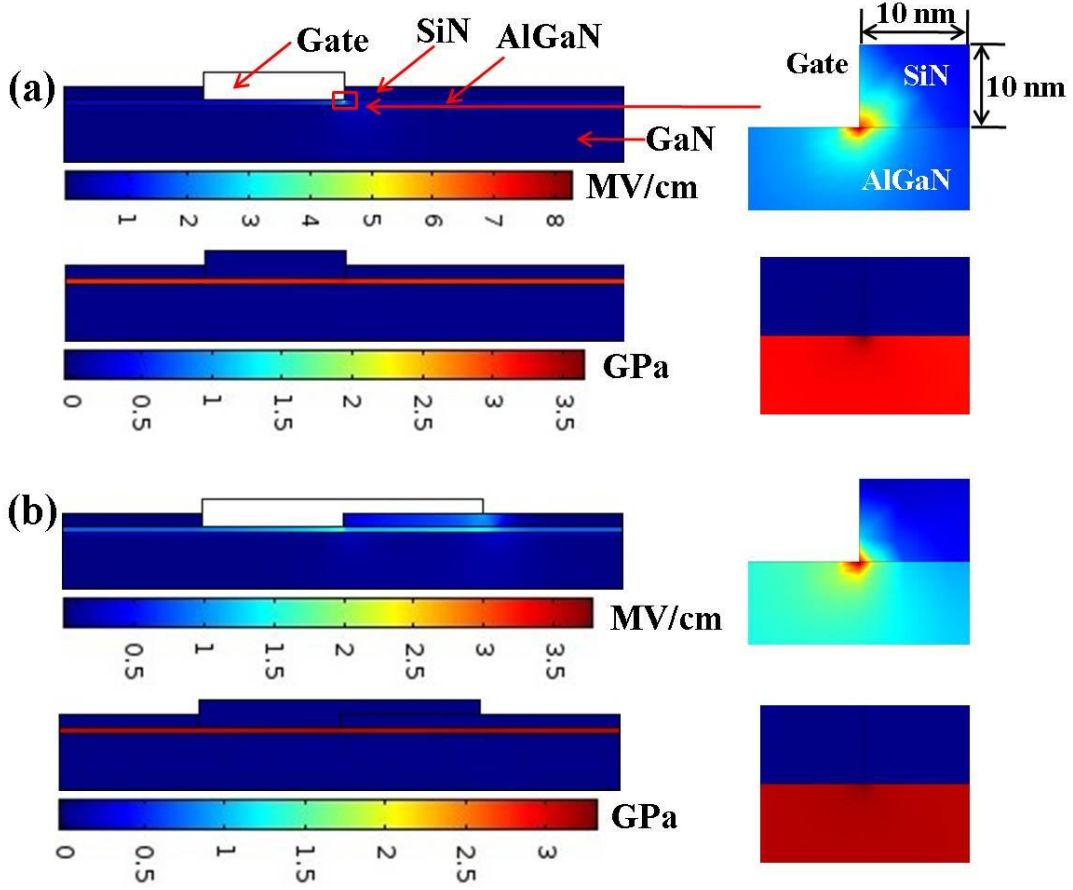


Figure 3-24: Simulated 2D distribution of electric field magnitude and first principal stress for standard gate (a) and field-plate gate (b) by using EM coupling method. The principal stress is the total value of built-in stress caused by the lattice mismatch in the epitaxial layers and converse piezoelectric stress (CPS). The biases are $V_{GS}=0$ V and $V_{DS}=10$ V. The right graphs show the enlarged area near the GEDS.

In order to view the results in Figure 3-24 clearly, 1D plots of $|E|$ and principle stress along the AlGaIn surface (horizontal direction) between the source and drain contact are showed in Figure 3-25(a) and across the whole AlGaIn barrier under GEDS (vertical direction) are showed in Figure 3-25(b).

Figure 3-26 shows simulated distribution of lattice temperature (T_L) and TS (also expressed as principal stress) for standard gate (a) and field-plate gate (b). We can see that two peaks of T_L appear at GEGS and field-plate edge of the drain side for the device with field-plate gate (Figure 3-26(b)), while one peak

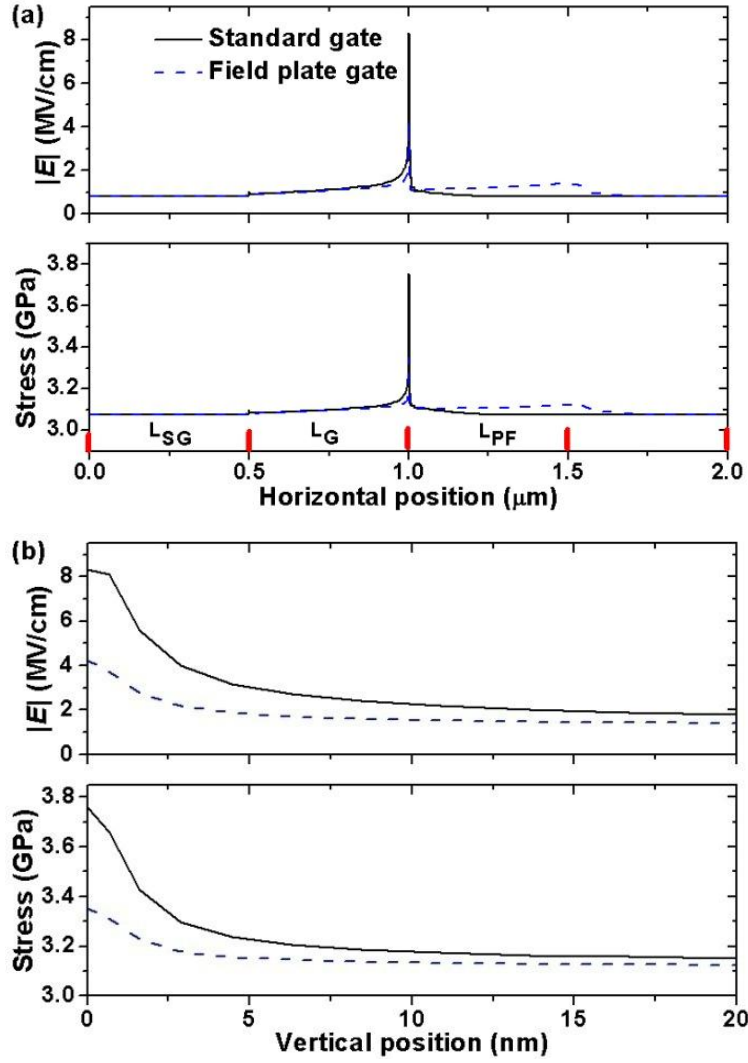


Figure 3-25: 1D plots extracted from Figure 3-24. Distribution of electric field magnitude and principal stress on the AlGaIn surface between the source (at 0 μm in x-axis) and drain contact (at 2 μm in x-axis) (a) and across the whole AlGaIn layer under the GEDS from the AlGaIn surface (at 0 nm in x-axis) to the AlGaIn/GaN interface (at 20 nm in x-axis) (b).

of T_L appears at GEDS for the device with standard gate (Figure 3-26(a)). TS has a complicated distribution and its peak values appear at the gate edges due to the difference of thermal expansion coefficient of different layers and the large temperature gradient there. Its maximum value is around 1 GPa with T_L being around 210 $^{\circ}\text{C}$ for the standard gate. The maximum T_L has a slight decrease of 14 $^{\circ}\text{C}$ for the field-plate gate, but the maximum TS is almost unchanged.

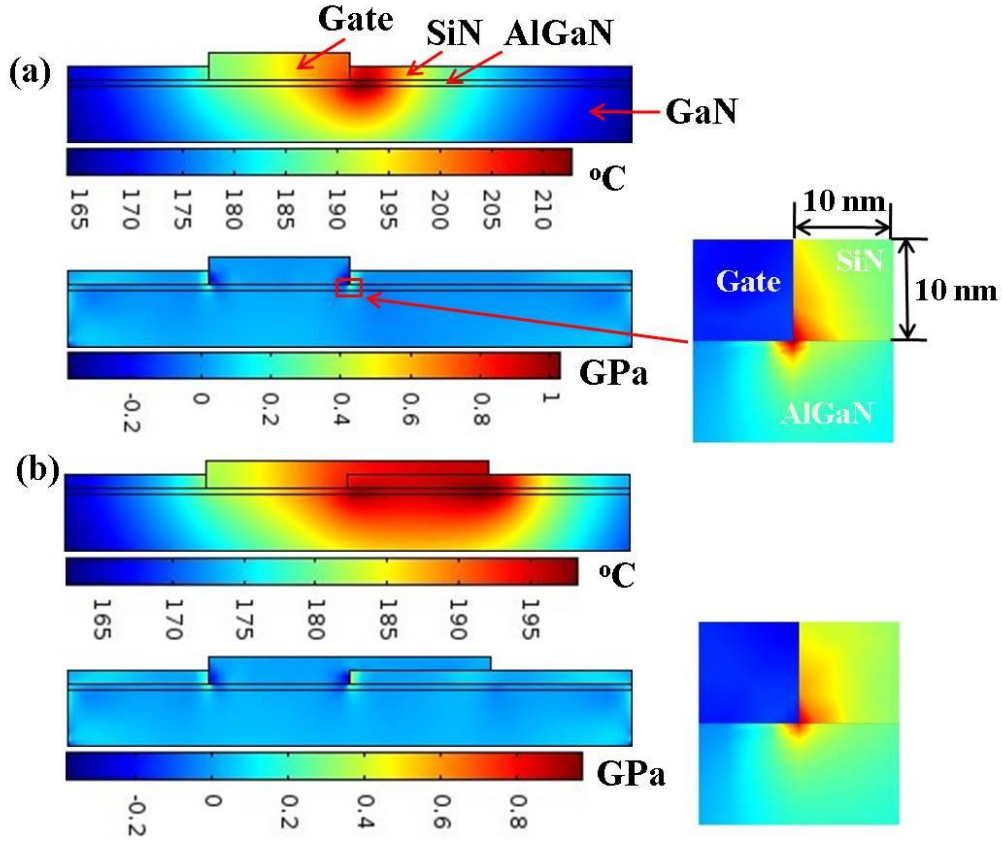


Figure 3-26: Same as Figure 3-24, but for the simulated lattice temperature and thermal stress (TS, also expressed as principal stress) by using ET coupling method. Also device with standard gate (a) and field-plate gate (b) were compared.

Figure 3-27 shows the simulated first principal stress by using a full ETM coupling method. Note that the principal stress showed in the graph is the total value of the built-in stress, TS, and CPS. The maximum values of the stress are 3.9 GPa and 3.5 GPa for standard gate and field-plate gate, respectively. Since the reported yield strength in GaN was 10-15 GPa [82], these principal stress seems cannot lead to the device mechanical failure [83]. However, if there has pre-existing pits/cracks where the maximum principal stress can tend to 13 GPa. These pits/cracks can be propagated in the AlGaN barrier so that the mechanical failure can be induced [79]. Moreover, the 2DEG density can be reduced by the relative big pits/cracks, which therefore leads to the decreased drain current [79].

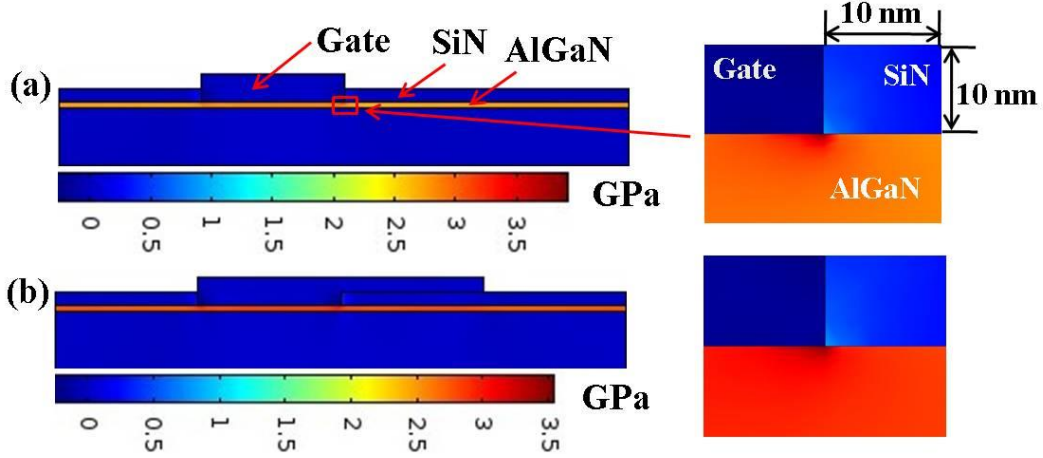


Figure 3-27: Same as Figure 3-24, but the simulations were performed by using full ETM coupling method. The principal stress is the total value of the built-in stress, TS, and CPS. Also device with standard gate (a) and field-plate gate (b) were compared.

3.4 Summary

The basic device electrical behaviors including the electrical potential, electric field, electron conduction band edge, and quasi-Fermi level were presented. With V_{DS} increasing or V_{GS} decreasing, the electrical potential drop tends to concentrate at the GEDS but be negligible in the access regions when the device biased in the $I_{DS,Sat}$ region. Correspondingly, the generated heat also concentrates at the GEDS, forming the hot spot. For the position between the source and drain contact (horizontal direction in the GaN buffer near the interface), the quasi-Fermi energy level is higher than the electron conduction band edge in the access region due to the high density of 2DEG. The decreased electron density by relative low V_{GS} leads to the lower quasi-Fermi level in the channel. The electron velocity saturates near the GEDS due to the relative high electric field there.

The device breakdown behavior influenced by the over-hanging and slant field plate were simulated. The maximum electric field in the channel was reduced significantly by these two types of field-plate and thus the device

breakdown voltage can be enhanced.

Simulations of C_{FR} and C_{FP} for the device in steady-state were carried out. When the device is in off-state ($V_{GS} < V_{th}$), C_{FP} has a main contribution for the parasitic capacitance before the depletion of 2DEG under the field-plate. After this depletion, C_{FR} is the main contribution for the parasitic capacitance. Also C_{FR} becomes more significant for the highly scaled device and it is dependent on the permittivity and thickness of the dielectric.

Electro-thermal simulations and electrical characterizations for the device geometry impact on the self-heating were performed. Long L_{GD} was found to decrease the channel temperature due to the less heat concentration at the GEDS, which would be an optimized way to suppress the device self-heating. Note that for specific applications, a compromise way should be considered for the optimized parameters, e.g., long L_{GD} can also increase the device on resistance as well as access capacitances and thus limits its applications in switches. Besides the device geometry, T_{amb} was also found to be relevant to the thermal resistance, mainly due to the temperature-dependent thermal conductivity of the substrate.

The EM, ET, and ETM coupling method were developed and used to analyze CPS, TS, and total stress in the device, and these stresses for two gate structures (standard gate and field-plate gate) were compared. EM simulations illustrated that the electric field and CPS can be reduced by the field plate. ET simulations illustrated that the TS has a complicated distribution in the device and also has the maximum value at the GEDS due to the different thermal expansion coefficient of the gate metal, the AlGaIn barrier, and the passivation. At last, the total stress including the built-in stress in the AlGaIn barrier, the CPS, and the TS was simulated by ETM full coupling method. It was illustrated that this total stress seems too small to cause the mechanical failure (here the mechanical failure means that when the total stress exceeds a critical value, pits/cracks will be formed at GEDS). However, relative high stress could

present in the pre-existing pits/cracks at GEDS, which might be high enough to cause mechanical failure.

The experimental work in Section 3.2.1 and 3.2.2 were performed by S. Mart ín-Horcajo. Most of contributions in Section 3.2.1 were published as an article, see [46]. Most of contributions in Section 3.2.2 were accepted to be published, see [84].

4 Simulation of heat spreading by integrated diamond with GaN HEMT

Apart from the current collapse induced by traps and the reliability concerns, self-heating presents a limitation for the practical applications of GaN HEMTs [85], which mandates a good thermal management in order to improve the device performance.

To reduce the channel temperature, several approaches have been reported. To incorporate a high thermal conductivity substrate, the epitaxial layers of the transistor structure were atomically bonded [86][87] or directly grown on a diamond substrate [88][89], which acted as a heat sink to extract the generated heat in the channel. However, there were size limitations for the diamond substrate and bonding layer limitations to the III-Nitride material. Furthermore, considerable thermal boundary resistance (TBR) exists at the substrate interface with epitaxially grown GaN [90]. Thus, any high thermal conductivity substrate would have to overcome the thermal resistance of the GaN buffer and TBR which has been found to have sizeable effect on the self-heating in devices on SiC substrate [90]. Standard flip-chipping method has also been used to extract the heat by bonding a heat sink on top of the device [91]. Other methods which focus on integrating a heat spreading layer on top of the device have also been attempted. One such material, nanocrystalline diamond, possesses a high thermal conductivity [92] and is expected to be a suitable candidate for a heat spreading material. Seelmann-Eggebert *et al* [93] and Alomari *et al* [94] have reduced the electrical degradation of HEMT devices using CVD diamond directly overgrown on a fully processed GaN-HEMT wafer. Tadjer *et al* adopted a different method termed “diamond-before-gate,” in which the diamond film is grown mid-process prior to gate deposition. Reduced channel temperature compared to a reference HEMT with only a

dielectric passivation layer and no diamond cap was also reported [95]-[97]. Anderson *et al.* reported diamond as gate contact for heat spreading [98] and Meyer *et al.* reported improved RF performance due to reduced source access resistance afforded by the diamond film [99]. Yang *et al.* used graphene-graphite quilts transferred around the drain contact, obtaining a reduced channel temperature and improved drain current [100].

Previous simulation work on the topic has only focused on temperature analysis by thermal simulation [95] [100]. In this chapter we expand this effort to include the device electrical characteristics in both DC and pulsed conditions. We will conduct a full electro-thermal coupling simulation to investigate the potential reduction of self-heating effects induced by the capped diamond layer. The steady-state simulation (DC) focuses on illustrating the mechanism of the heat spreading effect and optimizing its design, while the transient simulation demonstrates the advantages of the proximity of diamond layer to heat source by considering several factors, such as substrate and TBR. We will perform simulations for two cases:

- 1) The diamond layer is integrated on top of the device (top-side heat spreading), and
- 2) The diamond is integrated in the vias etched in the substrate (back-side heat spreading).

4.1 Top-side heat spreading

4.1.1 Steady-state simulation

The two simulated device geometries shown in Figure 4-1, control and diamond-capped HEMTs, have a 300 μm lateral device dimension, and include a 300 μm thick Si substrate (unless stated otherwise), a 2 μm thick GaN buffer layer, and a 20 nm thick $\text{Al}_{0.26}\text{Ga}_{0.74}\text{N}$ barrier layer. The gate length is 1 μm and

the distance between source and drain is 15 μm . This wider gate-drain spacing arose from experimental concerns. In the literature [95], the need for thermal characterization (Raman and infrared thermography) meant that a wider section of the channel had to be available, which limited the maximum current of these devices. Our simulations conform to the mask designs of the experimental work presented in the literature [95], with the understanding that future improvements in processing will likely lead to more conventional device dimensions for both experiment and simulation. A lower drain current and higher on-state resistance did not alter the basic findings in this work. The

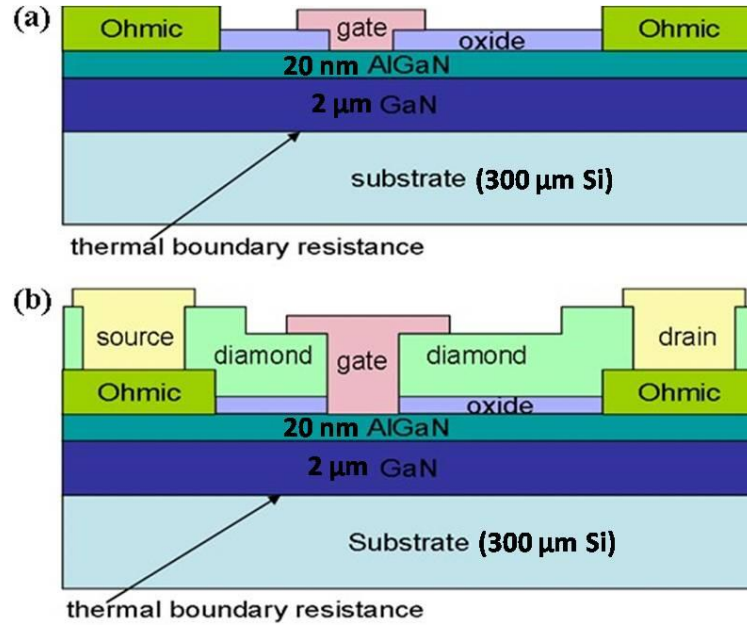


Figure 4-1: Simulated structure of (a) control (C-HEMT) and (b) diamond-capped HEMT (D-HEMT).

Material	Thermal conductivity k (W/K $\cdot$ m)	Heat capacitance c (J/kg $\cdot$ K)	Mass density ρ (k/m ³)
GaN/AlGaN	$160 \cdot (300/T)^{1.4}$	490	6150
Si	$148 \cdot (300/T)^{1.3}$	705	2329
SiC	$400 \cdot 300/T$	581	3100
Diamond	1000	520	3515
SiN	20	700	3000
Metal (Au)	300	128	19200

Table 4-1: Parameters used in the simulations

control HEMT (C-HEMT) had 100 nm thick SiN passivation layer, whereas the diamond capped HEMT (D-HEMT) had a 10 nm thick SiN and additional diamond layer whose thickness will be appointed for the specific simulation. The value of the TBR was not included unless pointed out specifically. Some required parameters in the simulation are given in Table 4-1.

A. Temperature distribution

Figure 4-2 shows the distribution of temperature and heat flux (denoted by red arrows) near the gate for a C-HEMT (a) and a D-HEMT (b) with 1 μm additional diamond layer. The devices work in the saturation region ($V_{GS}=1$ V and $V_{DS}=20$ V) where the generated heat highly concentrates at the GEDS, which increases the temperature and its high gradient in a very small area (hot spot). As the arrows show, for a D-HEMT most of the generated heat at the gate edge is spread by the diamond layer than the GaN epilayer, due to the high thermal conductivity of diamond compared to GaN (see Table 4-1). For C-HEMT, due to the low thermal conductivity of the SiN passivation layer, most heat is spread by the GaN and the substrate. Thus SiN passivation cap has negligible influence on the channel temperature distribution. Note that both the C- and the D-HEMT models only have been heat sunk at the bottom of the substrate, meaning that the heat ultimately gets extracted through the substrate.

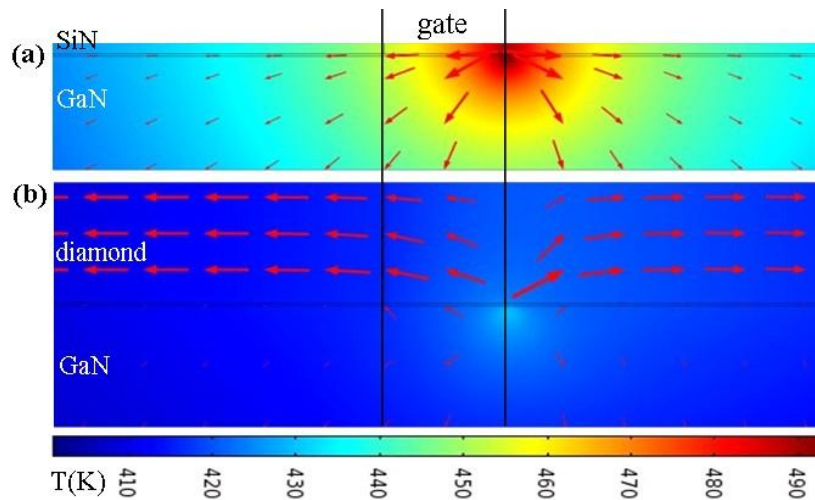


Figure 4-2: Distribution of heat flux (denoted as red arrows) and temperature near the gate for C- (a) and D- HEMT (b) with 1 μm diamond layer.

Figure 4-3 shows the horizontal temperature profile along the channel 1 nm below the AlGaIn/GaN interface for the C- and D-HEMT. The maximum temperature reduction of 55 K (30.4%) happens at the GEDS, and it decreases rapidly with the position departing there, having a relative low value of 3.6 K (3.4%) and 11 K (9.2%) at the drain and source contact, respectively. This is due to the highly concentrated heat, so that the temperature reduction happens in a very small area near the heat source. Considering the exponential relationship between the mean time to failure (MTTF) and the channel temperature, this might alleviate the thermally-induced degradation and improve the device reliability.

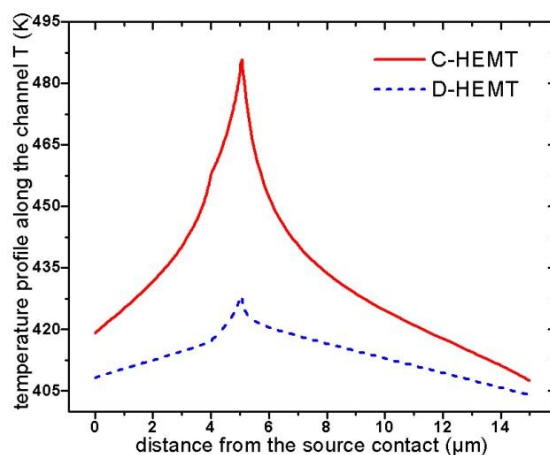


Figure 4-3: Temperature profile along the channel below the AlGaIn/GaN interface 1 nm for C- and D-HEMT, the latter with a 1 μm diamond layer.

B. Diamond layer thickness

Figure 4-4 shows the drain current I_{DS} (a) at $V_{GS}=1$ V and reduced peak channel temperature ΔT (b) as a function of the diamond thickness. I_{DS} for D-HEMT with 0.5, 5, and 20 μm thick diamond layer were compared with that of the C-HEMT. ΔT was simulated at $V_{GS}=1$ V, $V_{DS}=12$ and 20 V for the diamond thickness ranging from 0.5 to 20 μm. Figure 4-4(a) shows I_{DS} for the D-HEMT with a 5 μm diamond layer has a 10.5% improvement at $V_{DS}=20$ V, and this improvement should be more pronounced for higher V_{DS} . As Figure 4-4(b) shows, ΔT increases faster with the diamond cap thickness increasing

from 0.5 to 10 μm , relatively slower from 10 to 20 μm , and shows a saturation behavior for a thicker diamond layer. The saturation value is limited by the diamond thermal conductivity. For $V_{DS}=20$ V (red line), ΔT was about 80 K for D-HEMT for the first 5 μm of diamond cap thickness, and just ~ 12 K higher for a diamond cap thickness of 20 μm . Therefore, it is the initial few microns of the diamond layer which plays the main role on reducing the channel temperature. This is encouraging, as the growth rate is only about 0.1 micron per hour. Also the relative high growth temperature of diamond leads to considerable thermal stress due to mismatch of the thermal expansion

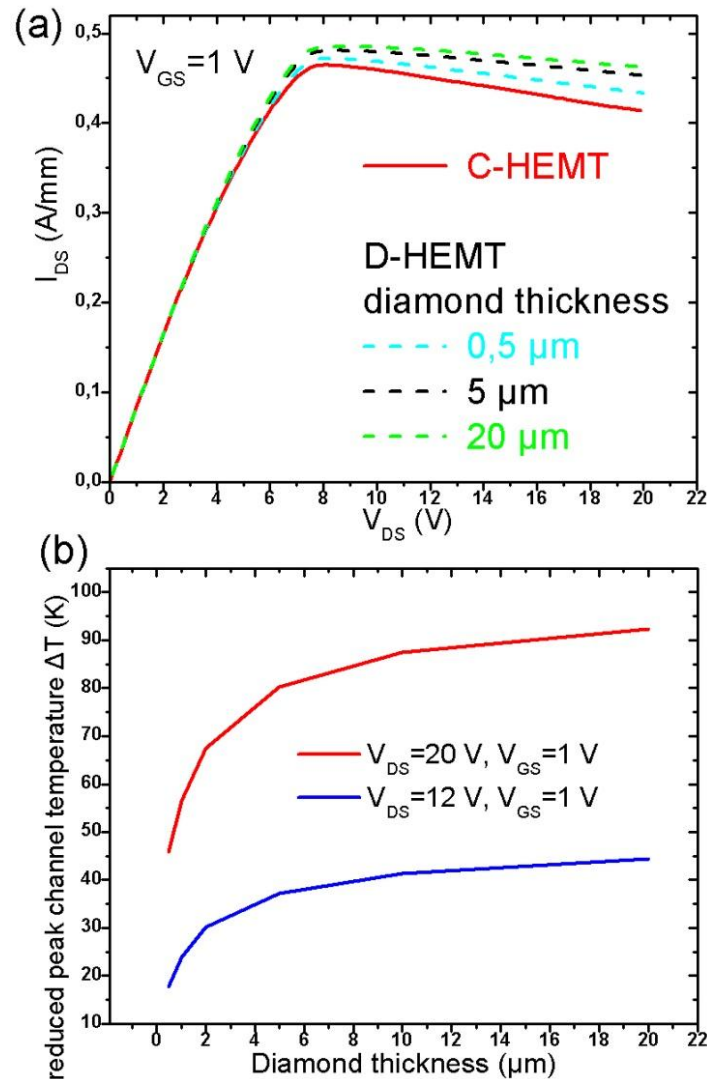


Figure 4-4: Simulated I_{DS} (a) and reduced peak channel temperature ΔT (b) as a function of the diamond thickness. I_{DS} has a comparison of C- and D-HEMT with 0.5, 5, 20 μm diamond layer. ΔT is simulated at $V_{GS}=1$ V, $V_{DS}=12$ V and 20 V.

coefficient for diamond and substrate, which cannot allow relative thick diamond growth. Here the simulation was performed using Si as a substrate material. Similar thermal simulation has shown that the heat spreading effect with graphene-graphite would be more pronounced for sapphire substrate due to its relative low thermal conductivity [100]. However, it was not feasible to grow diamond on sapphire substrate due to the exfoliation induced by large stress as a result of the large thermal mismatch between them [95].

C. Distance between diamond and heat source

Figure 4-5 shows a simulation case which illustrates how the proximity of diamond to the gate edges affects the channel temperature. The diamond did

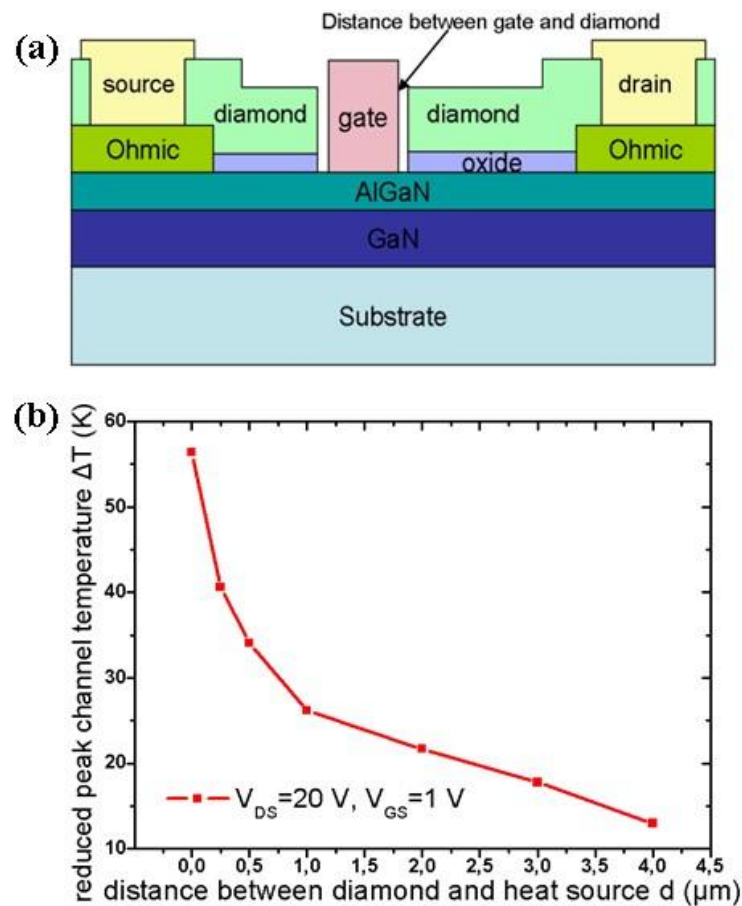


Figure 4-5: Simulated device structure (a) and reduced peak channel temperature ΔT (b) as a function of the distance between diamond and heat source.

not contact the heat source, while its thickness was kept unchanged (Figure 4-5(a)), the distance d between the gate and the diamond layer was varied. ΔT was simulated as a function of this distance, as shown in Figure 4-5(b), and increases from 13 K for $d=4\text{ }\mu\text{m}$ to 57 K for the case when the diamond was in contact with the gate edge ($d=0\text{ }\mu\text{m}$). In other words, ΔT depended on the small volume of diamond closer to the heat source, and was less affected by the total volume of grown diamond. Note that because diamond can be grown in a dopant-free reactor resulting in very low conductivity films [101], the device design can allow a physical contact to the gate. Considering the importance of the distance between the heat source and heat spreading layer, this might be an advantage over using other conductive materials such as graphene-graphite [100].

D. Anisotropy of diamond thermal conductivity

Kohn *et al* [102] have studied the thermal conductivity of integrated diamond layer on InAlN/GaN HEMTs. They showed that it was anisotropic, and its lateral component was much lower than the vertical component. In addition, it was influenced by the diamond layer thickness and growth mode. Based on their experimental results, we have analyzed how the diamond thermal conductivity would influence heat spreading ability. In the model, k was divided into lateral k_H and vertical k_V components. Figure 4-6 shows ΔT as a function of k_V (blue line, k_H is fixed as $1000\text{ W/K}\cdot\text{m}$) and k_H (red line, k_V is fixed as $1000\text{ W/K}\cdot\text{m}$). As the graph shows, ΔT increased by 1.4 and 17.6 K with k_V and k_H increasing from 300 to $1000\text{ W/K}\cdot\text{m}$, respectively. Therefore, k_H had a much more significant impact than k_V for the case of lateral heat spreading considered here. This can be explained by the fact that the thermal boundary on the device surface is almost adiabatic so that the heat mainly can be spread horizontally, as the heat flux graph shown in Figure 4-2(b).

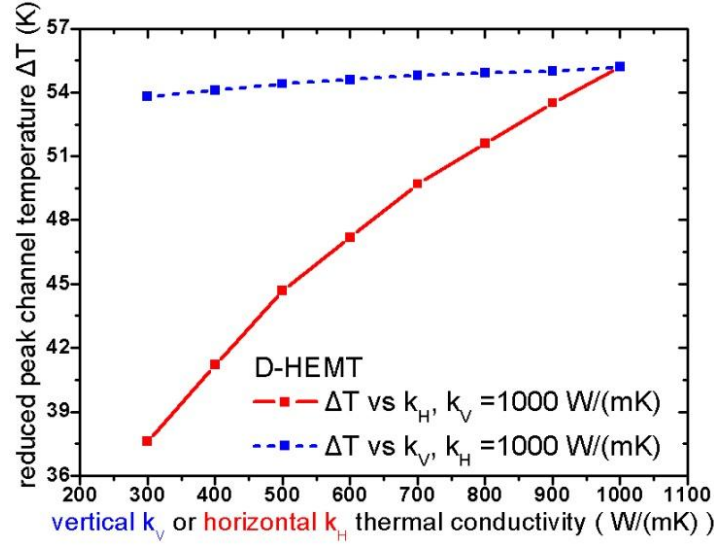


Figure 4-6: The effective diamond thermal conductivity k is divided into vertical k_V and horizontal k_H component. This graph shows the reduced peak channel temperature ΔT as a function of k_V (k_H is fixed as 1000 W/mK) and k_H (k_V is fixed as 1000 W/mK) for D-HEMT with 1 μm diamond layer.

E. Comparison with AlN heat spreading

Tsurumi *et al.* has reported that the AlN passivation can also be used for the heat spreading [103]. AlN and diamond integrated with the device have the same mechanism for the heat spreading except the difference of their thermal conductivity: around 300 W/mK for AlN compared to 1000 W/mK for diamond.

Figure 4-7 shows simulations on comparison of heat spreading ability by AlN

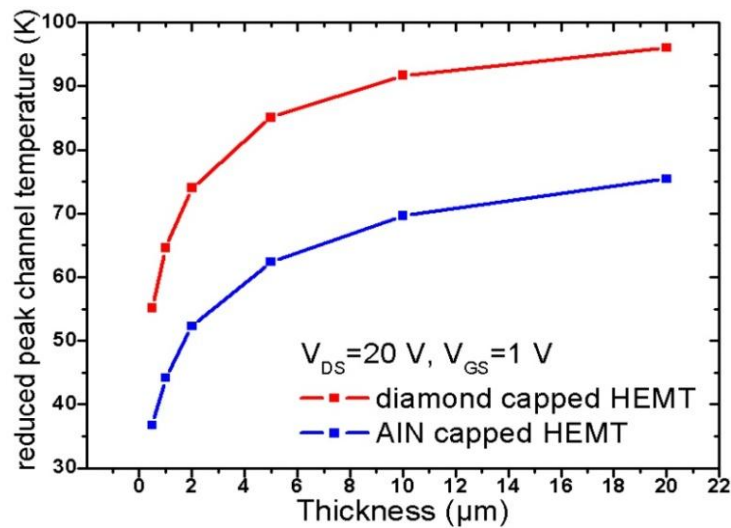


Figure 4-7: Simulations on comparison of heat spreading ability by AlN and diamond.

and diamond. As the graph shows, the reduction of T_{ch} by diamond heat spreading is larger than that achieved by AlN due to the former's higher thermal conductivity. Again, we note that the thermal conductivity is the determinate factor for the heat spreading ability when the reduction of T_{ch} tends to saturate (the thickness tends to 20 μm as shown in the graph).

4.1.2 Transient simulation

The thermal dynamics of the device is critical for their performance, and some experimental studies have been conducted in this regard. Kuzmik *et al* [104][105] have developed electrical characterization method and transient interferometric mapping to obtain the channel temperature responding with the pulse width, and Kuball *et al* [106] and Riedel *et al* [107] used time-resolved Raman thermography to directly measure the transient channel temperature and illustrated the influence of substrate on the device thermal dynamics. It is expected that the self-heating effect can be eliminated with short enough pulse heating. To the best of the authors' knowledge, no experimental work has been reported so far to show how short a pulse should be used to avoid the self heating. This is partially due to the difficulty in experimentally separating the effects of self-heating and charge trapping, both of which would result in a slump in drain current even under short-pulse conditions. In this section, transient simulations were performed to compare the thermal dynamics for C- and D-HEMT focusing on the above issues. The D-HEMT has a 5 μm diamond layer capping for the following simulations.

A. Self-heating effect suppression

Figure 4-8(a) and (b) show I_{DS} and T_{ch} responding to the gate pulse, respectively. V_{GS} is pulsed from -3 V (close to pinch off state) to 1 V while keeping V_{DS} constant at 12 V or 20 V. The transient self-heating effect was

compared for C- and D-HEMT. As Figure 4-8(a) shows, I_{DS} evolution can be divided into two phases. The first one is when it increased fast on a nanosecond scale with the channel transitioning from the off to the on state as the gate voltage pulse rises. The second one is different for the simulations including or not self-heating effect. With self-heating, I_{DS} decreases slowly after reaching its maximum value due to its dependence on temperature, which increases relatively slowly with time; without self-heating, I_{DS} did not degrade after reaching to its maximum value since the channel temperature will keep the same as the ambient temperature all the time (solid green line, Figure 4-8(a)). Here note that the trap effects' impact on the current degradation was not considered in order to identify self-heating effect clearly. For times shorter than ~ 4 ns, I_{DS} is same for both C- and D-HEMT with V_{DS} of 12 and 20 V due to T_{ch} being almost constant with such a short channel heating. As a result, no self-heating effect was observed. After about 4 ns, I_{DS} and T_{ch} start to appear

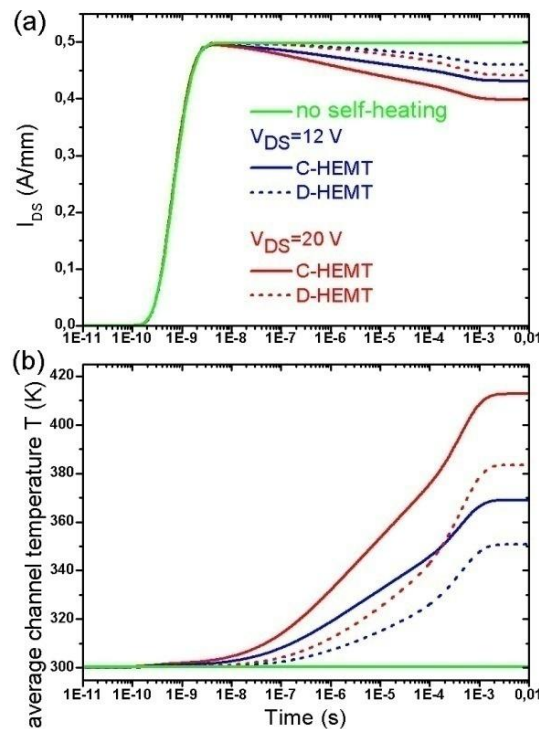


Figure 4-8: Simulation of drain current I_{DS} (a) and channel temperature T (b) responding to the gate pulse for C- and D-HEMT. The gate voltage V_{GS} is pulsed from -3 V (close to pinch off state) to 1 V while the drain voltage V_{DS} keeping as 12 V or 20 V.

difference. D-HEMT had higher I_{DS} and lower T_{ch} , and with increasing time the difference becomes larger until reaching the steady state at ~ 2 ms. Note that when appearing self-heating effect, I_{DS} at 12 V is always higher than that at 20 V for both C- and D-HEMT (reduced current degradation). This phenomenon has been illustrated experimentally in the studies of transient self-heating effect for AlGaIn/GaN HEMTs [104]. It is interesting that when compared to the simulation of the case without self-heating, the D-HEMT has no current degradation until to ~ 100 ns (dashed red line in Figure 4-8(a)), i.e., self-heating effect for the gate pulse width of less than 100 ns can be omitted due to the fact that most of heat generating in this time scale is spread away by the additional heat escaping channel of diamond layer. While for the C-HEMT, only for the gate pulse width less than several nanoseconds we can omit the self-heating effect (solid red line in Figure 4-8(a)), which is also consistent with the reported simulation [108]. Overall, the results show that the self-heating effect can be suppressed for relatively wide pulse by diamond heat spreading effect, which can improve the device performance in the RF regime.

Figure 4-9 shows more details with the comparison of I_{DS} as a function of

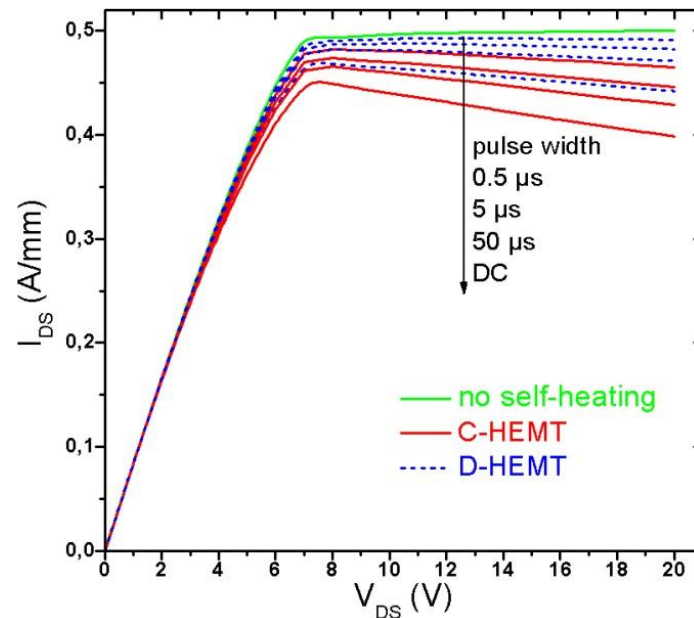


Figure 4-9: I_{DS} as a function of V_{DS} for C- and D-HEMT in DC and pulse operation condition. V_{DS} is fixed, while V_{GS} is pulsed from -3 V to 1 V, and keep its width lasting for 0.5, 5, 50 μ s and infinity (DC)

V_{DS} for C- and D-HEMT in DC and pulse operation condition. V_{GS} pulse widths are 0.5, 5, 50 μ s, and infinity (DC). As the graph shows, I_{DS} has no difference for V_{DS} lower than ~ 3 V, since there is not enough power to heat the channel to a relative high temperature, and no self-heating effect happens. As V_{DS} increasing higher than ~ 3 V, I_{DS} can be improved significantly for all the operation conditions mentioned above thanks to the heat spreading effect. For 0.5 μ s pulse width, D-HEMT has already shown very little self-heating effect.

B. Delayed influence by the substrate

In general, without top-side heat extraction, the heat in the channel would flow through the GaN and substrate into the bottom-side heat sink. Transient electrical behavior of the device should be locally determined by the thermal parameters of the epitaxial layers before the heat flowing into the substrate, as shown experimentally by Riedel *et al* [107]. However, the substrate plays a role.

Figure 4-10 shows the maximum temperature T_{max} at the GaN/substrate interface as a function of time for C- and D-HEMT. T_{max} was higher than the ambient temperature (300 K), meaning heat has already flown into the substrate, indicating that the device electrical behavior will be affected by the substrate.

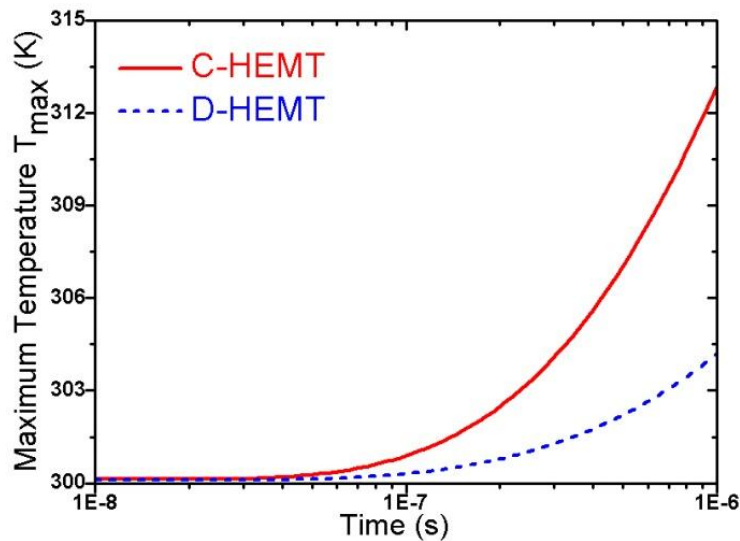


Figure 4-10: Maximum temperature T_{max} at the GaN/substrate interface as a function of time for C- and D-HEMT.

We can see that for C-HEMT T_{max} rises at ~ 40 ns, while for D-HEMT, it rises at ~ 70 ns exhibiting a delay due to most of the heat flowing into the diamond layer. The simulation results are basically in agreement with those of experimental works performed on the ungated AlGaN/GaN device [107].

After illustrating the mechanism of how diamond layer delays the heat flowing into the substrate, the corresponding electrical characteristics for the device on different substrates have also been simulated. Figure 4-11 shows I_{DS} as a function of time for both the C- and D-HEMT on Si and SiC substrate. Consistently with the above illustration, I_{DS} should have no difference until to ~ 40 ns and ~ 70 ns for C- and D-HEMT before heat flowing into the two different substrates, respectively. But if we set this difference within a relatively small but reasonable tolerance such as 1%, I_{DS} of D-HEMT shows this difference up to $10\ \mu\text{s}$, which presents a large delay compared to C-HEMT up to $0.6\ \mu\text{s}$. This result suggests that the thermal resistance of the substrate has no impact on the current degradation for wider pulse with the top-side heat spreading effect. Here we note that under this set difference the heat has slightly flown into the substrate.

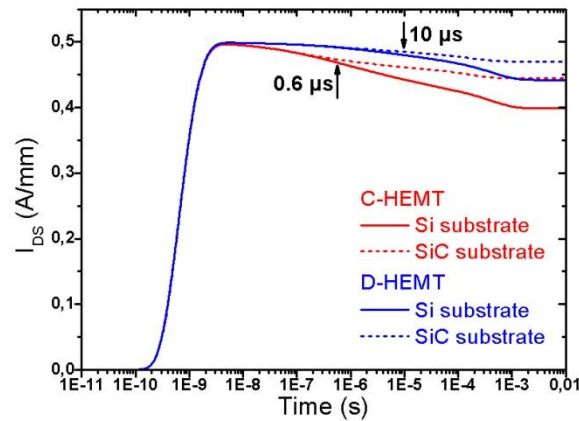


Figure 4-11: I_{DS} as a function of time for C- and D- HEMT on Si or SiC substrate.

C. Reduced influence by TBR

The thermal boundary resistance (TBR) at the GaN/substrate interface,

induced by the nucleation layer, dislocations, impurities, etc., has been shown experimentally to present a significant influence on the thermal behavior of the AlGaIn/GaN HEMTs, contributing up to 50% of the channel temperature for a device on a SiC substrate. In the process of the heat flowing through a thermal boundary, the channel temperature can be increased substantially [109], reducing the RF performance of the device. Figure 4-12 shows a simulation of I_{DS} as a function of time for HEMTs on SiC substrate including the case with and without a TBR. The reported TBR values in the literature show a large variation, and they are dependent on the measurement methods and the interface properties between the GaN and substrate [110][111]. Here we used the value $5.5 \times 10^{-8} \text{ m}^2\text{K/W}$ from the literature [110], and the GaN layer and SiC substrate thickness were set as 1 μm and 100 μm , respectively. As the graph shows for C-HEMT (red lines), in the process of heat flowing through TBR, I_{DS} dropped significantly (8.4%) in a relatively small time scale from 30 ns to 1 μs . This indicates that the TBR plays an important role on the electrical behavior of the device, which can lead to a significant I_{DS} degradation for a relative short pulse width. On the other hand, for a D-HEMT, the TBR only leads to 1.1% degradation of I_{DS} even for DC operation condition (blue lines). Therefore, TBR showed much reduced and delayed influence on I_{DS} due to this heat spreading effect.

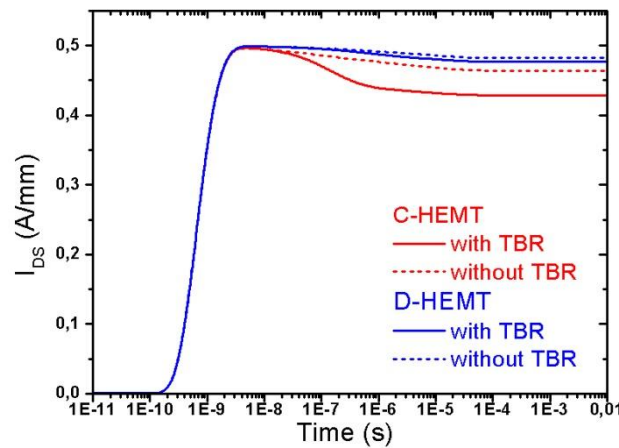


Figure 4-12: I_{DS} as a function of time for C- and D-HEMT on SiC substrate to show the impact of TBR.

4.2 Back-side heat spreading

In addition to integration on top of the device, diamond can also be deposited in vias etched in the substrate to form heat spreading path. Figure 4-13 illustrates top-side (a) and cross-sectional graphs (b) of the diamond-coated vias [112].

Simulated temperature and heat flow distribution are shown in Figure 4-14 for a control (a) and diamond-coated (b) AlGaIn/GaN HEMT on 100 μm Si substrate. The peak temperature is reduced by about 50 $^{\circ}\text{C}$ for the HEMT with the diamond-coated vias in the substrate, compared to a control HEMT. The heat flow in the coated diamond is significantly larger than that in the Si substrate due to the higher thermal conductivity of diamond, supporting the purpose to use of the diamond as a heat flow path (Figure 4-14(b)).

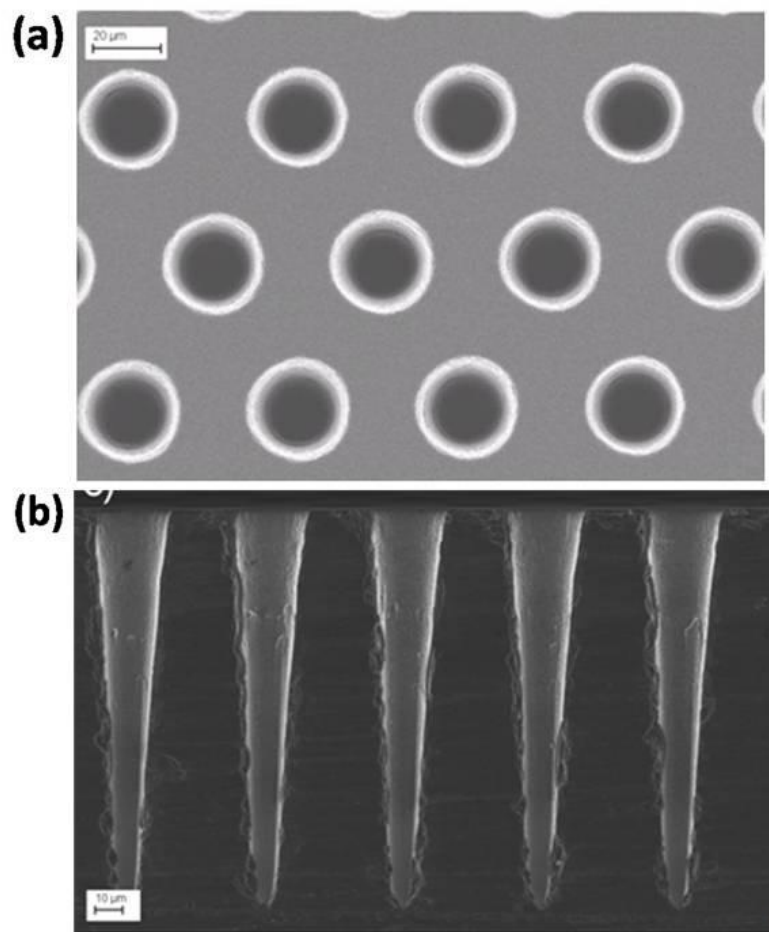


Figure 4-13: Diamond-coated vias in the substrate: top-side graph (a) and cross-sectional graph (b) [112]. The vias have 20 μm diameters.

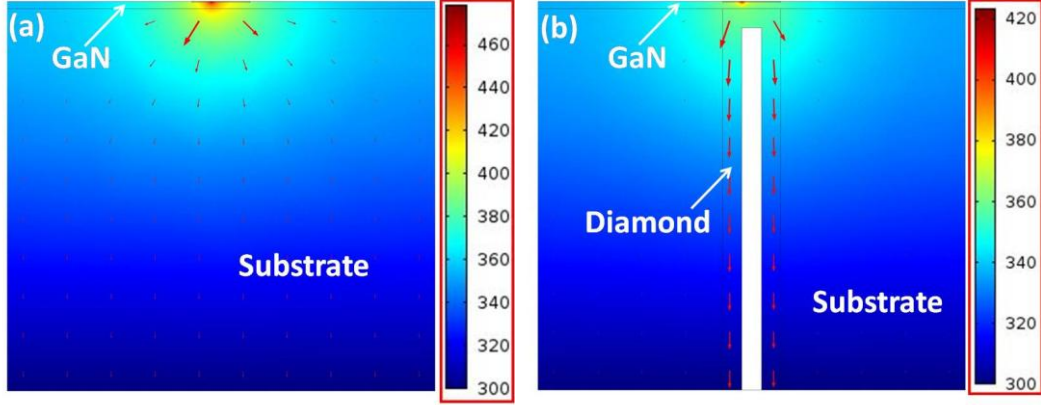


Figure 4-14: Simulated temperature and thermal flow distribution for a control (a) and diamond-coated vias AlGaN/GaN HEMTs (b) on 100 μm Si substrate. The simulated output power was 9W/mm.

Figure 4-15 shows the simulated I_{DS} for the control and diamond-coated HEMT with different diamond thickness. We can see that I_{DS} increased with the diamond thickness increasing. Note that there is a competing mechanism that influences the device temperature, i.e., the coated diamond would decrease the device temperature due to its higher thermal conductivity while the etched vias in the substrate would increase the device temperature due to the reduced heat spreading path. Therefore, as shown in the graph, due to the higher channel temperature, I_{DS} for the device with 0.5 μm coated diamond is lower than that for the control HEMT.

From a thermal standpoint, using a high thermal conductivity substrate (back-side heat spreading) or depositing a top-side heat spreading layer have the same purpose: to reduce self-heating effects. However, these two methods present substantial differences when accounting for the device in pulse operation condition. For the back-side heat spreading method, as shown in Figure 4-11, only for pulses wider than 0.6 μs , the higher thermal conductivity SiC substrate (compared to Si) causes I_{DS} to increase, which accordingly means that no benefits can be obtained for the pulse width shorter than 0.6 μs . This is because there is a distance between the heat source and the substrate induced

by the epitaxial layers. Another aspect is that any benefits of using a high thermal conductivity substrate might be eliminated by a simultaneously large TBR at the GaN/substrate interface. This is a major obstacle for the application of a diamond substrate [113]. While for the top-side heat spreading method, as shown in Figure 4-8(a), because the diamond layer is very close to the heat source, the increased I_{DS} can be obtained at the beginning of the pulse, showing a unique advantage over the back-side heat spreading method. So the top-side heat spreading method for thermal management might be more beneficial when the device operates in RF range. Overall, the essential reason for that is the different distance between the high thermal conductive path (SiC substrate, diamond capping layer) to the heat source.

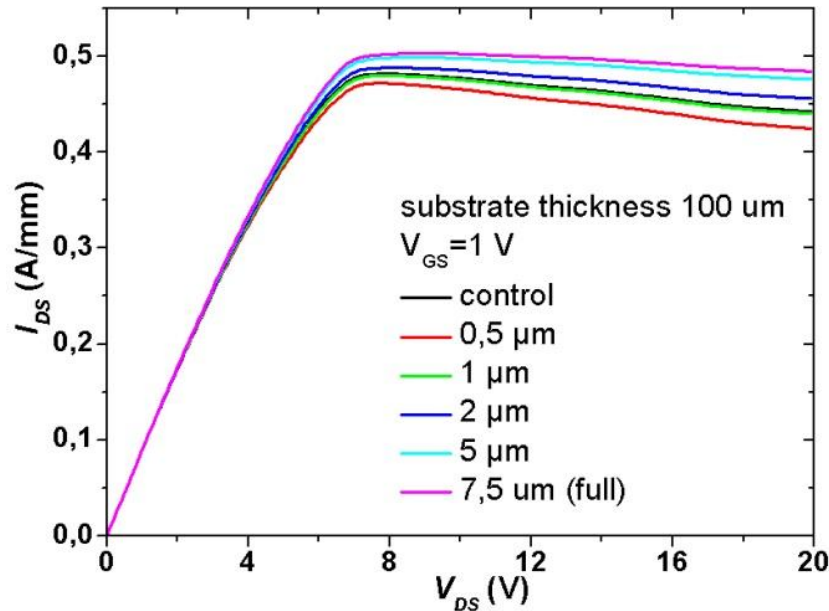


Figure 4-15: Simulated I_{DS} for the control and diamond-coated HEMT with different diamond thickness. Note that the diameter of the vias is 15 μ m, so the diamond thickness 7.5 μ m shown in the graph means the vias is full of diamond.

4.3 Summary

We performed steady state and transient electro-thermal simulations to study the thermal management of AlGaIn/GaN HEMTs with integrated diamond heat

spreading layers. The simulations focused on the comparison of thermal and electrical behavior for C- and D-HEMT operating in DC and pulse condition. It was shown that the device performance can be improved significantly by this top-side heat spreading. The diamond thickness and its thermal conductivity were discussed, which illustrated that the channel temperature reduction shows saturation when increasing the diamond thickness. Therefore, the initial deposited several micrometers thick diamond plays an important role on the heat spreading ability which also increases with increasing the diamond lateral thermal conductivity. This conclusion suggested that it is unnecessary to deposit a thick diamond layer, as the determining factor is the quality of the diamond film. Due to the generated heat concentrating at the drain side of gate edge, the channel temperature reduction depends on a small volume diamond near the heat source, and it is less affected by the total deposited diamond. The “diamond-before-gate” mid process fabrication method presents an advantage over other thermal management methods since the diamond cap is close to the heat source. With this thermal management procedure, the substrate and TBR have a delayed impact on the electrical behavior of the device in pulsed operation condition. The device with a 5 μm diamond layer can present 10.5% improvement of drain current, and the self-heating effect can be neglected for a 100 ns pulse width at 1 V gate and 20 V drain voltage. Our planned future work will address the investigation of this aspect. Overall, this work would provide useful guidelines to optimize the heat spreading layer design so as to make this method more feasible to the integration of device thermal management.

Most of contributions in Section 4.1 and 4.2 were published as articles, see [114] and [112], respectively.

5 . Impact of intrinsic stress in diamond capping layers on the device electrical behavior

It is well known that growing AlGa_N and Ga_N layer heteroepitaxially, such as on Si or SiC substrates, as well as deposition of a passivating dielectric layer, results in intrinsic stresses in the structure. The interaction of this intrinsic stress on the electrical properties of the device has been investigated previously, including the impact of substrate thinning on the 2DEG density n_s [115]. For films deposited at higher temperatures, such as nanocrystalline diamond (NCD), a significant stress will result from the large thermal expansion coefficient mismatch with the III-Nitride heterostructure [116] [117]. Therefore, an NCD film on either the device-side or the substrate-side of a Ga_N HEMT wafer will lead to changes in Q_p and n_s induced by the additional stress. The objective of this work is to quantify the additional stress a top-side NCD cap will induce on an AlGa_N/Ga_N HEMT.

Studies on how the device performance is influenced by mechanically deforming the heterostructure are well documented [118]-[120]. Experimental studies of the mechanical deformation have adopted two experimental paths. On one hand, an external force was applied on the whole heterostructure wafer, with or without fabricated HEMTs, and an increased conductivity of the 2DEG with external tensile was observed [118]. This effect was attributed to a change of 2DEG density induced by change of Q_p in both the AlGa_N and Ga_N layers, as well as a change in the electron effective mass. Consistent with this observation, an increased DC drain current in AlGa_N/Ga_N HEMTs was also observed with the application of external uniaxial tensile stress [120]. This change in electrical properties of the device has been applied to use AlGa_N/Ga_N HEMTs as pressure sensors [121][122]. The second approach focused on the change of the device electrical behavior caused by the internal stress in the device. This is to illustrate

the interaction of the intrinsic stress in the cap layer, built-in stress in the epitaxial AlGa_N/Ga_N layers, and residual stress in the substrate [123].

As studied in the previous chapter, HEMTs in high power operation condition generate considerable heat in the channel. This leads to elevated lattice temperature, furthermore reduces the electron mobility and thus the output drain current (self-heating effect). Diamond capped HEMTs were discussed above as a heat spreader to reduce the device self-heating due to its high thermal conductivity [93] [95]. The integration with HEMTs can be performed by direct deposition on the full processed HEMT [93] [94], or by the “diamond-before-gate” approach, in which the diamond deposition is prior to the gate fabrication [95] [96].

The experimental observations have previously been explained by an empirical tight-binding model, which assumed that the AlGa_N and Ga_N layers were subject to identical deformation [124]. Therefore, the stress resulting change in 2DEG density was only determined by the difference of the piezoelectric polarization coefficients of AlGa_N and Ga_N. Mechanical analysis has illustrated the non-uniform stress distribution in the channel induced by intrinsic stress in Si₃N₄ passivation, without considering its impact on the electrical transport properties [125]. In fact, the stress in the device does not only affect the 2DEG distribution, but introduces also structural degradations and reliability issues. This also requires the development of stress-related coupled model for the analysis of inverse piezoelectric and thermal stress.

The 2DEG density can be calculated by using a simple analytical expression with assumption that the strain only exists in the AlGa_N layer [80][81]. This method leads to uniform distribution of the polarization charge on the AlGa_N surface and at the AlGa_N/Ga_N interface, and also uniform distribution of the 2DEG close to the interface. However, interaction of the stresses in different parts of the device yields to a complex distribution of the stress in the epitaxial layers, and thus a non-uniform distribution of the 2DEG.

In this case, a numerical method would be necessary for the calculation of the 2DEG density. In this work, we present a model to couple the 2DEG in the channel with the stress in the device. This enabled us to analyze how an intrinsic stress, such as that induced by passivation and heat spreading layers, affected device electrical behavior. Based on this model, a finite element method (COMSOL) was used to perform the numerical calculations. Simulation results are complemented with experimental device data.

5.1 Experimental details

Transistor devices were fabricated on 25 nm $\text{Al}_{0.25}\text{Ga}_{0.75}\text{N}$ / 1.3 μm GaN HEMT structures grown by metal-organic CVD on high-resistivity (111) Si substrates. Details on transistor device processing, NCD growth, and NCD thermal conductivity characterization are provided elsewhere [95][126]. The control HEMT (A-HEMT) was passivated by a 100 nm thick SiN layer deposited by plasma-enhanced CVD (PECVD). For the diamond-capped HEMT (B-HEMT), a thin 10 nm PECVD SiN layer, followed by an additional 500 nm thick NCD cap, were grown after the mesa isolation and ohmic contact steps, but before Schottky gate metallization. NCD was grown at 750 °C (~100 nm/hr) and was unintentionally boron doped for p⁻ type conductivity. Prior to gate metal liftoff, O₂-based and SF₆-based inductively-coupled plasma (ICP) was used to etch the diamond and SiN layers, respectively. Plasma damage was minimized by reducing the O₂ ICP power as the AlGa_{0.25}N surface was approached. Process monitoring was performed via Hall measurements and SEM imaging. The device structure of the devices and some key electrical parameters are showed in Figure 5-1 and Table 5-1, respectively. The reduced electron mobility shown in Table 5-1 is due to the lattice damage in the process of dielectric and diamond growth. The change of other parameters will be explained in the following sections. Raman spectroscopic methods were employed for stress

characterization in the GaN heterostructure. An Ar-ion laser with excitation wavelength of 488 nm and a spot size of was 0.5 μm , and the measured Raman wavenumbers were averaged over the GaN layer depth. The details of micro-Raman spectroscopy analysis can be found in [53] and [127].

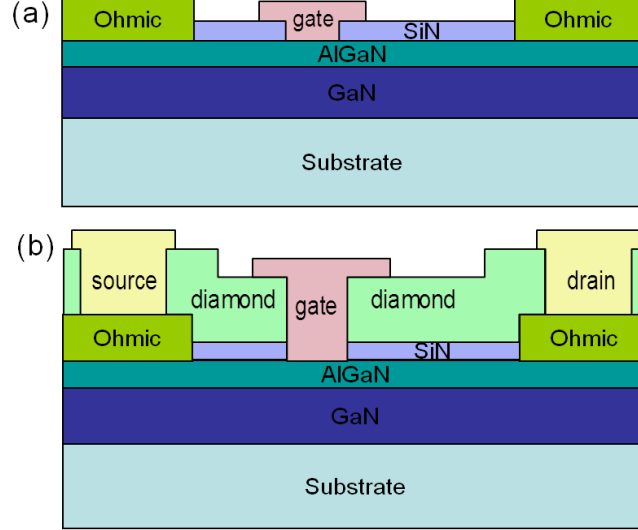


Figure 5-1: The device structure of a control (a) and diamond capped (b) AlGaN/GaN HEMTs. The gate width is 100 μm , gate length is 1 μm , and distance between gate and source/drain contact are 2.3 μm /10 μm .

	I_{DSS}	I_{OFF}	$G_{m,max}$	V_{TH}
	(A/mm)	(mA/mm)	(mS/mm)	(V)
A-HEMT	0.414	0.019	114	-3.53
B-HEMT	0.432	0.199	103	-3.96
	R_{ON}	R_{SH}	μ_n	n_s
	(Ω -mm)	(Ω /sq.)	(cm^2/Vs)	(cm^{-2})
A-HEMT	10.49	674	1660	5.59×10^{12}
B-HEMT	11.35	536	1080	1.08×10^{13}

Table 5-1: Device parameters of the control HEMT (A-HEMT) and the diamond-Capped HEMT (B-HEMT). Maximum drain current I_{DSS} was measured at $V_{GS}=1$ V, $V_{DS}=20$ V; Off state current I_{OFF} at $V_{GS}=-5$ V, $V_{DS}=0.1$ V; maximum transconductance $G_{m,max}$ and threshold voltage V_{TH} at $V_{DS}=0.1$ V; on resistance R_{ON} at $V_{GS}=1$ V. Sheet resistance R_{SH} , mobility μ_n , and 2DEG density n_s measured by Hall using a 0.1 mA current source.

5.2 Model description

A coupled electro-thermo-mechanical model was developed as described in the following using the parameters given in Table 4-1 and Table 5-2. In this model, first the stress, Q_P , and the non-uniform distribution of n_S in the epitaxial layers were determined by numerically solving the electromechanical coupling equations. The resulting Q_P distribution was thus fed into the Poisson equation, which was solved simultaneously with the continuity equation and the thermal conductivity equation. By solving the three equations simultaneously, the electrical characteristics of the device were obtained. In this coupling procedure, the electrical characteristics of the device under bias did not affect the Q_P distribution, indicating that it was a one-way coupling method without considering inverse piezoelectric and thermal stress which have negligible influence on the device electrical behavior as shown in [83].

[80]	C_{11} (GPa)	C_{12} (GPa)	C_{44} (GPa)
GaN	367	135	95
Al_{0.25}Ga_{0.75}N	375	136	100
e_{31} (C/m ²)	e_{33} (C/m ²)	e_{15} (C/m ²)	P_{SP} (C/m ²)
-0.49	0.73	-0.3	-0.029
-0.52	0.92	-0.35	-0.043

Table 5-2: Piezoelectric parameters used in the simulation

Due to the large gate width/length aspect ratio, the simulations were restricted in two dimensions (see the simulated device structures shown in Figure 5-1 which are consistent with the fabricated ones). Therefore, a plane-strain approximation was used for the mechanical simulation. This approximation is illustrated in Figure 5-2. We can see that σ_{yy} (here mainly induced by the strain in x- (s_{xx}) and z-direction (s_{zz})) is significant compared to σ_{xx} (Figure 5-2(a)). However, the strain in the y-direction (s_{yy}) is negligible compared to s_{xx} and s_{zz} (Figure 5-2(b)).

For A-HEMT, the 2DEG is simulated by assuming uniform tensile strain in AlGa_N layer but no strain in GaN. Strain in AlGa_N layer can be expressed as $s_{AlGaN} = (a_{GaN} - a_{AlGaN})/a_{AlGaN}$. In this case, Q_p only appears on the surface of AlGa_N and at the AlGa_N/GaN interface, but not in the GaN bulk. In ideal conditions without considering traps or defects, the following polarization charge is presented: surface polarization charge $\sigma_s = P_{PE(AlGaN)} + P_{SP(AlGaN)}$ on the surface of AlGa_N layer, and interfacial polarization charge $\sigma_i = P_{PE(AlGaN)} + P_{SP(AlGaN)} - P_{SP(GaN)}$ at the AlGa_N/GaN interface, where P_{PE} and P_{SP} denote piezoelectric and spontaneous polarization vector, respectively (see the schematic of the charge distribution in Figure 5-3).

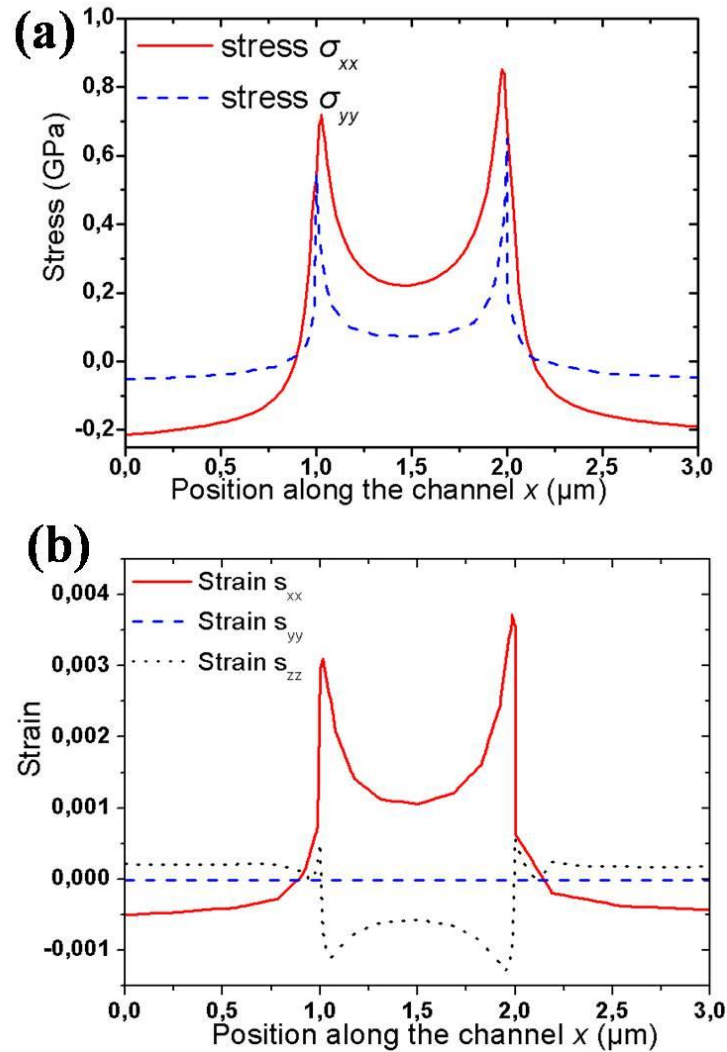


Figure 5-2: Comparison of simulated stress and strain in x-, y- and z-direction.

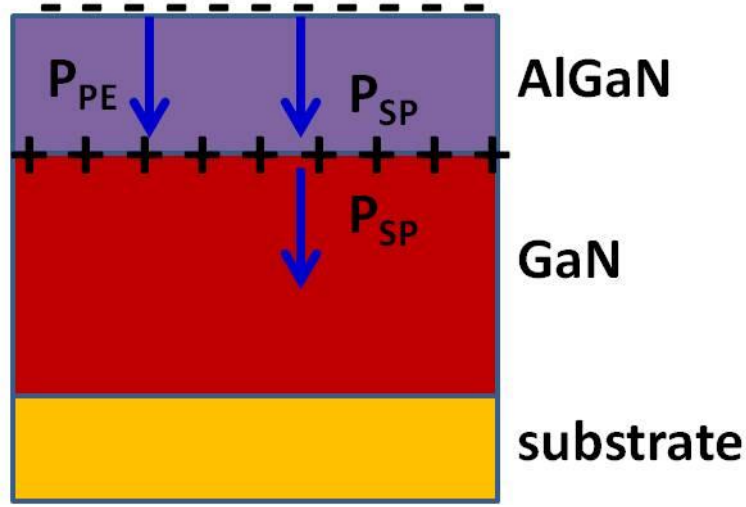


Figure 5-3: Schematic of the AlGaN surface and AlGaN/GaN interface polarization charge distribution

In the simulation, only the in-plane initial constant strain for AlGaN is provided, accordingly the out-of-plane strain will be numerically determined. For B-HEMT, Q_p for AlGaN and GaN is affected by the intrinsic stress in the cap layer since they are fully mechanically-coupled system, leading to the non-uniform distribution of Q_p on the AlGaN surface and at the AlGaN/GaN interface, as well as in the AlGaN and GaN bulk. This intrinsic stress is assumed as in-plane σ_{xx} along the gate length (x-direction) and σ_{yy} along the gate width (y-direction). Here we note again that since there is a the large gate width/length ratio, the change of the Q_p is only induced by a change of strain in x- and z- direction (vertical), while keeping the y-direction strain constant. With the initial strain in AlGaN and the intrinsic stress in diamond layer, we solve the coupled mechanical and electrostatic equations with the constitutive equations to obtain the electric displacement vector D_i as well as the 2DEG density n_s for A- and B-HEMT, respectively. The applied boundary conditions are mentioned in Chapter 2. In order to obtain the electrical characteristics of the device, we have also to solve the relevant semiconductor equations, i.e., Poisson, electron current continuity, and thermal conduction equations.

5.3 2DEG response to interfacial piezoelectric polarization charge

We follow with an analysis on how the 2DEG changes with Q_p induced by the intrinsic stress in the diamond layer. From the simulation point of view, Q_p can be determined completely by the stress, but to calculate 2DEG density the surface potential is needed. For A-HEMT, we do not care the negative surface charge of the AlGaIn layer (actually, no definite theory to describe it because of the surface states), but instead fixing the surface potential to make the 2DEG density similar to Hall measurement results.

As the device structure shows (Figure 5-1), the diamond layer is deposited on SiN but not directly on the surface of the AlGaIn layer, so both A- and B-HEMT should have same contact properties at the AlGaIn/SiN interface from the chemical point of view. However, the surface Q_p of AlGaIn should be different because of the additional Q_p induced by intrinsic stress in diamond layer for B-HEMTs (spontaneous polarization charge is not influenced by stress, so it was kept unchanged). Minding this difference, we still assume the surface potential of a B-HEMT is the same as that of a A-HEMT. Generally speaking, this assumption should be somewhat reasonable due to the very high density surface states always existing on the surface of AlGaIn, leading to Fermi level pinning. This was used to explain the 2DEG density dependence on AlGaIn thickness [1]. With this assumption in mind, the electron conduction band is schematically shown in Figure 5-4. The three curves represent three cases: the interface has unchanged, increased, and decreased Q_p . Using the Fermi-level (black) as reference energy level, we can see that the 2DEG would increase (decrease) with increased (decreased) interfacial Q_p . The magnitude of surface charge density of AlGaIn can be expressed as $|\sigma_{s(AlGaIn)}| = |\epsilon_{AlGaIn} dV/dx|$, where V is the conduction band potential and ϵ_{AlGaIn} is the AlGaIn permittivity. By analyzing the slope of V on the surface of AlGaIn, we can see that increased

(decreased) $|\sigma_{s(AlGaN)}|$ is induced for increased (decreased) interfacial Q_p , in the same way as the change of 2DEG density was induced. For B-HEMT, Q_p also exists in the bulk of AlGaN and GaN due to the non-uniform polarization there. Nonetheless, it can be neglected compared with the large Q_p at the interface due to the discontinuity of piezoelectric polarization coefficient. Here we denote the magnitude of change of the interfacial Q_p , AlGaN surface charge and 2DEG charge as $\Delta|\sigma_I|$, $\Delta|\sigma_{s(AlGaN)}|$, and $\Delta|\sigma_{2DEG}|$, respectively. So, according to the principle of charge conservation, $\Delta|\sigma_I| = \Delta|\sigma_{s(AlGaN)}| + \Delta|\sigma_{2DEG}|$. From the numerical calculations, $\Delta|\sigma_I|$ can induce 10% $\Delta|\sigma_{s(AlGaN)}|$ and 90% $\Delta|\sigma_{2DEG}|$, which means that change of the interfacial Q_p mainly causes the 2DEG other than the surface charge to compensate it. This presents the potential to improve the device performance by stress engineering.

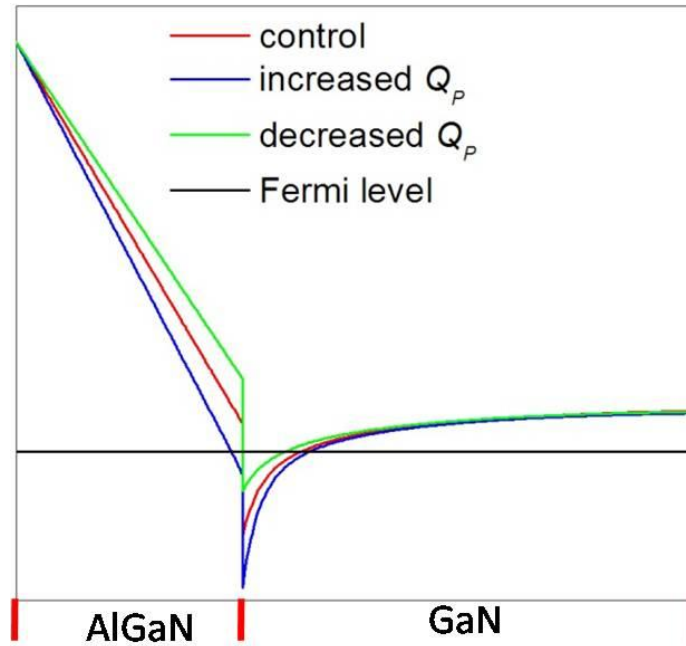


Figure 5-4: Schematic of electron conduction band for the AlGaN/GaN interface with unchanged (control), increased, and decreased piezoelectric polarization charge (Q_p). Fermi level is included as reference energy level. Surface potential is same for the three cases, meaning Fermi level pinning.

5.4 Experimental validation of simulated stress

Raman spectroscopy was used to determine the stress induced by the diamond capping layer. The Raman spectra were obtained at a distance of about 4 μm from the NCD edge near the drain side of the gate. The phonon frequency shift of the GaN layer in the device structure as a function of temperature is shown in Figure 5-5(a). From the phonon frequency difference, the additional stress induced by the diamond layer (assumed to be biaxial) was determined using a $-1.25 \text{ (GPa}\cdot\text{cm)}^{-1}$ deformation potential. The result is shown in Figure 5-5(b).

To validate the model, thermo-mechanical simulations were performed to obtain the stress as a function of temperature. In the simulations, the heat dissipation in the channel was calculated as $\mathbf{j}_n \cdot \mathbf{E}$, where $\mathbf{j}_n$ is the current

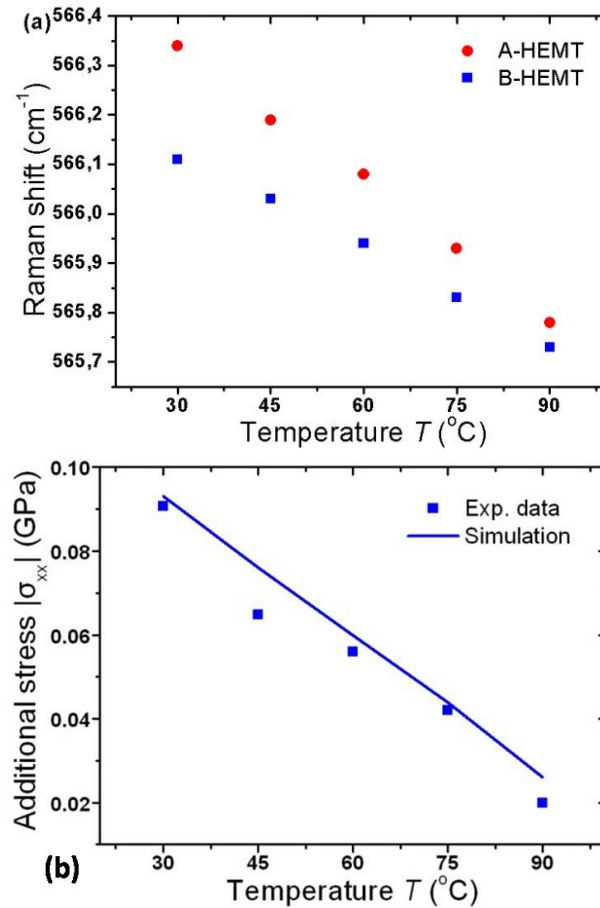


Figure 5-5: Comparison of Raman shift for A- and B-HEMT (a), and additional stress induced by the diamond cap layer determined by the Raman shift and thermo-mechanical simulations as a function of temperature (b).

density, and $\mathbf{E}$ is the electric field. Then, the temperature and thermal stress can be determined by this heat dissipation with the thermal conductivity and mechanical parameters. Figure 5-5(b) compares the measured additional stress in the GaN layer with simulated additional stress at the drain edge of the gate. The measured additional stress had indicated tensile stress, whereas the simulated one was compressive ($\sigma_{xx} < 0$ for $x < 1.0 \mu\text{m}$ and $x > 2.0 \mu\text{m}$ in Figure 5-7(a)). The reason for the discrepancy lies in the exposure of the SiN nucleation layer to the high temperature of NCD growth, thus changing its stoichiometry and increasing its contribution to the tensile profile in the access region. Future experiments will be performed to improve dielectric thermal stability.

Interestingly, the temperature dependence of both simulated and measured additional stress was consistent (Figure 5-5(b)). The temperature dependence of the additional stress was attributed to the difference of the thermal stress caused by the different thickness and thermal expansion coefficient of the SiN (A-HEMT) and diamond (B-HEMT) cap layer.

Figure 5-6 shows 2D distribution of stress σ_{xx} in A- and B-HEMT. We can see that the discontinuity of the diamond layer at the gate edge leads to non-uniform and relative large σ_{xx} for the B-HEMT.

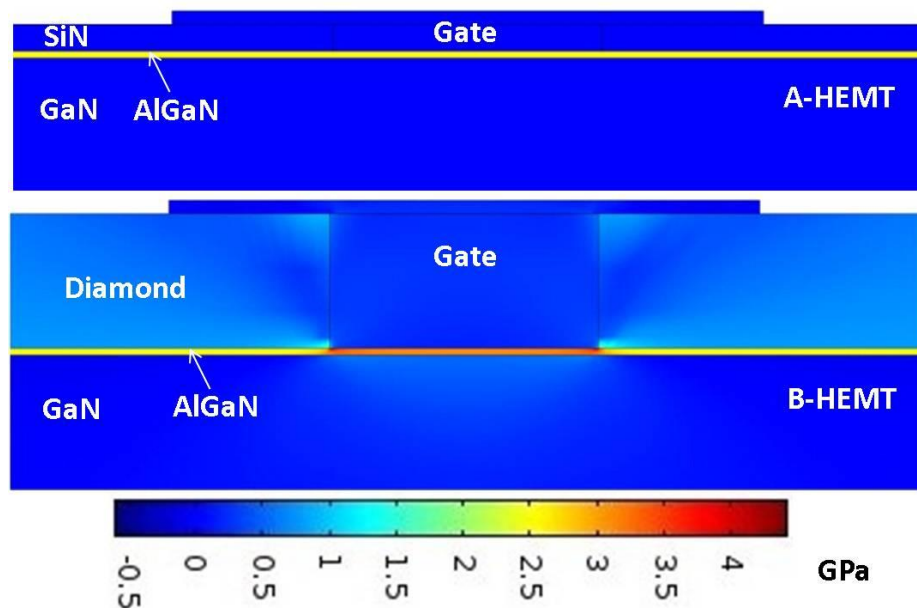


Figure 5-6: 2D distribution of stress σ_{xx} in the A- and B-HEMT

Figure 5-7 exhibits the 1D distribution of the additional stress σ_{xx} (a), the piezoelectric polarization P_{pz} (b), and the 2DEG density n_s (c) along the channel for varied thickness of diamond layer. As the graphs show, the A-HEMT has no stress and P_{pz} because of the assumption of zero strain in the GaN buffer (black line in (a) and (b)). As a result, the 2DEG density was uniformly distributed (black line in (c)). For the B-HEMT, the intrinsic tensile stress in the diamond layer induced additional non-uniform tensile stress, and thus the increase in 2DEG density under the gate up to the gate edge. For the source/drain region, additional compressive stress and thus decreased 2DEG density are induced. The simulation results show that integration of the 2DEG between the source and drain contacts ($1.41 \times 10^{10} \text{ cm}^{-1}$) is the same for both the A- and B-HEMT. This means that whereas the 2DEG was redistributed across the region between source and drain contact (SDC) induced by the stress in the diamond layer, the total concentration remains unchanged, since the mechanical coupling between the diamond layer and the heterostructure is internal. Note that this is different from the case where a change in the 2DEG density is induced by external bending on the whole wafer [15], in which the 2DEG would change in the same way across the whole region between SDC, and its integration from source to drain contacts would be changed with the external stress.

The asymmetrical distribution of 2DEG under the gate was attributed to the different geometrical spacings between the gate and the source/drain contacts (GSDC). A large change of the 2DEG happens in the area near the gate edge, compared with a relatively small change for the S/D regions. This can be explained that in addition to the large normal stress, a noticeable shear stress is also induced at the gate edge due to the discontinuity of the diamond layer. By reducing the gate length, the two gate edges become closer, leading to higher additional 2DEG under the gate. Figure 5-7(d) shows that the average additional stress and the average 2DEG density in the channel under the gate increase as the diamond thickness increases. This effect goes away for diamond layers thicker

than 2 μm . This case has also been illustrated experimentally for the device with SiN passivation [123].

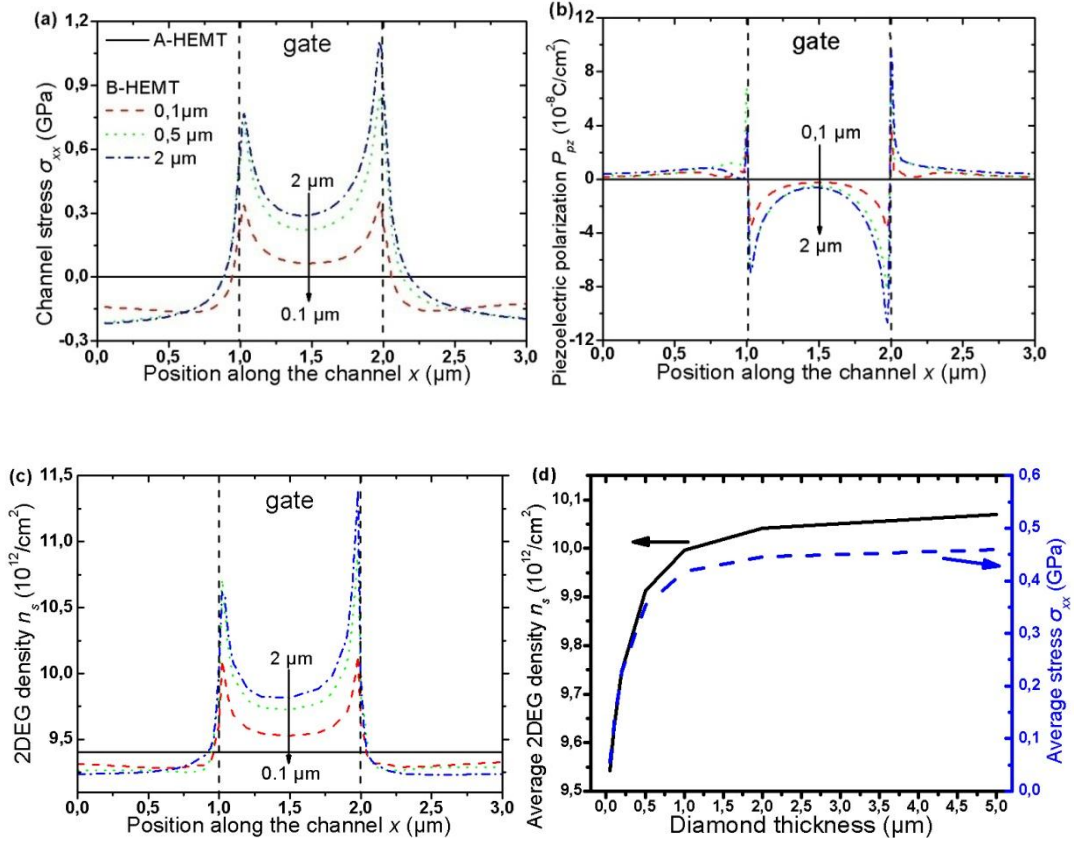


Figure 5-7: Distribution of non-uniform additional stress σ_{xx} (a), piezoelectric polarization P_{pz} (b), 2DEG density n_s (c) along the channel, and average 2DEG density (left axis) and average additional stress (right axis) in the channel under the gate (d) for varied thickness of diamond layer. σ_{xx} and P_{pz} were extracted 1 nm below the AlGaIn/GaN interface.

5.5 Influence of additional stress on the electrical behavior

Figure 5-8 compares electrical measurements and simulations of the drain current I_{DS} as a function of the drain voltage V_{DS} (a) and gate voltage V_{GS} (b), for both the A- and B-HEMT. In the case of the B-HEMT, the simulations include the coupling with the intrinsic stress in the diamond layer. Figure 5-8(a) shows that the simulation results were in very good agreement with the I_{DS} - V_{DS}

experimental data for the operating conditions considered. Figure 5-8(b) shows that the simulated I_{DS} of B-HEMT was higher than that of A-HEMT, which is contrary to the experimental result at around $V_{GS}=1$ V ($V_{DS}=0.1$ V). The reason is that we did not consider gate leakage current in the simulation. In addition, the inset graph shows that the threshold voltage was negatively shifted due to the increased 2DEG density under the gate.

The inset of Figure 5-8(b) shows how the average 2DEG density under the gate depends on V_{GS} . Compared to a A-HEMT, the increment in n_s in a B-HEMT

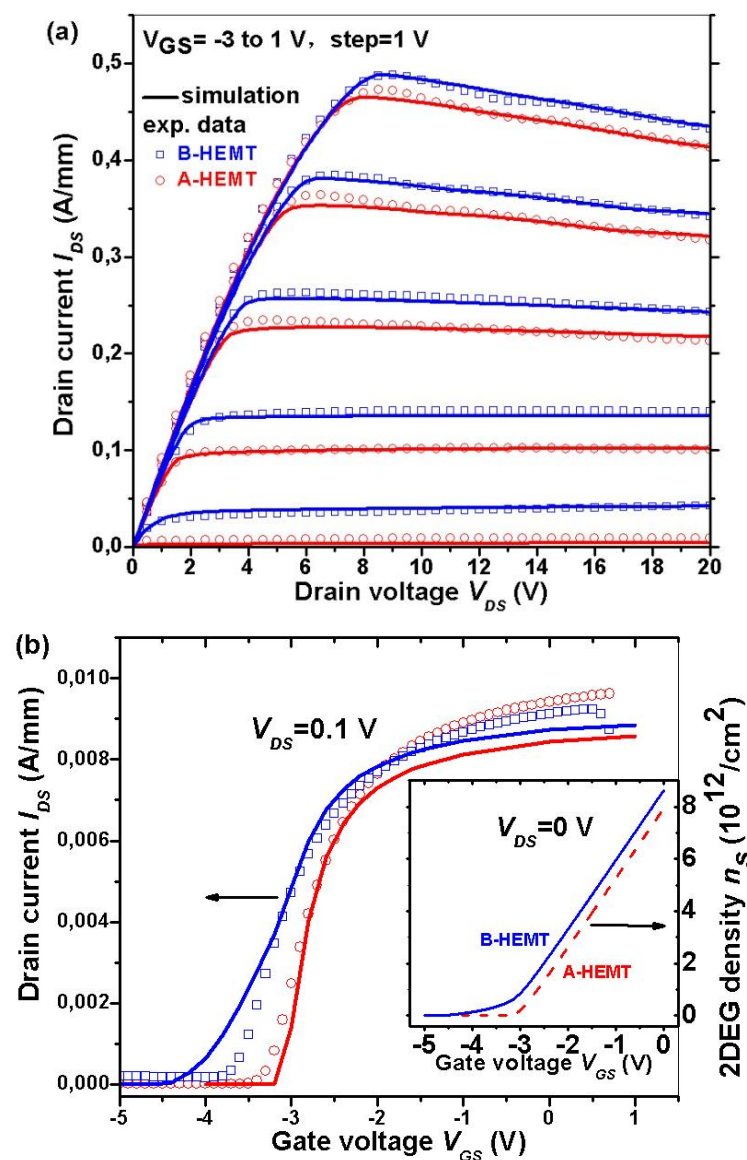


Figure 5-8: Comparison of experimental and simulation results for I_{DS} as a function of V_{DS} (a) and V_{GS} (b). Inset of (b) shows the average 2DEG density n_s under the gate as a function of V_{GS} for A- and B-HEMT.

does not change with V_{GS} for $V_{GS} > V_{th}$ as shown in the graph such that n_s has a relatively large increase at low V_{GS} , which further yields to a relative large I_{DS} increase at low V_{GS} as shown in Figure 5-8(a). The reason is that change of the 2DEG only depends on the additional polarization charge induced by the diamond layer, and at the same time V_{GS} has no influence on the polarization charge (as far as we are not considering the inverse piezoelectric polarization effect). This phenomenon has been equivalently illustrated in pressure sensors based in AlGaIn/GaN HEMTs, as I_{DS} is more sensitive to pressure at lower gate voltage [121][122].

In accordance with our experimental work, we only discussed the intrinsic stress from the diamond layer impact on the device electrical behavior. However, this model should be suitable to generalize the effect of any capping layer on the stress profile in the device, such as a SiN passivation layer traditionally used to mitigate current collapse effects during RF operation [128][129]. The intrinsic stress in such passivation layers has been shown to be as high as 3 GPa [130], which could present significant reliability concerns in a scaled device.

In order to analyze how intrinsic stress in the capping layers affects device scaling, we extended our simulations with a 3GPa intrinsic stress using the verified model for a long-channel HEMT. Figure 5-9(a) shows the maximum σ_{xx} in the channel as a function of the gate length L_G and the distance between the gate and drain contact L_{GD} . As the graph shows, σ_{xx} increases as L_{GD} increases or L_G decreases. As severe reliability effects due to structural stress near the gate have been reported [17] it has become clear that a multilayer device capping strategy could be engineered to minimize the total stress in the heterostructure. The stress contribution of a novel material such as NCD can be calibrated on a HEMT structure of known intrinsic stress. Consequently, stress induced by the NCD cap could be mitigated when a cap layer stack such as NCD/SiN is employed.

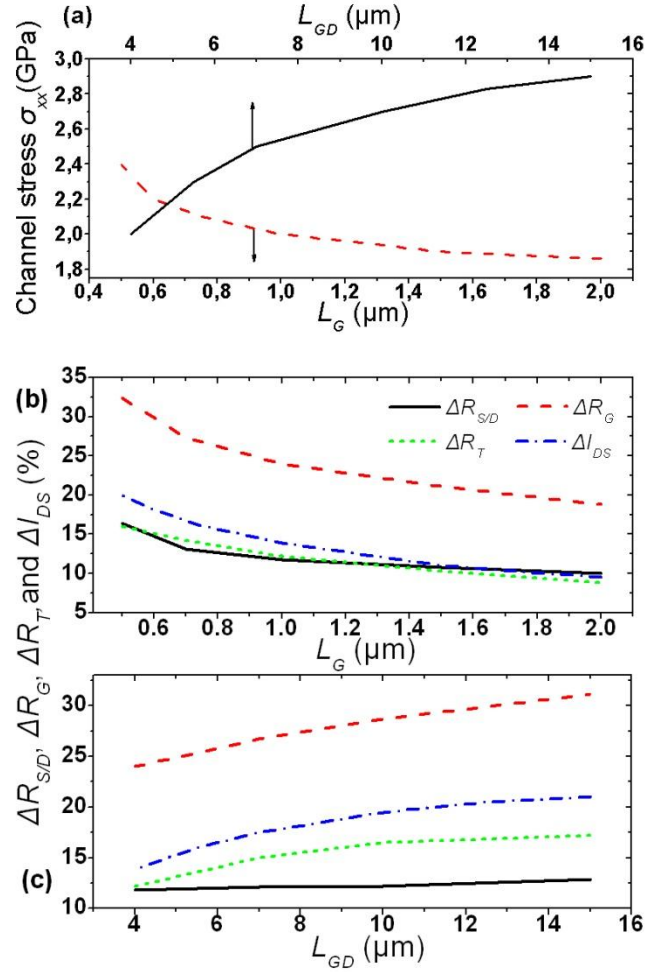


Figure 5-9: (a) shows simulation results for the maximum additional stress σ_{xx} in the channel (1 nm below the AlGaIn/GaN interface) induced by the cap layer with intrinsic stress $\sigma_{xx} = \sigma_{yy} = 3$ GPa as a function of the gate length L_G and the distance between the gate and drain contact L_{GD} . (b) and (c) show GSDC resistance increase $\Delta R_{S/D}$, gate resistance decrease ΔR_G , total resistance decrease ΔR_T , and drain current increase ΔI_{DS} compared to A-HEMT as a function of L_G and L_{GD} , respectively. In all the graphs, $L_{GD}=L_{GS}=4$ μm when L_G is varied, and $L_{GS}=4$ μm and $L_G=1$ μm when L_{GD} is varied.

To analyze the effect of sheet charge redistribution in the channel under the additional stress of the cap layer, we can consider I_{DS} as a function of the total resistance R_T between SDC. R_T would be divided into two constituent

resistances: $R_T = R_{S/D} + R_G$, where $R_{S/D}$ and R_G are the resistances between GSDC and under the gate, respectively. R_G is modulated by V_{GS} , whereas $R_{S/D}$ is independent on the operating conditions. As we discussed before, for B-HEMT the lateral redistribution of sheet charge between the SDC would lead to the higher $R_{S/D}$ (lower 2DEG density) and lower R_G (higher 2DEG density) compared to A-HEMT. Here, the GSDC resistance $\Delta R_{S/D}$, gate resistance ΔR_G , total resistance ΔR_T , and drain current ΔI_{DS} as a function of L_G and L_{GD} of the B-HEMT, compared to the A-HEMT, are shown in Figure 5-9(b) and Figure 5-9(c), respectively. Due to gate modulation, ΔR_G dominates $\Delta R_{S/D}$, which leads to an overall lower R_T , and thus a higher I_{DS} . We also note that I_{DS} has significantly increased by 20% and 21% for $L_G=0.5 \text{ }\mu\text{m}$ and $L_{GD}=15 \text{ }\mu\text{m}$, respectively.

Due to the lack of systematic studies on the dependence of surface potential on additional stress applied on the device, we assumed Fermi-level pinning on the surface of AlGaN and obtained reasonable agreement of simulation and experimental results. We attributed the intrinsic tensile stress in the diamond layer to the increase in I_{DS} . Based on this model, if the tensile stress is replaced with compressive stress, the decreased 2DEG density under the gate region (increased R_G), increased 2DEG density between GSDC (decreased $R_{S/D}$), and therefore the decreased saturation I_{DS} would be obtained. This shows a similar behavior to that observed experimentally, in which AlGaN/GaN HEMTs were capped with diamond-like carbon layer with the $\sim 6 \text{ GPa}$ high compressive stress [131].

In general, mechanical failure such as cracks/pits always happen at the gate edge of the drain side, where highly localized heat dissipation, high thermal stress, and high inverse piezoelectric stress are present, and these aspects have been proposed to explain the mechanical degradation [132][133]. Our work

shows that the intrinsic stress in the cap layer gives also highly localized additional stress at the gate edge (Figure 5-7(a)), which therefore presents a probability to aggravate or mitigate the mechanical degradation. This relation has been illustrated by the simulation, showing that the compressive/tensile stress in the passivation layer can decrease/increase the total stress at the gate edge [79]. Combined with our conclusions, this means that tensile stress in the cap layer would improve the device electrical behavior, but aggravate the mechanical degradation; conversely, compressive stress in the cap layer would mitigate the mechanical degradation, but reduce the electrical performance.

5.6 Summary

We presented an electro-thermo-mechanical coupled model to analyze how the intrinsic stress in the capped layer affects the electrical behavior of AlGaIn/GaN HEMTs. The simulation results were verified by the experimental data from Raman spectroscopy and I-V characterization for the diamond capped device. The stress in the capped layer induces non-uniform 2DEG distribution in the channel. A change of the AlGaIn/GaN interfacial Q_p mainly causes the 2DEG other than the AlGaIn surface charge to compensate it, which may potentially improve the device performance by stress engineering. Due to the discontinuity of the capped layer at the gate edge, unlike the whole wafer bending by external stress, intrinsic stress in the capped layer would induce the 2DEG under the gate showing opposite variation to that in other regions between SDC. Since mechanical coupling between the device and capped layers is internal, the 2DEG would be redistributed by the additional stress from the cap layer while remaining its total concentration between SDC unchanged. For the device operating in saturation condition, the change of R_G is the dominating factor of change of R_T , so the decreased R_G induced by tensile stress in the capped layer leads to decreased R_T and therefore increased drain current.

Extended simulations with this validated model showed that a significant improvement of the device electrical behavior would be achieved.

Most of contributions in this chapter 5 were published as an article, see [134].

6 Surface, buffer, and barrier traps effect

I_{DS} collapse in GaN HEMTs under pulsed bias or DC-RF dispersion is still a limitation for commercial applications. Generally, this deleterious phenomenon is attributed to the presence of deep-level traps, which in principle can be located mainly on any of the following: on the AlGaN surface, in the GaN buffer, or within the AlGaN barrier [135]. In this chapter, we will separately simulate the effects of the traps in these three positions. The mechanisms of trapping/detrapping and the correspondent simulation method will be described. Especially, the mechanisms of electric field and temperature-dependent detrapping for the barrier traps and relevant reverse gate current will be comprehensively illustrated.

6.1 Surface traps effect

Intensive studies including transient/pulse measurements and simulations have been devoted to study the AlGaN surface traps effects on I_{DS} collapse or DC-RF dispersion [26] [136] [137]. The basic principle of these traps effects is that the filled traps on the AlGaN surface near the GEDS can reduce the 2DEG density in the correspondent position of the drain-side access region and therefore yields the I_{DS} collapse. The surface area with filled traps is usually called virtual gate. In this section, we will study how the device geometry influences the I_{DS} collapse when the device under pulsed bias and explain that by simulations on the surface traps effects.

The studied device has the following structure: 1 nm GaN cap/22 nm $\text{Al}_{0.29}\text{Ga}_{0.71}\text{N}$ barrier/GaN buffer/SiC substrate (see the schematic of the device structure shown in Figure 6-1). Figure 6-2 shows pulsed measurement results, in which the gate lag ratio (GLR) was extracted as a function of the pulse width

τ_{on} variation from 1 μ s to 10 ms. The duty cycle of the pulse is 1%. GLR was calculated as $I_{DS,Pulse}/I_{DS,DC}$, where $I_{DS,Pulse}$ and $I_{DS,DC}$ are the drain current under pulsed and DC bias, respectively. Different device geometries, i.e., L_G variation from 3 μ m to 6 μ m with $L_{GD}=15$ μ m (Figure 6-2(a)) and L_{GD} variation from 10 μ m to 20 μ m with $L_G=3$ μ m (Figure 6-2(b)), were studied. As shown in the graphs, the smaller τ_{on} the lower GLR (larger I_{DS} collapse), which can be

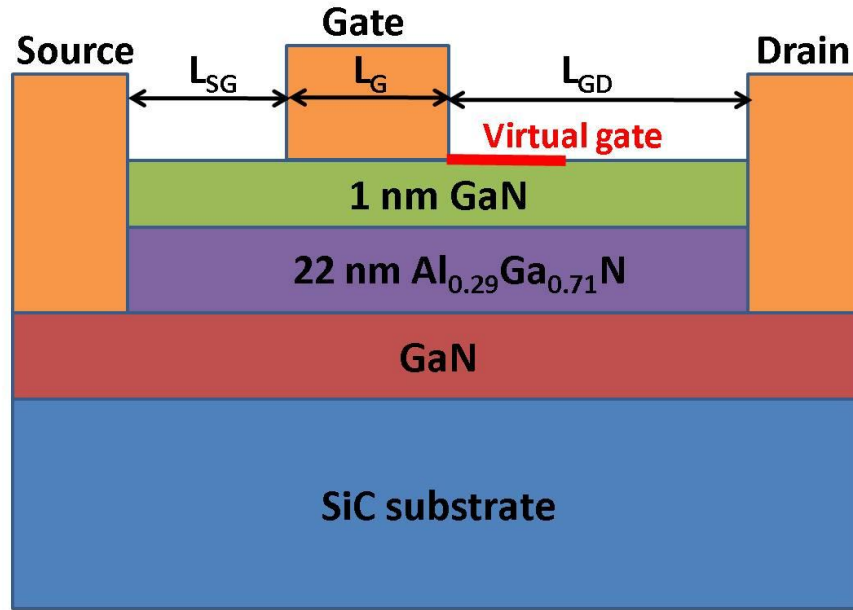


Figure 6-1: Schematic of the studied device structure and the virtual gate.

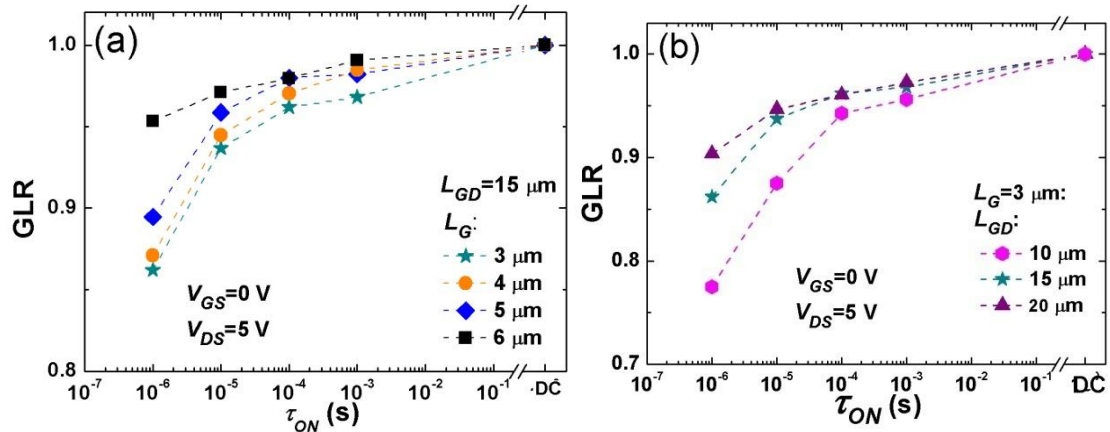


Figure 6-2: Experimental data of the gate lag ratio (GLR) as a function of L_G variation from 3 μ m to 6 μ m with fixed $L_{GD}=15$ μ m (a), and L_{GD} variation from 10 μ m to 20 μ m with fixed $L_G=3$ μ m. τ_{on} represents the pulse width. These experimental results were taken from S. Martin-Horcajo's work (to be published).

explained as less amount of traps were detrapped for narrower pulse width. The interesting results were that GLR was influenced by the device geometry as shown in Figure 6-2. In detail, GLR increased when increasing L_G (Figure 6-2(a)) and L_{GD} (Figure 6-2(b)).

In order to explain the experimental results, we performed the following simulations. We did not consider the detailed volume charge distribution since the GaN cap was very thin (1 nm), but considered it as surface charge on the AlGaIn surface. In consistent with the studies in [1], ionized donor-like traps (positive charge) were assumed on the AlGaIn surface which can neutralize partial negative polarization charge. Note that the total surface charge density is a summation of positive charge of the ionized donor-like traps and the negative charge of the polarization. In the simulations, we set fixed polarization charge of $10 \times 10^{12} \text{ q C/cm}^2$ at the AlGaIn/GaN interface and total charge of $-1 \times 10^{12} \text{ q C/cm}^2$ on the AlGaIn surface to produce reasonable 2DEG density of $9 \times 10^{12} / \text{cm}^2$ near the interface. When the device was biased in OFF-state, the donor-like traps near the drain-side gate edge were filled by the electrons tunneling from the gate metal. These filled traps were not charged and therefore lead to higher density of the total negative surface charge and lower 2DEG density, usually termed “virtual gate” phenomenon.

Figure 6-3 shows the distribution of the simulated electric field magnitude $|E|$ on the AlGaIn surface as a function of L_G variation from 3 μm to 6 μm with fixed $L_{GD}=15 \text{ }\mu\text{m}$ (a) and L_{GD} variation from 10 μm to 20 μm with fixed $L_G=3 \text{ }\mu\text{m}$ (b). The maximum $|E|$ indicates the position of the drain-side gate edge. The devices were biased in off-state with $V_{GS}=-6 \text{ V}$ ($<V_{TH}$) and $V_{DS}=5 \text{ V}$. In the simulations, a beveled gate edge was used in order to avoid singularity of the electric field there, which was expected to be similar to the practical shape [138]. As the graphs show, no matter varying L_G or L_{GD} , the maximum $|E|$ and its distribution keep almost unchanged. As we mentioned before, the surface traps were filled by the electrons tunneling from the gate metal. This tunneling

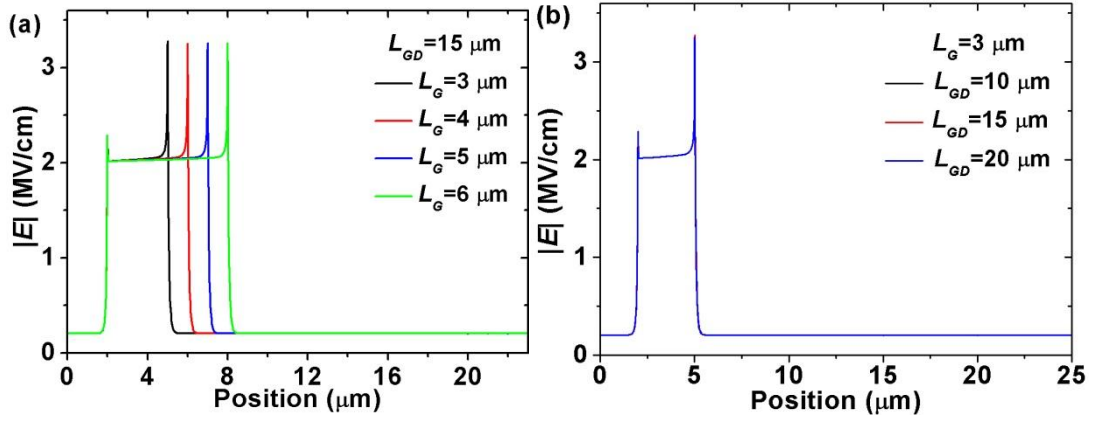


Figure 6-3: Electric field magnitude $|E|$ distribution on the AlGaIn surface as a function of L_G variation from 3 μm to 6 μm ($L_{GD}=15 \mu\text{m}$) (a) and L_{GD} variation from 10 μm to 20 μm ($L_G=3 \mu\text{m}$). Note that the curves in (b) are overlapping. The maximum value of $|E|$ indicates the position of the GEDS. The device was in off-state with the biases $V_{GS}=-6 \text{ V}$ ($<V_{TH}$) and $V_{DS}=5 \text{ V}$.

current might be determined by $|E|$. It was unknown yet exactly about the electron transport mechanisms on the barrier surface, therefore we will not address this and also the transient process of trapping and detrapping in the simulations. However, due to the unchanged $|E|$, we would assume that the devices with different geometries (different L_G or L_{GD}) have same virtual gate length with same probability of traps occupancy. (see the indicated virtual gate shown in Figure 6-1). Specifically, in the simulations, we set the virtual gate length of 0.4 μm with the total negative charge density of $-9 \times 10^{12} \text{ q C/cm}^2$.

Figure 6-4 shows the simulation results for the electron density of the device with and without virtual gate (the device was in on-state with $V_{GS}=0 \text{ V}$ and $V_{DS}=5 \text{ V}$). As expected, the electron density under the virtual gate was reduced significantly.

In order to explain the simulation results, we will do the following mathematic analysis. For the device without virtual gate, $I_{DS,DC}$ can be calculated as $I_{DS,DC}=V_{DS}/R_T$, where R_T is the total resistance between the source and drain contact. For the device with virtual gate, $I_{DS,pulse}$ can be calculated as $I_{DS,pulse}=V_{DS}/(R_T+\Delta R_{VG})$, where ΔR_{VG} is the increased resistance caused by the virtual gate. So GLR can be expressed as

$GLR = I_{DS,Pulse}/I_{DS,DC} = 1/[1+(\Delta R_{VG}/R_T)]$. We note that R_T will increase with increasing V_{DS} when the device is biased in the non-linear or saturation region due to the pinch-off behavior, while ΔR_{VG} will not change with V_{DS} and also the device geometry because we set the fixed virtual gate length and the fixed charge density.

Figure 6-5 shows the simulated I_{DS} for the device with and without virtual gate. Note that we did not consider the pulse width that can determine the probability of traps occupancy. Therefore, $I_{DS,Pulse}$ and $I_{DS,DC}$ were simulated for the device with (traps completely filled) and without (traps completely emptied) the virtual gate, respectively. As the graph shown, I_{DS} was decreased ($GLR < 1$), especially in the knee region of the I-V curves, because ΔR_{VG} was comparable to R_T (i.e., $\Delta R_{VG}/R_T$ was not much less than one). When $V_{DS} > 20$ V in the saturation region, R_T was increased (it mainly concentrated at the GEDS) so that $\Delta R_{VG}/R_T$ was much less than one and therefore $GLR \approx 1$ (see the GLR expression mentioned before), i.e., there is no significant I_{DS} collapse as shown in the graph.

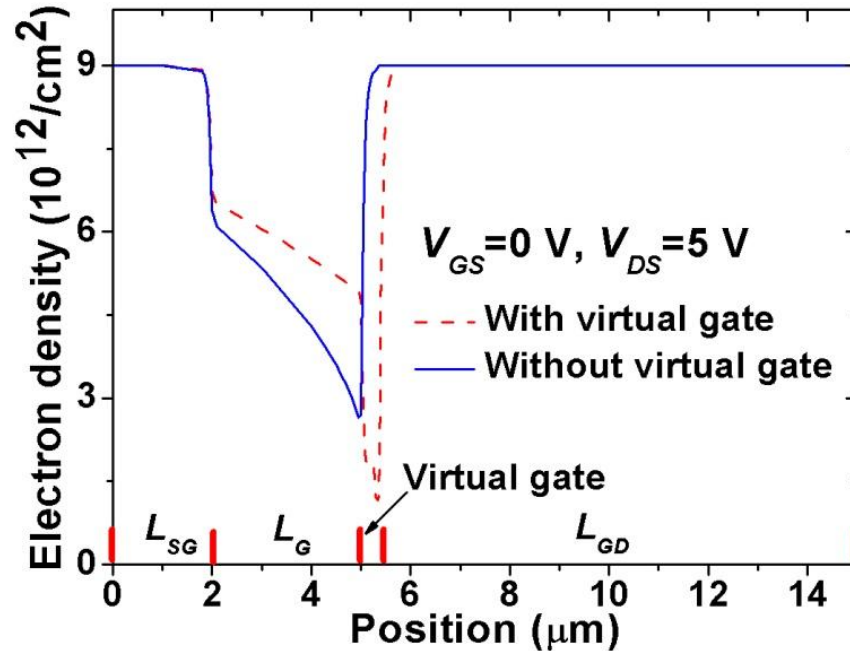


Figure 6-4: Comparison of electron density in the channel for the device with and without virtual gate. $L_{SG} = 2.2 \mu m$, $L_G = 3 \mu m$, $L_{GD} = 10 \mu m$, and the virtual gate length is $0.4 \mu m$.

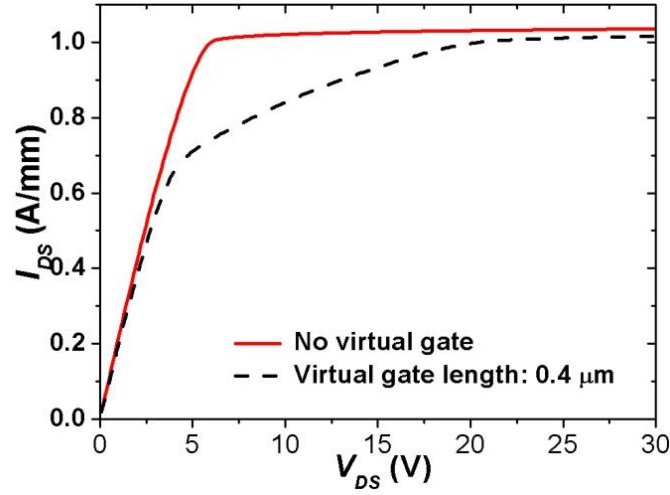


Figure 6-5: Simulation of I_{DS} collapse caused by the virtual gate. The virtual gate length is $0.4 \mu\text{m}$.

Figure 6-6 shows the simulated GLR as a function of L_G (a) and L_{GD} (b). As the graphs show, GLR increased with increasing L_G or L_{GD} , showing a similar tendency compared to the experimental results displayed in Figure 6-2. The explanations are presented in the following. We note that the devices were biased in the knee region of I_{DS} ($V_{GS}=0 \text{ V}$ and $V_{DS}=5 \text{ V}$, same as the experimental biases), in which the access resistances between the gate and source/drain contact influenced by L_{SG} and L_{GD} are comparable to the channel resistance under the gate influenced by L_G . In other words, R_T does not mainly concentrate at the GEDS. Therefore, devices with larger L_G or L_{GD} had larger R_T , smaller $\Delta R_{VG}/R_T$ (ΔR_{VG} was kept unchanged as mentioned before), and thus larger GLR (lower I_{DS} collapse).

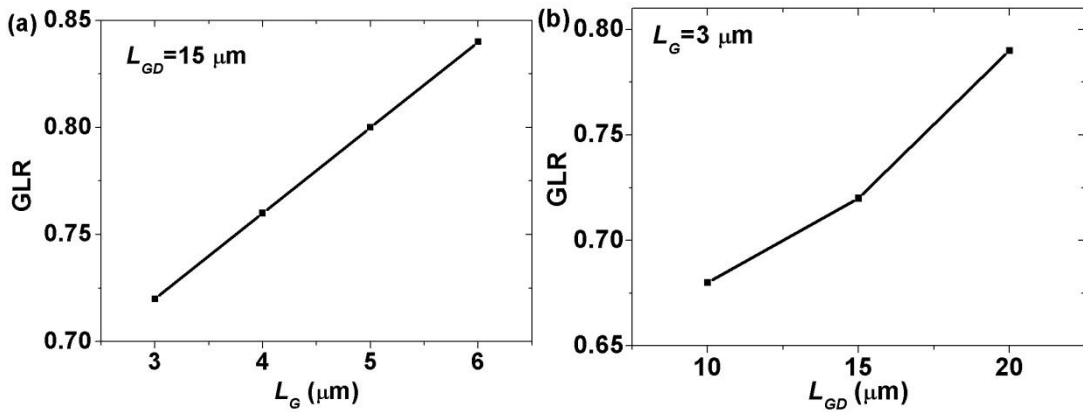


Figure 6-6: Simulated GLR as a function of L_G (a) and L_{GD} (b).

6.2 Buffer traps effect

In this section, we will simulate the effect induced by the traps in the GaN buffer and analyze the self-heating impact on the transient I_{DS} . The buffer traps can be deep-level acceptors, such as C or Fe doping [139]. The transient electron capture and emission process have to be included in the transient I_{DS} simulation. Besides the relevant semiconductor equations (Eqs. 2.1, 2.2, and 2.7), the following rate equation (here consider acceptor-like traps and only electrons exchange between the traps and the electron conduction band) has to be solved in a full coupling manner,

$$\frac{\partial N_A^-}{\partial t} = C_n (N_A - N_A^-) n - e_n N_A^-, \quad (6.1)$$

$$e_n = V_T \sigma_n N_C \exp\left(-\frac{E_T}{kT}\right), \quad (6.2)$$

$$C_n = V_T \sigma_n, \quad (6.3)$$

where t is time, N_A is the acceptor-like traps density, N_A^- is the ionized acceptor-like traps density, n is the electron concentration, C_n is the capture rate, e_n is the electron emission rate, V_T is the electron thermal velocity, σ_n is the capture cross-section, T is temperature that equals to the device channel temperature $T_{channel}$, and E_T is the ionization energy. If the device is in steady state, the term $\frac{\partial N_A^-}{\partial t} = 0$ in Eq. 6.1, therefore

$$N_A^- = \frac{C_n N_A n}{e_n + C_n n}. \quad (6.4)$$

The DC current degradation for the device under long time electrical or thermal stress might be attributed to the generation of traps in the device. Figure 6-7 shows an example of such type of simulations comparing I_{DS} for the device with or without buffer traps. We can see that I_{DS} has a significant collapse due to the reduced 2DEG in the channel. Higher ambient temperature increases e_n and thus decreases N_A^- , so that I_{DS} has less difference for the simulations with and without traps (see for example the results for ambient temperature 700K in Figure 6-7). As illustrated in the previous simulations, the surface traps mainly

led to I_{DS} collapse in the knee region of the I-V curves, while here the simulations illustrated that the buffer and barrier traps mainly led to I_{DS} collapse in the saturation region and threshold voltage shift.

The reported buffer traps simulations did not include the self-heating impact [31] [139]. In the following, we will analyze this impact on e_n and transient I_{DS} .

Figure 6-8 shows steady state simulations of the device self-heating impact on the characteristic time of electron emission ($\tau_n = 1/e_n$). The buffer traps density always depends on the lattice mismatch between the epitaxial layers and substrate and the growth technology. Here a typical value $2 \times 10^{18} \text{ cm}^{-3}$ was used. T_{ch} and τ_n were simulated as a function of V_{DS} for two gate biases ($V_{GS}=1 \text{ V}$ and -2 V). As the graph shows, for relative high power bias, e.g., $V_{DS}=20 \text{ V}$ and $V_{GS}=1 \text{ V}$, τ_n was significantly decreased due to the elevated T_{ch} .

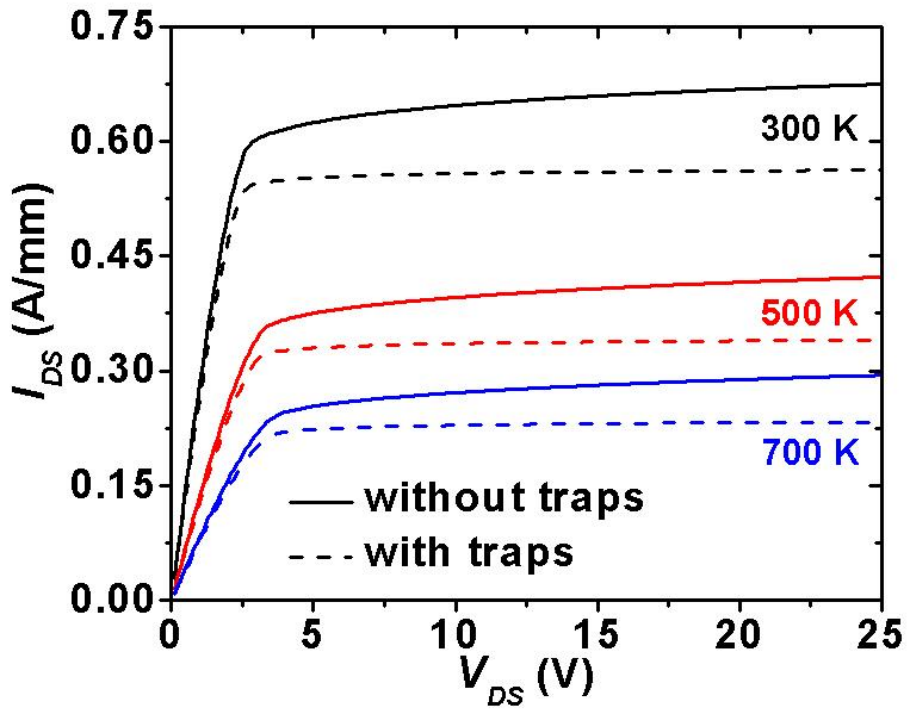


Figure 6-7: Simulation of DC I_{DS} collapse caused by the buffer traps under ambient temperature 300 K, 500 K, and 700 K. Self-heating was not included in the simulations.

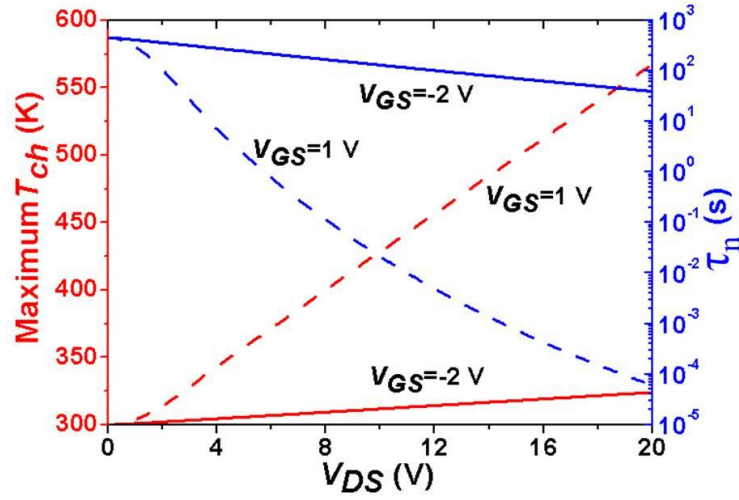


Figure 6-8: Steady state simulations of the self-heating impact on τ_n . The graph shows T_{ch} (left axis) and τ_n (right axis) as a function of V_{DS} for $V_{GS}=1$ V and -2 V.

Figure 6-9 shows the distribution of ionized acceptor-like traps in the GaN buffer. The device was biased in on-state ($V_{DS}=25$ V and $V_{GS}=1$ V) to illustrate the device self-heating impact on N_A^- (the traps density used in the simulation was $2 \times 10^{18}/\text{cm}^3$). Due to the relative high electric field under the GEDS, the electrons penetrate into the deep GaN buffer and therefore lead to high density of ionized traps there. Simulation including the self-heating shows less N_A^- due to the lower τ_n compared to that without the self-heating.

Figure 6-10 shows simulations on transient I_{DS} (a) and transient T_{ch} as well as τ_n (b) with and without the device self-heating. Results for two pulsed biases were compared: from quiescent voltage $V_{GS,Q}=-3$ V and $V_{DS,Q}=25$ V (trap filling) to $V_{GS}=0$ V, $V_{DS}=5$ V and 10 V (trap emptying). The faster electron emission when considering the self-heating manifests itself as a left shift of the transient I_{DS} (Figure 6-10(a)). Higher V_{DS} (higher power) leads to higher T_{ch} and therefore the faster electron emission. Note that the decrease of I_{DS} for the time less than 1 ms is due to the domination of self-heating effect in this time scale.

Figure 6-10(b) shows the maximum T_{ch} changed from 300 K (before the pulse) to 395 K (after the pulse tending to steady state, for $V_{DS}=10$ V). Correspondingly, τ_n had a significant variation (from 400 s to 0.2 s).

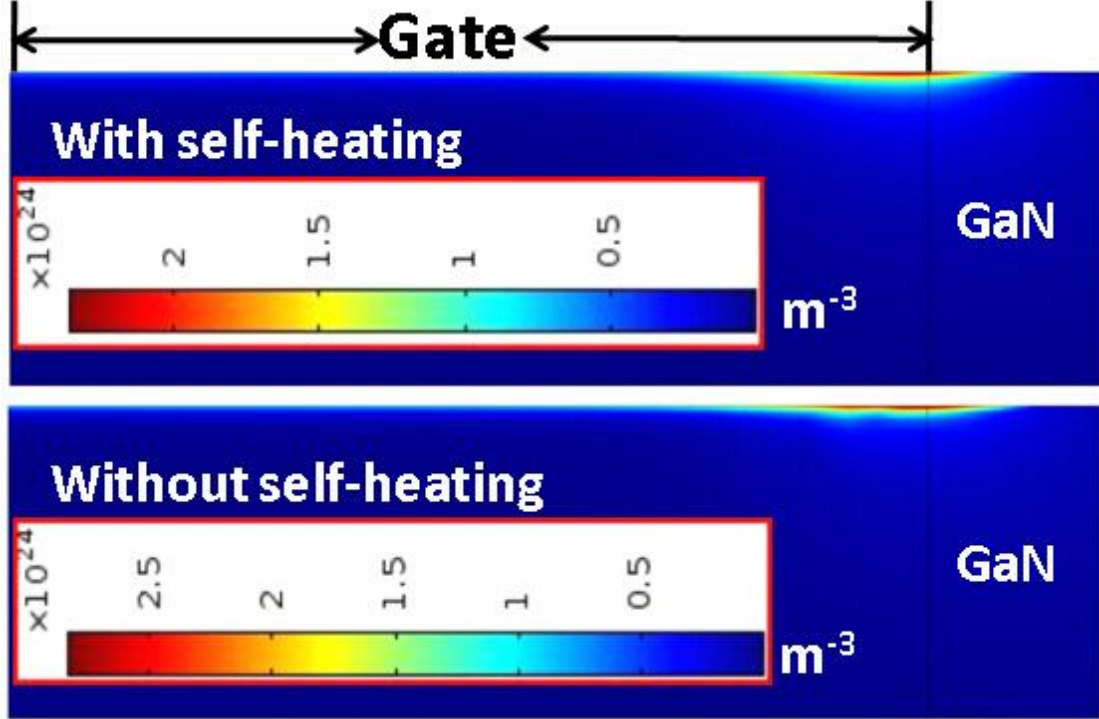


Figure 6-9: Comparison of ionized acceptor-like traps density in the GaN buffer (200 nm thick GaN under the AlGaN/GaN interface is showed) for the simulations with (top) and without self-heating (bottom). The device is in on-state with $V_{DS}=25$ V and $V_{GS}=1$ V.

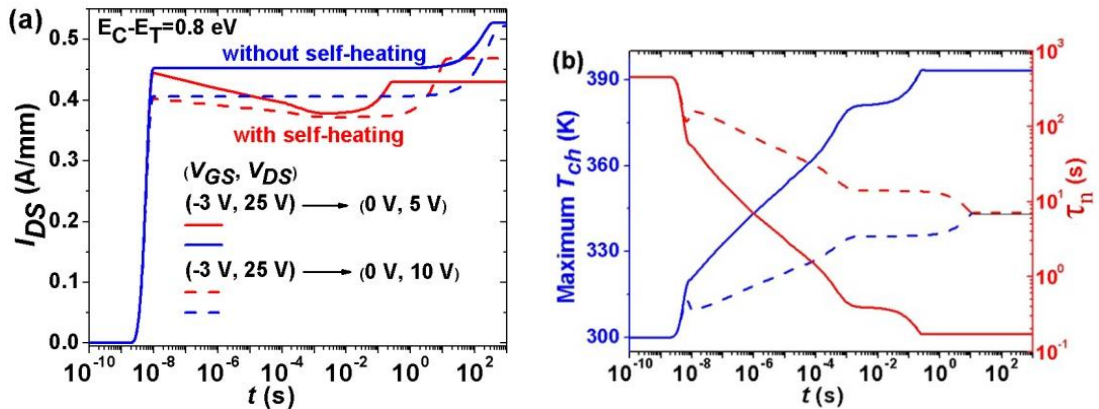


Figure 6-10: Comparison of transient I_{DS} for the simulations with and without the self-heating (a), and transient T_{ch} and τ_n (b). The biases were pulsed from quiescent voltage $V_{GS,Q}=-3$ V and $V_{DS,Q}=25$ V to two different biases: $V_{GS}=0$ V, $V_{DS}=5$ V and 10 V.

6.3 Barrier traps effect and reverse gate current mechanism

In the previous sections, we have dealt with the surface and buffer traps effects. In this one, we will illustrate the barrier traps effects and the mechanism of the relevant reverse gate current.

Mechanisms of electrons emission from the traps can be thermal activation and quantum tunneling with the presence of electric field. If the traps are charged when empty, the emission can be enhanced by Poole-Frenkel (PF) effect [140]. For the following barrier traps simulation, the electric field effect on enhancement of the electron emission will be included.

In addition causing current collapse, traps in the AlGa_N barrier were also found to play an important role in reverse gate current ($I_{G,Rev}$) via mechanisms of trap-assisted tunneling [141]-[143], one-dimensional variable-range hopping conduction [144], or direct tunneling [145]. The trap-assisted tunneling mechanism is determined by the balance in two competing processes: the trapping and detrapping, in which the slow detrapping in the barrier can influence the 2DEG density and thus be related to I_{DS} collapse. Simulation of $I_{G,Rev}$ based on this mechanism has been performed by only considering the direct tunneling for the detrapping determined by electric field [141]. In this work, to simulate $I_{G,Rev}$ and its temperature dependence, we completely considered their impact on the detrapping process, which is described by phonon-assisted tunneling (PAT) [146].

Intensive studies have been devoted to the traps characterization of AlGa_N/Ga_N HEMTs by electrical techniques, such as frequency-dependent capacitance and conductance [147] as well as pulse and drain current-transient measurement [148], in which the drain current-transient is a popular technique because of its simplicity and fewer restrictions on device scaling [149]. The basic principle of this technique is that the trapping/detrapping can be

manifested as a gradual change of transient I_{DS} when the device under pulsed bias. Numerical methodologies, such as polynomial fitting, multi-exponential fitting, or stretched exponential fitting [27], are needed to fit the transient I_{DS} so as to extract τ_n , and further extract the traps ionization energy (E_T) and the capture cross-section from the Arrhenius plot. For this technique, in addition to the numerical fitting methodologies, the influence of the self-heating and electric field on τ_n demands more detailed treatments, which have been investigated separately in the literature [150]-[152]. However, there is a lack of investigations which consider the impact of both, as this would be difficult to achieve experimentally. Our simulations address this issue.

Simulations of the transient I_{DS} have been performed to study the current collapse phenomena, in which only the emission mechanism by pure thermal activation was addressed [31][139][153], without considering the electric field impact. It is well known that AlGaIn/GaN HEMT suffers a strong self-heating, especially for the device operation in $I_{DS,sat}$ region, leading to elevated channel temperature [46]. High electric field exists in the barrier due to the presence of large polarization charge at the AlGaIn/GaN interface and on the AlGaIn surface even the device is not biased [154]. So, it is necessary to include both of these two factors in the transient I_{DS} simulation. In this work, based on the detrapping by PAT, mechanism of device bias via self-heating and electric field impact on τ_n and transient I_{DS} was simulated, and further insights were presented for the current-transient characterization technique.

6.3.1 Model of trapping/detrapping in the barrier

A threshold voltage shift under pulsed bias indicates the presence of traps in the barrier [27]. In the following, we will present a model of trapping/detrapping in the barrier (see Figure 6-11). Trapping is described by direct tunneling of electrons from the gate metal into the AlGaIn barrier.

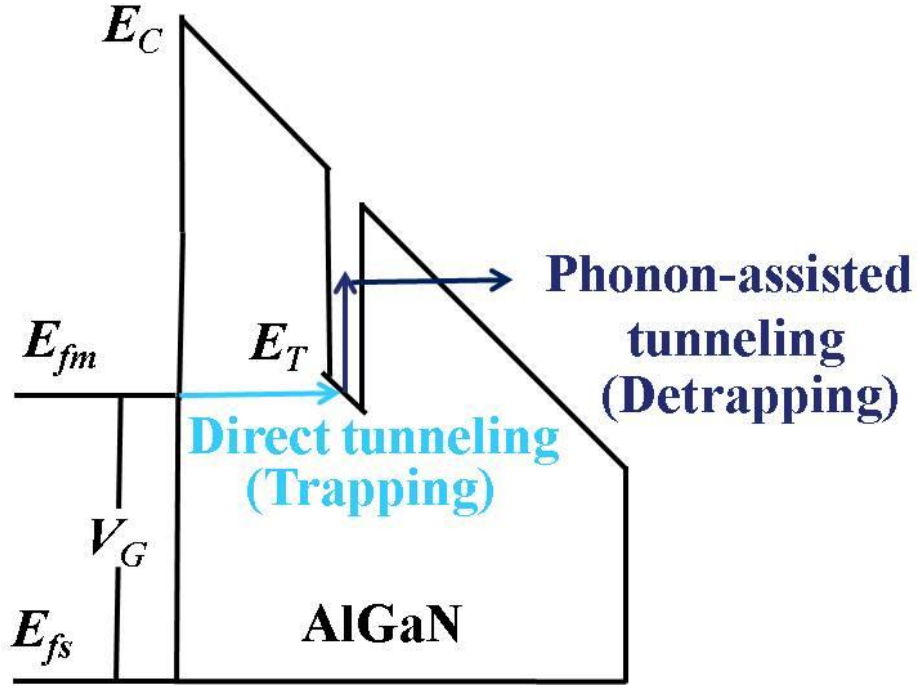


Figure 6-11: Schematic of trapping and detrapping mechanism in the AlGaIn barrier. The trapping is the electron direct tunneling from the gate metal and the detrapping is the electron emission into the AlGaIn conduction band by phonon-assisted tunneling (PAT), as labeled in the graph.

Electron capture from the AlGaIn conduction band was neglected due to the low electron concentration in the barrier. Conversely, detrapping is the electron emission into the AlGaIn conduction band, modeled by PAT theory. In a PAT process, electrons are first activated to pseudo-levels by the lattice phonons, and then tunnel into the conduction band (Figure 6-11) [140]. Note that the direct tunneling and pure thermal emission are already included in the PAT. Therefore, we can conclude the rate equation for the barrier traps as

$$\frac{\partial(F_T N_T)}{\partial t} = C_{tun} (1 - F_T) N_T - F_T N_T e_n. \quad (6.5)$$

In Eq.(6.5), F_T is the occupancy probability for the traps and N_T is the traps density. C_{tun} is a coefficient relevant to the electron tunneling from the gate metal, expressed as [155]

$$C_{tun} = P \left(\frac{m_M}{m_S} \right)^{5/2} \frac{16\pi q \phi_1^{3/2}}{3h\sqrt{\phi_t - \phi_1}}, \phi_1 = 0.2 \text{ V}, \frac{m_M}{m_S} = 0.75, \quad (6.6)$$

where P is the electron tunneling probability from the gate metal into the traps in the AlGaIn barrier. It can be calculated by the WKB approximation:

$$P = \exp(-2 \int_0^x k(x) dx). \quad (6.7)$$

In Eq. 6.7, x is the tunneling point in the AlGaIn barrier and $k(x)$ is the wave number. The electric potential is linear if the effect of the image force and the space charge in the AlGaIn barrier layer are not considered. Then P has an analytical expression for the WKB approximation [155]:

$$P = \exp[-\frac{8\pi\sqrt{2m_e^*q}}{3hE}(\phi_B^{3/2} - \phi_t^{3/2})], \quad (6.8)$$

where ϕ_B and ϕ_t are the Schottky barrier height and traps depth, respectively, m_e^* is the electron effective mass, and E is the electric field.

In Eq. (6.5), e_n is the electron emission rate described by PAT, expressed as [146]

$$e_n = e_{n0}(1 + \Gamma_n^{Dirac}), \quad (6.9)$$

$$\Gamma_n^{Dirac} = \frac{\Delta E_n}{kT} \int_0^1 \exp(\frac{\Delta E_n}{kT} u - K_n u^{\frac{3}{2}}) du,$$

$$K_n = \frac{4}{3} \frac{\sqrt{2m_e^* \Delta E_n^3}}{q(\frac{h}{2\pi})|E|},$$

where e_{n0} is the electron emission by pure thermal activation (Eq.6.2) and Γ_n^{Dirac} is the enhancement factor by PAT which has the following analytical approximation [156]:

$$\Gamma_n^{Dirac} = \sqrt{\pi} \frac{E}{E_{trap}} \exp\left[\frac{1}{3}\left(\frac{E}{E_{trap}}\right)^2\right] \left(2 - \operatorname{erfc}\left[\frac{1}{2}\left(\frac{E_{trap}}{E} - \frac{E}{E_{trap}} \cdot \frac{\Delta E_n}{kT}\right)\right]\right), \quad \frac{E}{E_{trap}} \leq \sqrt{\frac{\Delta E_n}{kT}}$$

$$\Gamma_n^{Dirac} = \sqrt{\pi} \frac{E}{E_{trap}} \left(\frac{\Delta E_n}{kT}\right)^{\frac{1}{4}} \exp\left[-\frac{\Delta E_n}{kT} + \frac{E}{E_{trap}} \left(\frac{\Delta E_n}{kT}\right)^{\frac{1}{2}} + \frac{1}{3} \frac{E_{trap}}{E} \left(\frac{\Delta E_n}{kT}\right)^{\frac{3}{2}}\right] \\ \times \operatorname{erfc}\left[\left(\frac{E}{E_{trap}}\right)^{\frac{1}{2}} \left(\frac{\Delta E_n}{kT}\right)^{\frac{1}{4}} - \left(\frac{E_{trap}}{E}\right)^{\frac{1}{2}} \left(\frac{\Delta E_n}{kT}\right)^{\frac{3}{4}}\right]. \quad \text{elsewhere}$$

$$E_{trap} = \frac{\sqrt{8m_e^* (kT)^3}}{q \frac{h}{2\pi}}$$

The reader can see more details about ΔE_n in [156]. To include the PF effect, Eq. (6.9) becomes

$$e_n = e_{n0}(\chi_F + \Gamma_n^{Coul}), \quad (6.10)$$

$$\chi_F = \exp\left(\frac{\Delta E_{fp}}{kT}\right)$$

$$\Gamma_n^{Coul} = \frac{\Delta E_n}{kT} \int_{\Delta E_{fp}/\Delta E_n}^1 \exp\left(\frac{\Delta E_n}{kT} u - K_n u^2 \left[1 - \left(\frac{\Delta E_{fp}}{u \Delta E_n}\right)^{5/3}\right]\right) du$$

$$\Delta E_{fp} = \left(\frac{q^3 |E|}{\pi \epsilon}\right)^{1/2}$$

where χ_F and Γ_n^{Coul} are the enhancement factors by the PF effect and PAT with PF effect, respectively [157]. ΔE_{fp} is the barrier lowering induced by PF effect.

Figure 6-12 shows the electric field-enhanced factor e_n/e_{n0} by PF effect and PAT with and without PF effect at different temperatures of $T=300$ K and $T=400$ K. The electric field strength ranging from 0.7 MV/cm to 2.2 MV/cm showed in the graph can represent the general values in the barrier for typical AlGaIn/GaN HEMTs working in on- and off-state. At $T=300$ K, the electron

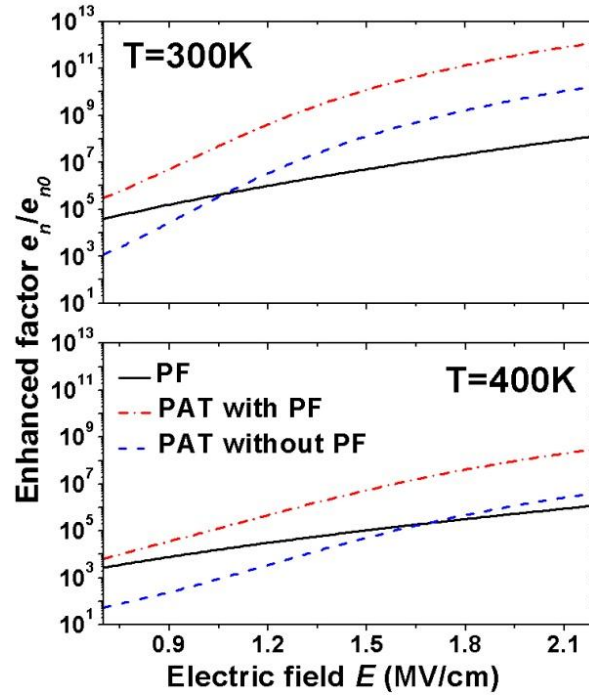


Figure 6-12: Electric field-enhanced factor e_n/e_{n0} by sole PF effect and PAT with/without PF effect at different temperature $T=300$ K and $T=400$ K.

emission is significantly enhanced by the PAT or PF effect. This enhancement is reduced at higher temperature $T=400$ K due to the increased contribution of thermal activation. Around same magnitude is enhanced for varied electric field when comparing the PAT with and without PF effect.

6.3.2 Simulation details

Figure 6-13 shows the vertical (E_y) and horizontal (E_x) electric field distribution along the middle of the barrier beneath the gate for the device in on-state. E_y is much higher than E_x . Here note that PAT theory was obtained in a one-dimensional approach [146]. In accordance with that, only the impact of E_y on e_n was considered in the following simulations.

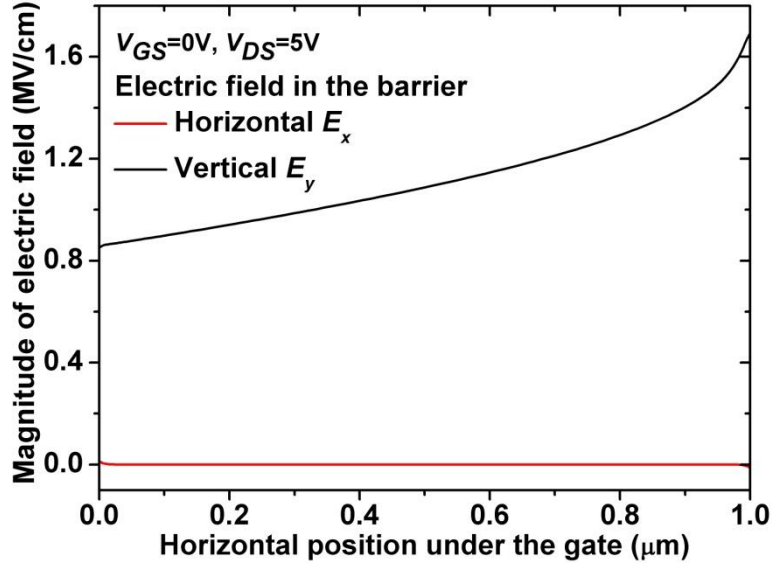


Figure 6-13: Vertical (E_y) and horizontal (E_x) electric field distribution along the middle of the AlGaIn barrier under the gate for the device in on-state.

A. Simulation details of $I_{G,Rev}$

This section is devoted to describe the simulation method of $I_{G,Rev}$ based on the presented trapping/detrapping model. With the traps as an intermediate medium, $I_{G,Rev}$ is formed by the electron direct tunneling from the gate metal

and then emission into the AlGaN conduction band by PAT. In steady state, in Eq. (6.5) the term $\frac{\partial(F_T N_T)}{\partial t} = 0$ and then F_T is derived as

$$F_T = \frac{C_{tun}}{C_{tun} + e_n}. \quad (6.11)$$

Therefore, $I_{G,Rev}$ can be calculated by an integral in the barrier, expressed as

$$I_{G,Rev} = \int q F_T N_T e_n dx \quad (6.12)$$

We implemented this model for the device under reverse bias because the relatively high electric field in the barrier can enhance electron emission significantly. For the device under forward bias, thermionic emission would dominate the gate current [158].

B. Simulation details of τ_n and transient I_{DS}

Generally, the traps are filled for the device in off-state and emptied for the device in on-state. Collapse in I_{DS} happens if the emptying process cannot follow the bias signal. The following transient I_{DS} simulation describes this phenomenon. Neglecting the edge area in the barrier, the occupied traps should be uniformly distributed in the horizontal direction when V_{GS} is lower than the threshold voltage, due to the uniform distribution of horizontal electric field. For simplicity, the trapping process was neglected for the device in on-state in this transient simulation. Thus, in Eq. (6.1), $C_{tun} = 0$ and we have

$$\frac{\partial(F_T N_T)}{\partial t} = -F_T N_T e_n, \quad (6.13)$$

whose analytical solution is $F_T = C \cdot \exp(-e_n t)$. Here we note that $e_n = 1/\tau_n$ is coupled with the temperature (related to self-heating) and electric field. The trapped charge density determined by $F_T N_T$ will be integrated into the Poisson equation. The characteristic time of adjustment of electron density in the channel (which mainly happens during pulse rise process) is much shorter than that of traps and thermal effect, so the transient behavior of I_{DS} in the pulse rise process was neglected. Holes were not included in the simulations. A drift and diffusion model was applied for the electron transport within a 200 nm thick

active region below the AlGaIn/GaN interface. A thermal conduction equation was numerically solved within the entire device, coupling with the drift-diffusion model, when considering the self-heating effect. More details of the simulation process can be found in Chapter 2.

A typical AlGaIn/GaN HEMT with 1 μm gate-source spacing, 1 μm gate length, 2 μm gate- drain spacing, 20 nm AlGaIn/1 μm GaN heterostructure, and 300 μm Si substrate was used in τ_n and transient I_{DS} simulations. In order to avoid the electric field singularity at GEDS, a beveled gate edge was used, which was expected to be in accordance with the practical shape [138]. Fixed polarization charge density at the AlGaIn/GaN interface and on the AlGaIn surface were set as $9 \times 10^{12} \text{q/cm}^2$ and $-2 \times 10^{12} \text{q/cm}^2$, respectively, to produce reasonable 2DEG density of $7 \times 10^{12} \text{q/cm}^2$ in the channel. A large variation of E_T in GaN-based material and device was reported [27]. Here typical deep-level value of $E_T=0.9 \text{ eV}$ and electron capture cross-section $1 \times 10^{-15} \text{ cm}^2$ were used. Also, the simulations were performed using the commercial FEM software (COMSOL).

6.3.3 Mechanism of reverse gate current

Figure 6-14 shows the $I_{G,Rev}$ ($V_{DS}=0 \text{ V}$) simulations including the PF effect and the results were compared to the experimental data at 100 K and 300 K (ambient temperature). We can see that the results can well fit to the experimental data (see the fitting parameters in the graph). The fitting parameter $m^*=0.35m_e$ is higher than the reported value $0.22m_e$ [159]. Φ_b , Φ_B , and N_T are in reasonable range [27][158]. Note that $I_{G,Rev}$ tends to saturate when V_G is lower than the threshold voltage due to the complete depletion of the 2DEG in the channel, resulting in a constant the electric field in the barrier, which was also reported elsewhere [144][154]. $I_{G,Rev}$ at 300K is higher than that

at 100 K due to the increase of e_n with temperature. Overall, based on the presented model, the mechanism of $I_{G,Rev}$ and its temperature dependence can be attributed to the electric field and temperature-dependent e_n in the barrier.

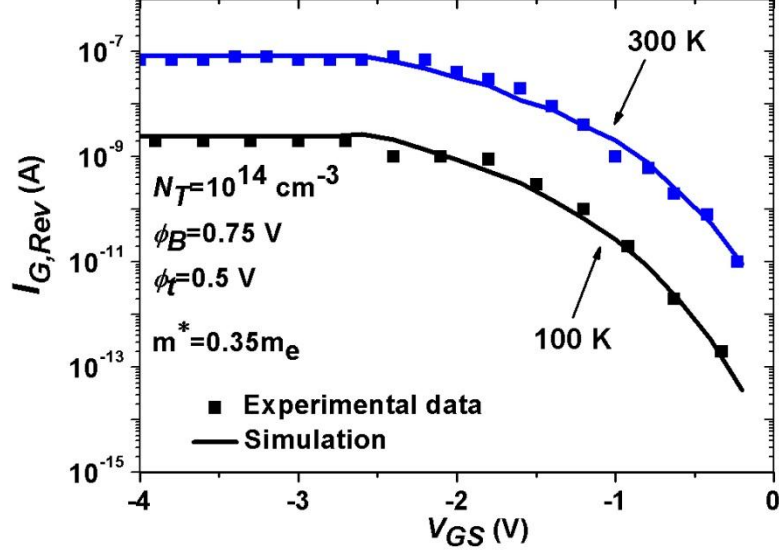


Figure 6-14: Comparison of simulation and experimental results of $I_{G,Rev}$ at 100 K and 300 K. The experimental data were taken from [141].

6.3.4 Stretched exponential characteristic of transient drain current

Experimental results of the transient I_{DS} always show a stretched exponential characteristic [27] [148], which cannot be replicated by the simulations with only considering the thermal activation for the electron emission. In the following, an explanation for this will be presented.

The spatial distributions of E_y (Figure 6-15(a)), temperature T (Figure 6-15(b)), and τ_n along the middle of the barrier beneath the gate (Figure 6-15(c)), and the normalized transient I_{DS} for V_{DS} in $I_{DS,Lin}$ and $I_{DS,Sat}$ region (Figure 6-15(d)), were simulated. When self-heating and traps effects were included in the simulation, the transient I_{DS} decreased (Figure 6-15(d) for $V_{DS}=4$ V) for the pulse width less than 100 μ s due to the domination of

self-heating effect in this time scale, as illustrated experimentally in [149]. It was clear that E_y had a non-uniform distribution in the barrier for $V_{DS}=4$ V in $I_{DS,Sat}$ region, where E_y has a relatively high value at the GEDS due to the impact of I_{DS} pinch-off. This leads to a non-uniform τ_n distribution, with a decreasing gradient going from the source to drain, indicating a faster (slower) electron emission towards the drain (source) and thus the stretched exponential characteristic of transient I_{DS} . For $V_{DS}=1$ V in $I_{DS,Lin}$ region, E_y and τ_n tend to have a uniform spatial distribution, presenting the convenience for exponential fitting of the experimental transient I_{DS} in order to obtain the Arrhenius plot. τ_n for $V_{DS}=4$ V is lower than that for $V_{DS}=1$ V due to the higher T induced by self-heating and higher E_y , similar to the measurements in [151] and [152].

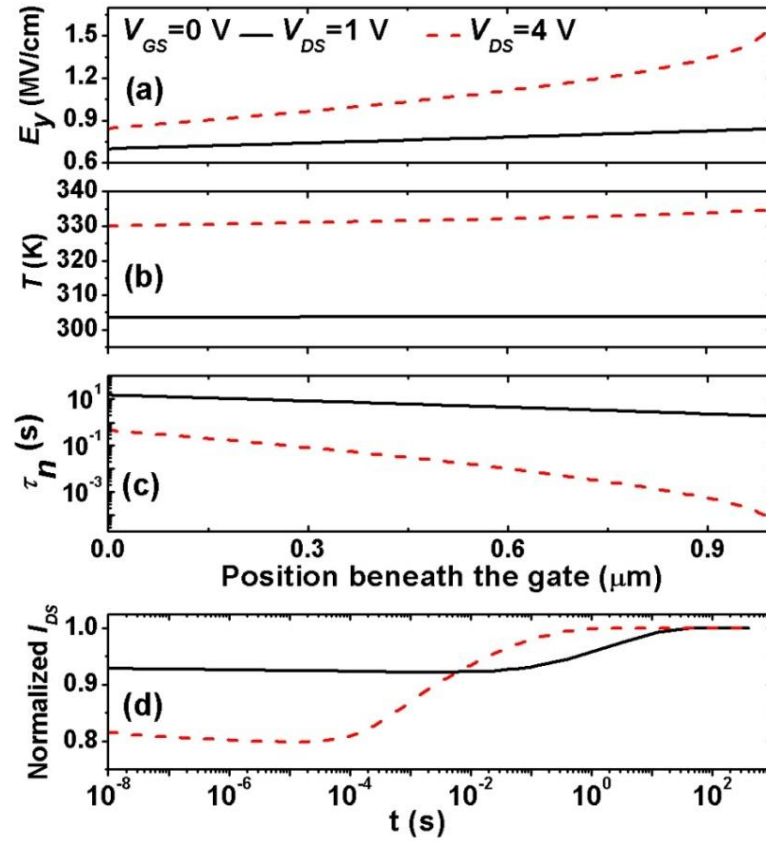


Figure 6-15: Simulation results of E_y (a), T (b), τ_n (c) distribution along the middle of the AlGaIn barrier beneath the gate at $t=100$ s, and the normalized transient I_{DS} (d) for $V_{DS}=1$ V in $I_{DS,Lin}$ region and $V_{DS}=4$ V in $I_{DS,Sat}$ region ($V_{GS}=0$ V).

We note that, in addition to the non-uniformly distributed electric field, stretched exponential I_{DS} also can be attributed to other mechanisms, such as the presence of traps with discrete or continuous ionization energy [27] (different τ_n). To illustrate this, transient I_{DS} simulations were performed for three different discrete ionization energies: $E_{T1}=0.45$ eV, $E_{T2}=0.5$ eV, and $E_{T3}=0.55$ eV (the traps are in the AlGaIn barrier), as shown in Figure 6-16. We can see that, for the relative small E_T the transient I_{DS} is left shifted due to the relative high e_n . When all of the traps with the three different traps ionization energies were included in the simulations, as expected, the transient I_{DS} has a stretched exponential characteristic.

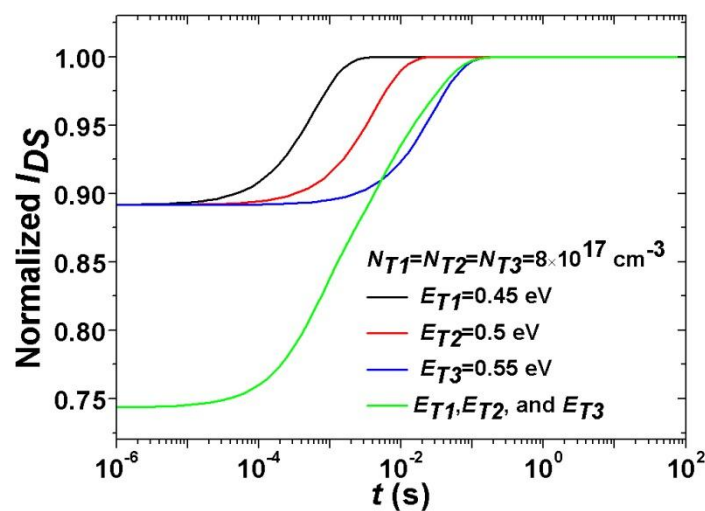


Figure 6-16: Simulations to illustrate the stretched exponential characteristic of transient I_{DS} caused by the traps in the AlGaIn barrier with three discrete ionization energies: $E_{T1}=0.45$ eV, $E_{T2}=0.5$ eV, $E_{T3}=0.55$ eV, separately and simultaneously.

6.3.5 Bias impact on electron emission characteristic time

Traps effect characterization is important for both the $I_{DS,Lin}$ and $I_{DS,Sat}$ region which are related to the device on-resistance [160] and power output, respectively. Also we know that the device working in these two regions has big difference of the thermal and electrical behavior. Therefore, the following simulations will be performed separately for these two different regions and

simulations with and without self-heating will be compared to distinguish its impact.

A. In $I_{DS,Lin}$ region

Figure 6-17 shows E_y (Figure 6-17(a)) and T (Figure 6-17(b)) distribution, and τ_n averaged in the middle of the AlGaIn barrier beneath the gate (Figure 6-17(c)) with V_{DS} variation in $I_{DS,Lin}$ region. As the graph shows, with increasing V_{DS} , E_y is significantly enhanced, whereas T increased by only a few degrees because of the uniform heat distribution in the channel and the relatively low power dissipation for the device in the $I_{DS,Lin}$ region. Thus, the average τ_n has no significant difference for the simulations with and without self-heating (Figure 6-17(c)).

V_{GS} cannot only modulate I_{DS} and therefore the device temperature, but also it can influence electric field in the barrier. As shown in Figure 6-18(a) and (b), as V_{GS} increased, E_y and τ_n decreased and increased, respectively, while T had

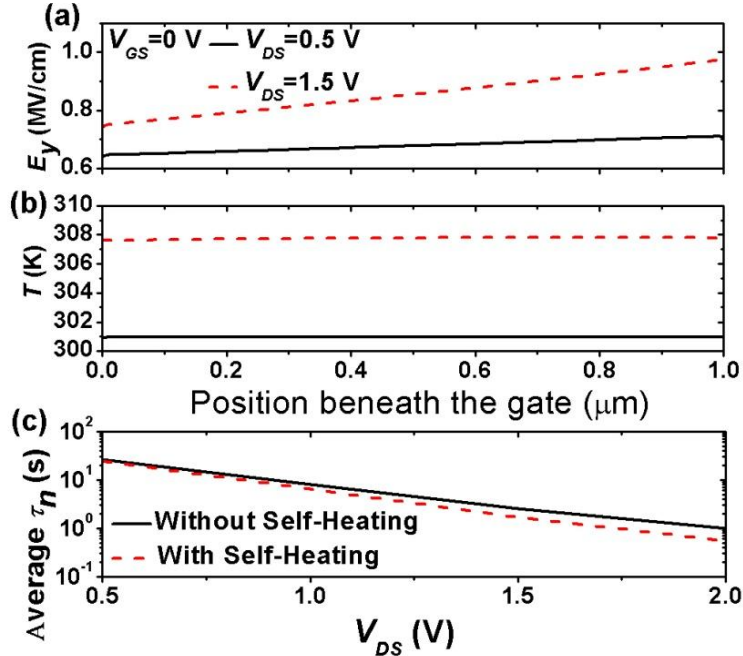


Figure 6-17: Simulation results of E_y (a) and T (b) distribution, and τ_n averaged in the middle of the AlGaIn barrier beneath the gate with V_{DS} variation in $I_{DS,Lin}$ region (c) at $t=100$ s. Simulations with and without self-heating effect were compared (c).

several degrees increase so that τ_n has negligible difference for the simulations with and without self-heating (Figure 6-18(c)). τ_n has a much bigger difference with V_{GS} variation (from around 10 μ s to 2 s with V_{GS} from -2 to 0 V), compared to V_{DS} variation. Over all, we concluded that for the $I_{DS,Lin}$ region, with V_{DS} or V_{GS} variation the electric field has dominant impact on the τ_n , while the self-heating impact was negligible.

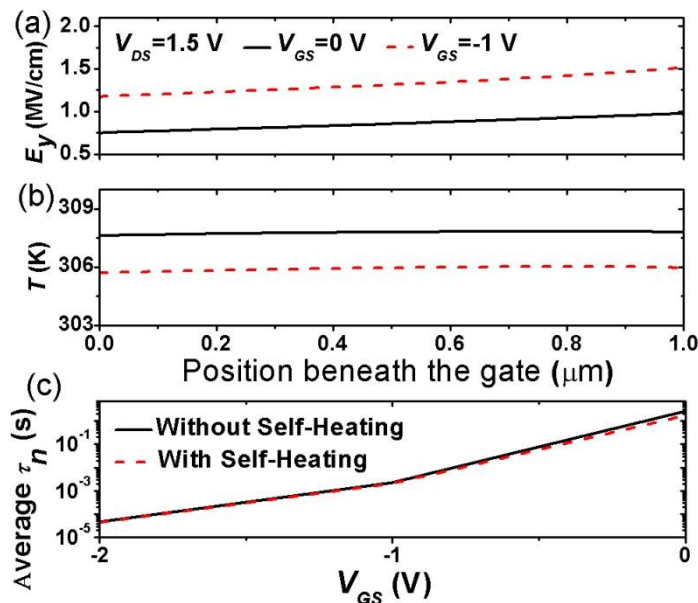


Figure 6-18: Same as Figure 6-17, but for V_{GS} variation.

B. In $I_{DS,Sat}$ region

Figure 6-19 shows the simulations for the device working in $I_{DS,Sat}$ region. As expected, self-heating became more pronounced (maximum T is increased by around 90 K with $V_{DS}=10$ and $V_{GS}=0$ V) for the $I_{DS,Sat}$ region compared to that of $I_{DS,Lin}$ region (Figure 6-19(b)), because of the concentrated heat source at GEDS and the relatively high power dissipation. With self-heating, τ_n had a significant change from 0.5 s to 0.01 s with V_{DS} variation from 3 V to 10 V, whereas, τ_n had a negligible difference without self-heating (Figure 6-19(c)). This means that self-heating was the dominant factor on τ_n with V_{DS} variation, which can be explained that a change in electric field caused by V_{DS} only influenced a small area near GEDS as shown in Figure 6-19(a), whereas a

change of device temperature appeared in the entire barrier (Figure 6-19(b)).

Figure 6-20 shows the simulation results of the V_{GS} impact on τ_n . When the self-heating was pronounced, with V_{GS} increasing, a competing mechanism affected τ_n , i.e., the decreased electric field (see Figure 6-20(a)) increased τ_n while increased self-heating (see Figure 6-20(b)) decreased τ_n . Figure 6-20(c)

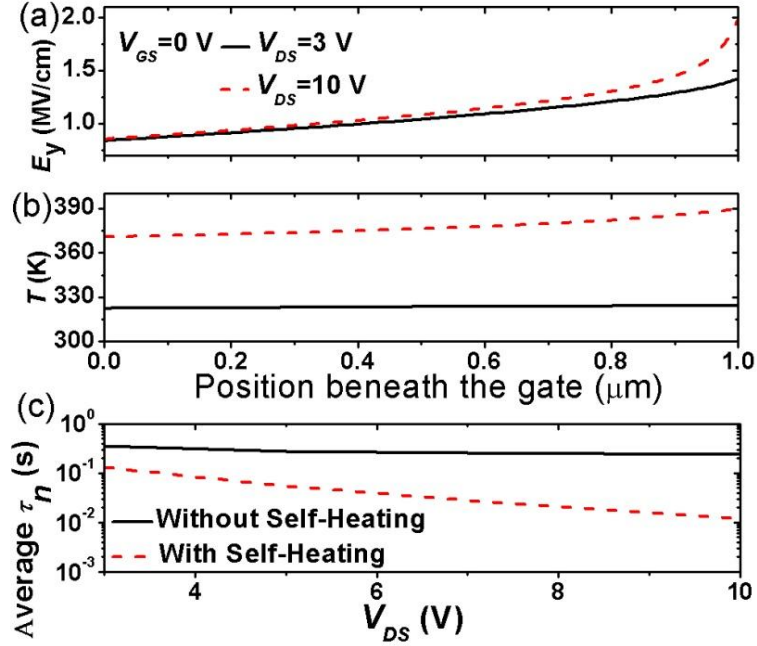


Figure 6-19: Same as Figure 6-17 but for V_{DS} variation in $I_{DS,Sat}$ region.

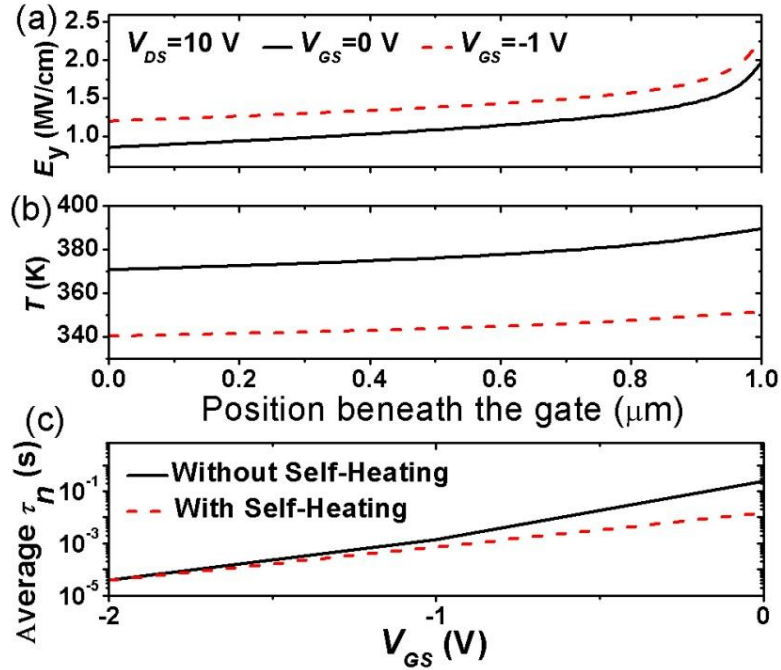


Figure 6-20: Same as Figure 6-17 but for V_{GS} variation in $I_{DS,Sat}$ region.

shows that the simulated τ_n including self-heating increased with V_{GS} increasing, meaning the electric field is the dominant factor for this competing mechanism.

6.3.6 Comparison with detrapping in buffer

A. Electric field and temperature effects on τ_n in the GaN buffer

Based on the presented trapping/detrapping model, we have limited our illustrations for the barrier traps which would be solely filled in off-state with $V_{DS}=0$ V in practical measurements. However, generally, positive drain field will cause traps in the GaN buffer and on the AlGaN surface near GEDS to be filled as well. Possible filling mechanism for the traps on the AlGaN surface is that the electrons tunnel from the gate metal and transport by hopping. Regarding the traps in the buffer, they are possibly filled in the deep buffer under GEDS by gate or drain leakage current described by drift-diffusion transport model. The detrapping mechanism can also be temperature and electric field-dependent. Here, we performed simulations on both of these two factors impact on τ_n (also described by PAT) for the GaN buffer when the device is in steady-state, without considering the detailed filling mechanisms and transient process of detrapping. Since the filled traps mainly located near GEDS where the electric field along the horizontal direction ($|E_x|$), we simulated the distribution of $|E_x|$, T , and τ_n in the GaN buffer, as shown in Figure 6-21(a-c). We can see that τ_n was non-uniformly distributed and had relative low values near GEDS, due to the relative high $|E_x|$ and T .

Figure 6-22 shows the simulated minimum τ_n in the buffer as a function of V_{DS} (a) and V_{GS} (b). As Figure 6-22(a) shows, similar to the detrapping in the barrier, the self-heating and electric field significantly decreased τ_n for the device in $I_{DS,Sat}$ region (see for $V_{DS}>3$ V). As Figure 6-22(b) shows, according to the aforementioned competing mechanism, the self-heating should be the

dominant factor because the simulated τ_n including the self-heating presented decreased value with V_{GS} increasing, different from the detrapping in the barrier (see Figure 6-20(c)).

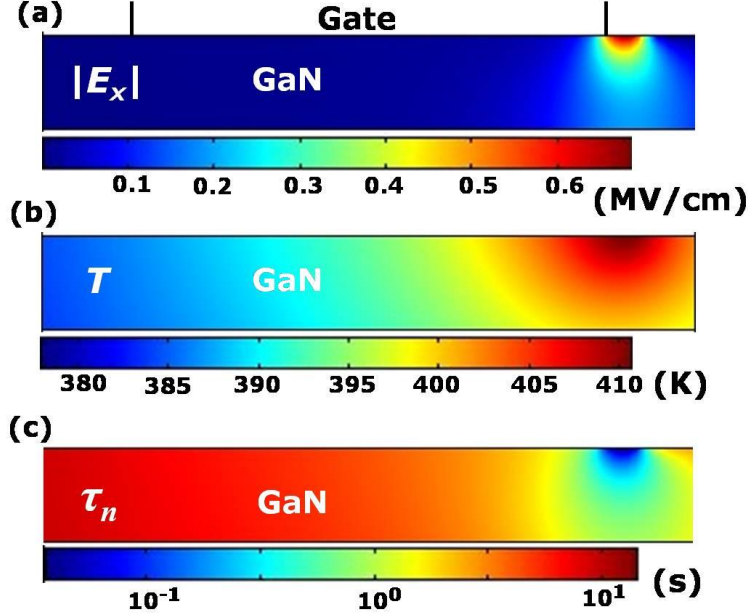


Figure 6-21: Simulations of $|E_x|$ (a), T (b), and τ_n (c) distribution in the GaN buffer (200 nm thickness under the AlGaIn/GaN interface was showed). Note that τ_n displayed in the graph was Log10 scaled. The device biases were $V_{DS}=10$ V and $V_{GS}=0$ V.

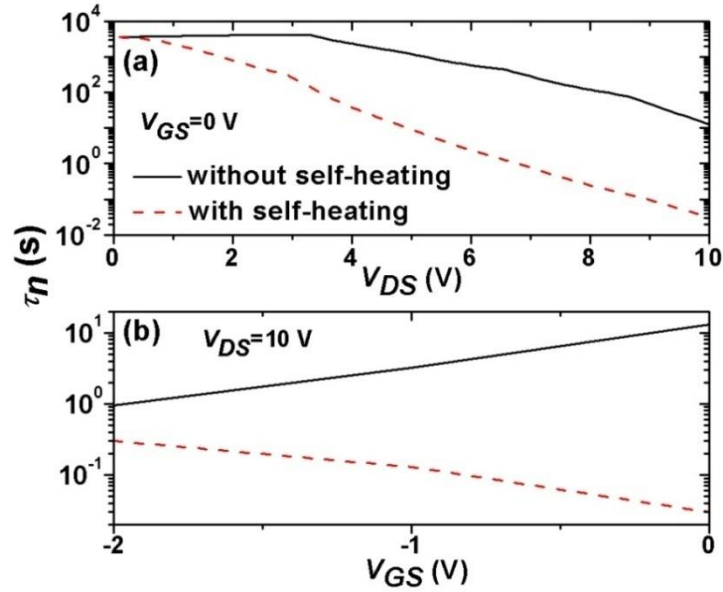


Figure 6-22: Simulated minimum τ_n in the GaN buffer as a function of V_{DS} (a) and V_{GS} (b). Simulations with and without self-heating were compared.

B. Insights into the current-transient method for traps characterization

We have the following insights into the current-transient method for the traps characterization in the framework of temperature and electric field-dependent detrapping mechanism (note that an additional load is needed in order to probe the transient I_{DS} in practical measurements so that V_{DS} also varies in this transient process). The device would be biased in the $I_{DS, Lin}$ region, because the influence of self-heating can be avoided and τ_n has relative uniform distribution in the barrier. However, we note that the unobvious current collapse in $I_{DS, Lin}$ region (Figure 6-22(d)) probably cannot clearly manifest the detrapping process.

The detrapping mechanism described by PAT illustrates that the electron thermal emission can be enhanced at a higher temperature. The corresponding impact of the tunneling effect induced by the electric field on electron emission was reduced (see Eq. (6.9) and more details in [146]). In order to reduce the bias impact on this characterization method, the device would work under high ambient temperature, because this can not only reduce the tunneling effect but also possibly the self-heating impact (device under higher ambient temperature has lower power output and therefore less self-heating if the device thermal resistance does not increase significantly with temperature).

6.3.7 Impact of traps-dependent gate current on drain current collapse

As mentioned before, the presence of deep-level traps in the device was proposed to the explanation of the I_{DS} collapse. In this section, we will compare three mechanisms of traps-dependent $I_{G, Rev}$ and discuss their impact on I_{DS} collapse.

Based on the fact that the AlGaIn surface has high density of donor-like traps, the $I_{G, Rev}$ mechanism can be explained as follows: the electrons in the

gate metal first are thermally activated into these surface traps and then tunnel into the conduction band [161](see Figure 6-23(a)). This manner can increase the electron tunneling probability due to the thinner tunneling barrier, compared to the direct tunneling from the gate into the conduction band. Because the surface traps directly contact the gate metal, change of the traps occupancy (or charge density) is compensated by electrons in the gate metal, but not the 2DEG in the channel, so that the surface traps filled by $I_{G,Rev}$ cannot affect the I_{DS} collapse. This is different from the case that the filled surface traps near GEDS (usually termed virtual gate) reduces the 2DEG density in the correspondent position of the drain-side access region and thus yields I_{DS} collapse. Hashizume *et al.* reported that donor-like traps can concentrate in the AlGaIn barrier near the gate metal due to the nitrogen vacancy probably caused by the device processing [162][163]. These ionized donor-like traps form a thin surface barrier (TSB) with a relative high electric field beneath the gate which can also increase the electron tunneling probability. The $I_{G,Rev}$ mechanism can be explained as the electrons in the gate metal directly tunnel through this TSB into the conduction band (see Figure 6-23(b)). However, we note that, the TSB

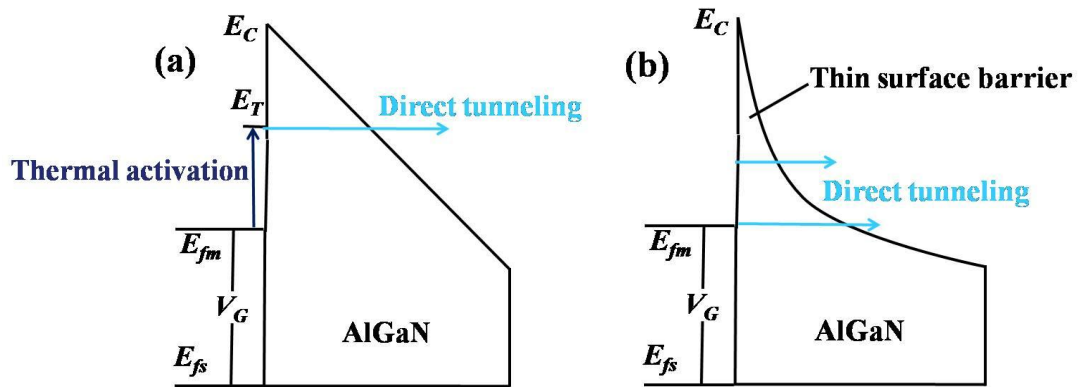


Figure 6-23: Schematic of $I_{G,Rev}$ mechanisms: (a) electrons in the gate metal first are thermally activated into the AlGaIn surface traps and then directly tunnel into the AlGaIn conduction band, and (b) electrons in the gate metal directly tunnel into the AlGaIn conduction band through a thin surface barrier (TSB) caused by high density of ionized donor-like traps near the AlGaIn surface.

requires the donor-like traps keep unfilled, so that $I_{G,Rev}$ cannot affect the I_{DS} collapse. According to the presented $I_{G,Rev}$ mechanism, the trapping/detrapping process happens in the barrier (see Figure 6-11). $I_{G,Rev}$ affects the occupancy probability of the barrier traps and therefore the collapse in I_{DS} .

6.4 Summary

The device geometry impact on the I_{DS} collapse expressed as the GLR was studied and the results were illustrated by correspondent simulations considering the “virtual gate” effect. It was shown that device has less current collapse (large GLR) for long L_G or L_{GD} . This can be attributed to low ratio of the increased resistance caused by the virtual gate to the device total resistance.

Buffer traps effects were also simulated. The simulations mainly focused on the device self-heating impact on the electron emission and the transient I_{DS} . The electron emission rate can be increased for the device operation in $I_{DS,Sat}$ region. The barrier and buffer traps led to the current collapse in the saturation region of I-V curves and the threshold voltage shift, while the surface traps led to the current collapse in the knee region of I-V curves.

A model describing electron trapping and detrapping processes in the AlGa_N barrier of AlGa_N/Ga_N HEMTs is presented to simulate the reverse gate current. We refer to trapping as the process of electrons directly tunneling from the gate metal, while detrapping is electron emission into the AlGa_N conduction band by phonon-assisted tunneling. With including the Poole-Frenkel effect, simulation results fit well to the experimental data. The mechanism of the reverse gate current and its temperature dependence can be attributed to the temperature and electric field-dependent electron emission in the barrier.

Furthermore, by fully coupling the electron emission characteristic time (τ_n) with the device thermal and electrical behavior, simulations were

performed to comprehensively analyze the contributions of device biasing, accounting for self-heating and electric field dependent electron detrapping. Results illustrated that the vertical electric field had a dominant impact on τ_n , while self-heating was negligible for the device biased in the linear region of the drain current. For the saturation region, the simulated transient drain current had a stretched exponential characteristic due to the non-uniform spatial distribution of E_y , conforming to reported experimental results. With V_{DS} variation, self-heating had a dominant impact on τ_n while E_y impact was negligible, because variation of E_y only appeared in a small area near GEDS, while the device temperature variation appears in the whole barrier. With V_{GS} variation, E_y has dominant impact on τ_n . At last, discussions on detrapping in the GaN buffer, the current-transient method for traps characterization, and impact of traps-dependent $I_{G,Rev}$ on I_{DS} collapse are presented based on these simulations. The simulations on detrapping in the AlGaIn barrier would provide useful insights into the current-transient method for traps characterization. The presented model can also account for the $I_{G,Rev}$ impact on I_{DS} collapse caused by the barrier traps.

Experimental work in Section 6.1 was performed by S. Mart ́n-Horcajo. Most of contributions in Section 6.1 and 6.3 were prepared as articles, see [164] and [165], respectively.

7 Conclusion and future work

7.1 General conclusions

A. Simulation methods

By using the commercial FEM software COMSOL, we have developed the simulation methods for GaN HEMTs with electro-thermal, electro-mechanical, and electro-thermo-mechanical coupling. We have applied these methods to study several topics on the performance and reliability of GaN-based HEMTs, including the device electrical, thermal and stress behavior, as well as the surface, buffer, and barrier traps effects.

B. Electrical behaviors

Regarding the electrical behavior, the distribution of electrical potential, electric field, quasi-Fermi level, electron density, etc., were simulated, which would make us for better understanding the device operation principle. The origin of the device field-plate and fringing capacitance were illustrated. It was shown that the higher permittivity and thicker for the dielectric the larger fringing capacitance. For the field-plate gate, the fringing capacitance is an important contribution for the device parasitic capacitance when the 2DEG under the field-plate is depleted. When the device is scaled, the fringing capacitance should play an important role in the device RF performance, while it is negligible for the device with long gate length.

C. Thermal effects

Regarding the thermal behavior, the substrate and geometry impact on the device self-heating have been studied by electro-thermal simulations and electrical characterizations. The simulated channel temperature and thermal resistance were compared to that extracted from the electrical characterizations.

These studies provide clear guidelines to optimize the device geometry for reduction of the self-heating effect, i.e., longer L_{GD} leads to less concentration of the device heat source at the drain-side gate edge and thus lower channel temperature. Also, simulations on heat spreading by integrated diamond were also performed. Two different places for the diamond were studied: on top of the device and in vias etched in the substrate. It was shown that the self-heating can be reduced by the heat spreading effect of the diamond placed in these two places. Transient simulations illustrated that the device RF performance can also be improved due to the very short distance between the heat source and the top-side diamond heat spreader. Note that the ability of heat spreading might not be influenced by the device scaling.

D. Stress effects

Regarding the stress behavior, local stress at the gate edge of the drain side including mainly the built-in stress in the epitaxial layers, thermal stress, and converse piezoelectric stress were simulated for standard gate and field-plate gate. Specially, the intrinsic stress in the capped diamond layer and its impact on the device electrical behavior were studied by micro-Raman spectroscopy characterization and simulations, respectively. The simulated stress and electrical output characteristics were compared to the correspondent experimental data. These studies show that the mechanical failure may be mitigated by the modulation of the local stress at the GEDS by the intrinsic stress in the capped layers. Improvements of the device power output could be achieved by stress engineering. When the device is scaled, the tensile stress and 2DEG under the gate will be increased, therefore more improvement of drain current can be achieved.

E. Traps effects

The surface traps effects are studied, including the influence of the relevant device design parameters L_G and L_{GD} . The simulations can explain the pulsed measurements regarding the gate lag ratio. Results show that relative large L_G or L_{GD} could reduce the current collapse due to the less ratio of the increased resistance induced by the virtual gate compared to the device total resistance. Buffer traps effects were also simulated, focusing on the self-heating impact on the transient electron emission rate and I_{DS} . For the barrier traps effects, a model about the trapping and detrapping in the barrier was presented. The reverse gate current was simulated based on this model and its mechanism was illustrated to be dependent on the electron emission rate influenced by the temperature and electric field, i.e., the higher ambient temperature and electric field (lower gate voltage) the higher reverse gate current. Furthermore, describing the detrapping in the barrier by means of a model of phonon-assisted tunneling, the influence of the device bias on the electron emission rate and the transient I_{DS} were simulated. Results showed that for the barrier traps the self-heating and electric field can increase the emission rate and the stretched exponential characteristic of the transient I_{DS} can be attributed to the non-uniform distribution of electric field in the barrier. These studies provide insights into the traps characterization by the current-transient technique and improve our knowledge of the effects of traps. Note that when the device is scaled, the short channel effect can influence the electric field distribution and thus electron emission rate as well as transient drain current. Our future work will address this.

In this dissertation, we have addressed the impact of the device geometry/structures on the thermal, stress, and traps effect. From the point of view of the design optimization, a compromise way should be considered for specific applications, e.g., long L_{GD} can reduce the self-heating, reduce traps effects, and increase the breakdown voltage while increase the device on

resistance and access capacitances, short L_G can have more significant performance improvement by stress in the cap layers while presents more traps effect, field plate can increase the breakdown voltage while also increase the parasitic capacitances, etc.

7.2 Suggestions for future work

As we described in Section 3.2.2, thermal simulation was a popular method for studying the device self-heating, because it was simple and could be performed in 3D with including the heat dissipation in the third dimension. The impact of substrate thickness, thermal conductivity, heat power, and gate width on the device temperature can be correctly taken into account by this method. However, the gate length and distance between the gate and drain contact impact on the device temperature cannot be properly evaluated, and the results are sensitive to the heat source size that is manually set. Therefore, an important aspect that can improve this thermal simulation method is to properly define the heat source size and build a relationship between the heat source size and the device geometry. This approach might need complicated mathematical derivations, and would be part of the future work.

We have simulated the temperature and electric field-dependent barrier traps effects. Further experimental work would be needed to study the device bias impact on the trapping for the device in the off-state, and detrapping in on-state, characterized by the transient I_{DS} . Also the detrapping on different substrates would be studied for considering the self-heating impact on the transient I_{DS} . Note that the defects/traps density influenced by the lattice mismatch between the epitaxial layers and the substrate would be an important issue for this study.

We have simulated the device stress behavior for mechanical failure analysis, and suggested possible power improvements. Future work might be

involved to electrical stress measurement for degradation/reliability study. Passivations with different materials could be studied, especially using dielectrics with high permittivity, which are expected to reduce the electric field and thus the converse piezoelectric stress at the gate edge of the drain side, and further the mechanical failure. The reduction of the electric field can also be expected to minimize the electron tunneling from the gate metal into the surface traps, and thus limit the traps effects. Standard gate, slant field plate gate, and over-hanging field plate gate present different electric field distributions in the device. Comparative studies on the device reliability and traps effects for these different gate structures should be performed.

Up to now, we have performed DC and transient simulations. The RF simulation including the traps, heat spreading, or stress effect would be also interesting topics for future work.

Article contributions

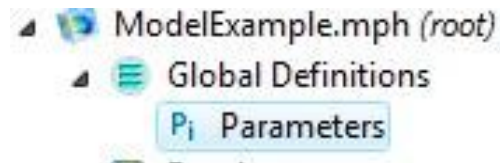
1. A. Wang, M. J. Tadjer, and F. Calle, "Simulation of thermal management in AlGaIn/GaN HEMTs with integrated diamond heat spreaders," *Semicond. Sci. Technol.*, vol. 28, p. 055010, 2013.
2. A. Wang, M. J. Tadjer, T. J. Anderson, R. Baranyai, J. W. Pomeroy, T. I. Feygelson, K. D. Hobart, B. B. Pate, F. Calle, and M. Kuball, "Impact of intrinsic stress in diamond capping layers on the electrical behavior of AlGaIn/GaN HEMTs," *IEEE Trans. Electron Devices*, vol. 60, p. 3149, 2013.
3. A. Wang, M. J. Tadjer, and F. Calle, "Simulation of temperature and electric field-dependent barrier traps effects in AlGaIn/GaN HEMTs," submitted to *Semicond. Sci. Technol.*, under review, 2014.
4. S. Martin-Horcajo, A. Wang, M.-F. Romero, M. J. Tadjer, and F. Calle, "Simple and accurate method to estimate channel temperature and thermal resistance in AlGaIn/GaN HEMTs," *IEEE Trans. Electron Devices*, vol. 60, p. 4105, 2013.
5. S. Martin-Horcajo, A. Wang, M. F. Romero, M. J. Tadjer, A. D. Koehler, T. J. Anderson, and F. Calle, "Impact of device geometry at different ambient temperatures on the self-heating of GaN-based HEMTs," *Semicond. Sci. Technol.*, Accepted, 2014.
6. S. Martin-Horcajo, A. Wang, M. F. Romero, M. J. Tadjer, A. D. Koehler, T. J. Anderson, and F. Calle, "Assessment of trapping effects for different geometrical HEMTs with AlGaIn and InAlN barriers," In preparation to *Semicond. Sci. Technol.*, 2014.
7. M. J. Tadjer, K. D. Hobart, T. I. Feygelson, A. Wang, T. J. Anderson, A. D. Koehler, F. Calle, B. B. Pate, F. J. Kub, and C. R. Eddy, "Diamond-coated high density vias for silicon substrate-side thermal management of GaN HEMTs," p. 283, *CS MANTECH Conference*, Denver, USA, 2014.
8. D. Cucak, M. Vasic, O. Garcia, J. Oliver, P. Alou, J. A. Cobos, A. Wang, S. Martin-Horcajo, M.-F. Romero, and F. Calle, "Physical model for GaN HEMT design optimization in high frequency switching applications," *ESSDERC/ESSCIRC*, Venice, Italy, 2014.

9. D. Cucak, M. Vasic, O. Garcia, Y. Bouvier, J. Oliver, P. Alou, J. A. Cobos, A. Wang, S. Martin-Horcajo, M.-F. Romero, and F. Calle, “Physical modeling and optimization of a GaN HEMT design with a field plate structure for high frequency application,” *ECCE*, Pittsburgh, PA, USA, 2014.

Appendix: Simulation of GaN HEMTs using COMSOL

In the following, the steps of building the models using COMSOL for GaN HEMTs simulations are presented.

1. Input parameters



As shown in the above graph, add **Parameters** in the **Global definitions** item.

Parameters			
Parameters			
Name	Expression	Value	Description
ls	0.1e-6	1.0000e-07	source and drain contact length
lsg	2.2e-6	2.2000e-06	distance between source and gate contact
lg	3e-6	3.0000e-06	gate length
lgd	15e-6	1.5000e-05	distance between gate and drain
dalg	20e-9	2.0000e-08	AlGaN barrier thickness
dgan	200e-9	2.0000e-07	simulated active region of GaN buffer
dsub	300e-6	0.00030000	substrate thickness
dwd	150e-6	0.00015000	half of horizontal length used in the simulation
dmetal	100e-9	1.0000e-07	gate metal thickness
dsin	50e-9	5.0000e-08	passivation thickness

Input the geometry parameters as shown in the above graph. The unit is meter (m). Note that the source/drain contact length in the simulations was 0.1 μm . The larger value will not influence the simulation results, because the current mainly concentrates at the contact edge.

Continue inputting the following basic parameters used in the simulations. Each parameter has the description of its physical meaning.

Name	Expression	Value	Description
x_Al	0.25	0.25000	Al mole fraction
mu_GaN	0.352	0.35200	GaN Poisson ratio
mu_AIN	0.22	0.22000	AlN Poisson ratio
mu_AlGaIn	$\mu_{AIN} \cdot x_{Al} + \mu_{GaN} \cdot (1 - x_{Al})$	0.31900	AlGaIn Poisson ratio
mu_dm	0.1	0.10000	diamond Poisson ratio
mu_SiN	0.27	0.27000	SiN Poisson ratio
E_GaN	181e9	1.8100e+11	GaN Young's modulus
E_AIN	308e9	3.0800e+11	AlN Young's modulus
E_AlGaIn	$E_{AIN} \cdot x_{Al} + E_{GaN} \cdot (1 - x_{Al})$	2.1275e+11	GaAlN Young's modulus
E_dm	1120e9	1.1200e+12	diamond Young's modulus
E_SiN	250e9	2.5000e+11	SiN Young's modulus
e15_GaN	-0.3	-0.3000	GaN piezoelectric coefficient
e31_GaN	-0.49	-0.4900	GaN piezoelectric coefficient
e33_GaN	0.73	0.73000	GaN piezoelectric coefficient
e31_AlGaIn	$-0.11 \cdot x_{Al} - 0.49$	-0.5175	AlGaIn piezoelectric coefficient
e33_AlGaIn	$0.73 \cdot x_{Al} + 0.73$	0.91250	AlGaIn piezoelectric coefficient

Name	Expression	Value	Description
e15_AlGaIn	$-0.18 \cdot x_{Al} - 0.3$	-0.3450	AlGaIn piezoelectric coefficient
eps11_AlGaIn	$-0.5 \cdot x_{Al} + 9.5$	9.3750	AlGaIn permittivity
eps33_AlGaIn	$0.3 \cdot x_{Al} + 10.4$	10.475	AlGaIn permittivity
ro_GaN	6150	6150.0	GaN mass density
ro_AlGaIn	6150	6150.0	AlGaIn mass density
ro_dm	3150	3150.0	diamond mass density
ro_SiN	3440	3440.0	SiN mass density
Psp_GaN	-0.029	-0.0290	GaN spontaneous polarization
Psp_AIN	-0.081	-0.0810	AlN spontaneous polarization
Psp_AlGaIn	$-0.052 \cdot x_{Al} - 0.029$	-0.0420	AlGaIn spontaneous polarization
C11_AIN	396e9	3.9600e+11	AlN elastic constant
C11_GaN	367e9	3.6700e+11	GaN elastic constant
C12_AIN	137e9	1.3700e+11	AlN elastic constant
C12_GaN	135e9	1.3500e+11	GaN elastic constant
C44_AIN	116e9	1.1600e+11	AlN elastic constant
C44_GaN	95e9	9.5000e+10	GaN elastic constant

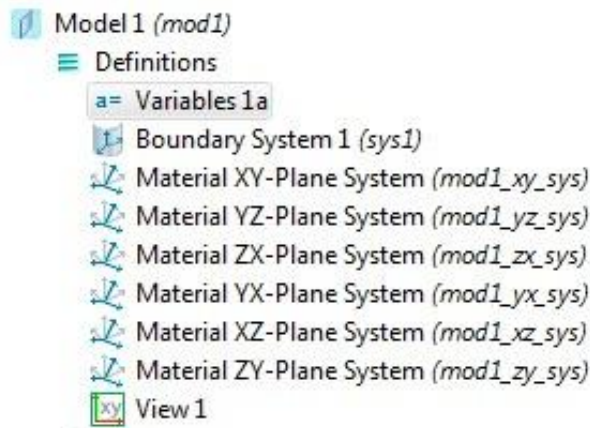
Name	Expression	Value	Description
C11_AlGaN	$C11_AlN \cdot x_{Al} + (1 - x_{Al}) \cdot C11_GaN$	3.7425e+11	AlGaN elastic constant
C12_AlGaN	$C12_AlN \cdot x_{Al} + (1 - x_{Al}) \cdot C12_GaN$	1.3550e+11	AlGaN elastic constant
C44_AlGaN	$C44_AlN \cdot x_{Al} + (1 - x_{Al}) \cdot C44_GaN$	1.0025e+11	AlGaN elastic constant
eps11_GaN	9.5	9.5000	GaN permittivity
eps33_GaN	10.4	10.400	GaN permittivity
q	1.6e-19	1.6000e-19	charge unit
k	1.38e-23	1.3800e-23	Boltzmann constant
T	300	300.00	Ambient temperature
ho	6.62605896e-34	6.6261e-34	Planck constant
mo	9.108e-31	9.1080e-31	static electron mass
eps0	8.854187817e-12	8.8542e-12	Vacuum permittivity
X_GaN	6.56e-19	6.5600e-19	GaN affinity
X_AlGaN	6.076e-19	6.0760e-19	AlGaN affinity
EgAlN	$6.23 - (1.799e-3 \cdot T^2) / (T + 1462)$	6.1381	AlN bandgap
EgGaN	$3.507 - (0.909e-3 \cdot T^2) / (T + 830)$	3.4346	GaN bandgap
EgAlGaN	$EgAlN \cdot x_{Al} + EgGaN \cdot (1 - x_{Al}) - 1....$	3.8667	AlGaN bandgap

Name	Expression	Value	Description
meGaN	$0.2 \cdot m_o$	1.8216e-31	GaN effective mass
meAlGaN	$(0.314 \cdot x_{Al} + 0.2 \cdot (1 - x_{Al})) \cdot m_o$	2.0812e-31	AlGaN effective mass
NcGaN	$2 \cdot (2 \cdot \pi \cdot meGaN \cdot k \cdot T / ho^2)^{1.5}$	2.2424e+24	GaN state density of CB
NcAlGaN	$2 \cdot (2 \cdot \pi \cdot meAlGaN \cdot k \cdot T / ho^2)^{1.5}$	2.7384e+24	AlGaN state density of CB
Vds	0	0	drain voltage
Vgs	0	0	gate voltage
Ndom	1.04e25	1.0400e+25	Dopping density in Ohmic con...
munogan	0.15	0.15000	Electron mobility
Vsatgan	1.5e5	150000	Electron saturation velocity
betn	2	2.0000	Coefficient used in variables
TrCsg	60e-9	6.0000e-08	mesh parameters
TrCg	30e-9	3.0000e-08	mesh parameters
TrCgd	150e-9	1.5000e-07	mesh parameters
gra	10	10.000	mesh parameters
sur	$9e16 \cdot q$	0.014400	interface charge density
surm	$-1e16 \cdot q$	-0.0016	surface charge density

alf_SiN	1.5e-6	1.5000E-6	SiNThermalExpansionCoefficient
alfa_Au	14.2e-6	1.4200E-5	AuThermalExpansionCoefficient
alfa11_GaN	5.59e-6	5.5900E-6	GaNThermalExpansionCoefficient
alfa33_GaN	3.17e-6	3.1700E-6	GaNThermalExpansionCoefficient
alfa11_AlN	4.2e-6	4.2000E-6	AlNThermalExpansionCoefficient
alfa33_AlN	5.3e-6	5.3000E-6	AlNThermalExpansionCoefficient
alfa11_AlGaN	$alfa11_AlN \cdot x_{Al} + alfa11_GaN \cdot (1 - x_{Al})$	5.2425E-6	AlGaNThermalExpansionCoefficient
alfa33_AlGaN	$alfa33_AlN \cdot x_{Al} + alfa33_GaN \cdot (1 - x_{Al})$	3.7025E-6	AlGaNThermalExpansionCoefficient
kmatal	300	300.00	GateMetalThermalConductivity
ksin	20	20.000	SiNThermalConductivity
kgan	160	160.00	GaNThermalConductivity
ksi	148	148.00	SiThermalConductivity

2. Input variables

Define a set of variables named **Variables 1a** as shown in the following graph.



Input the following variables, such as electron density, temperature and electric field-dependent mobility, etc. See the description of the physical meaning for each variable.

Geometric Entity Selection

Geometric entity level: Entire model

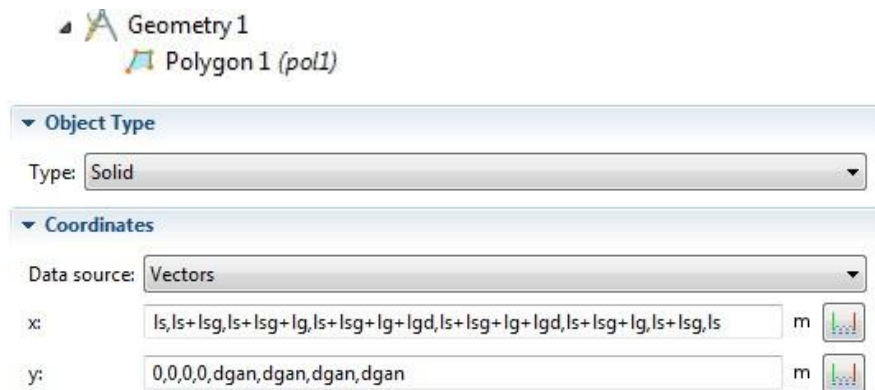
Variables

Name	Expression	Description
noGaN	$NcGaN / (\exp(-ntanoGaN) + 3 \cdot \sqrt{\pi/2}) \cdot (ntanoGaN + 2.13 + (\text{abs}(ntanoGaN - 2.13)^{2.4} + 9.6)^{5/12})^{-1.5}$	GaN electron density
ncGaN	$NcGaNI / (\exp(-ntanGaN) + 3 \cdot \sqrt{\pi/2}) \cdot (ntanGaN + 2.13 + (\text{abs}(ntanGaN - 2.13)^{2.4} + 9.6)^{5/12})^{-1.5}$	GaN electron density
ntanoGaN	$(q \cdot faio + X_GaN) / (k \cdot T)$	
ntanGaN	$(q \cdot fai + X_GaN - q \cdot fain) / (k \cdot T)$	
NcGaNI	$2 \cdot (2 \cdot \pi \cdot meGaN \cdot k \cdot TT / ho^2)^{1.5}$	GaN state density of CB
mun_gan1	$muno_gan \cdot (TT/300)^{-1.5}$	TemperatureDependentMobility
munGaNp	$mun_gan1 \cdot (1 / (1 + (mun_gan1 \cdot \text{abs}(faix) / Vsatgan)^{betn}))^{1/betn}$	ElectricFieldDependentMobility

3. Draw geometry

The steps of drawing the device geometry used in the simulations are described in the following.

3.1. Draw active region of the GaN layer for the electron transport



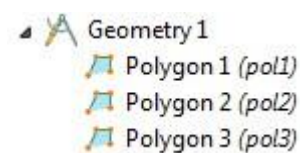
Input the parameters as shown in the above graph.

3.2. Draw the AlGaIn layer



Input the parameters as shown in the above graph.

3.3. Draw the GaN layer



▼ Object Type

Type: Solid

▼ Coordinates

Data source: Vectors

x: $-dwd, 0, 0, ls+ls_g+lg+lgd+ls, ls+ls_g+lg+lgd+ls, ls+ls_g+lg+lgd+ls+dwd, ls+ls_g+lg+lgd+ls+dwd, -dwd$ m

y: $dgan, dgan, 0, 0, dgan, dgan, -dGaN, -dGaN$ m

3.4. Draw the source Ohmic contact

▼ Object Type

Type: Solid

▼ Size

Width: ls m

Height: $dgan$ m

▼ Position

Base: Corner

x: 0 m

y: 0 m

▼ Rotation Angle

Rotation: 0 deg

Geometry 1

- Polygon 1 (*pol1*)
- Polygon 2 (*pol2*)
- Polygon 3 (*pol3*)
- Rectangle 5 (*r5*)

3.5. Draw the drain Ohmic contact

▼ Object Type

Type: Solid

▼ Size

Width: ls m

Height: $dgan$ m

▼ Position

Base: Corner

x: $ls+ls_g+lg+lgd$ m

y: 0 m

▼ Rotation Angle

Rotation: 0 deg

Geometry 1

- Polygon 1 (*pol1*)
- Polygon 2 (*pol2*)
- Polygon 3 (*pol3*)
- Rectangle 5 (*r5*)
- Rectangle 6 (*r6*)

3.6. Draw the gate metal



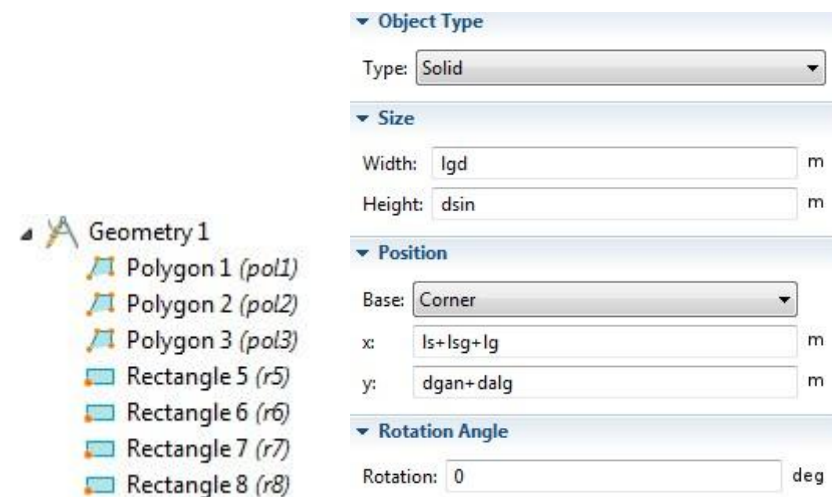
Object Type
Type: Solid

Size
Width: lg m
Height: dmetal m

Position
Base: Corner
x: ls+ls m
y: dgan+dalg m

Rotation Angle
Rotation: 0 deg

3.7. Draw the drain-side passivation



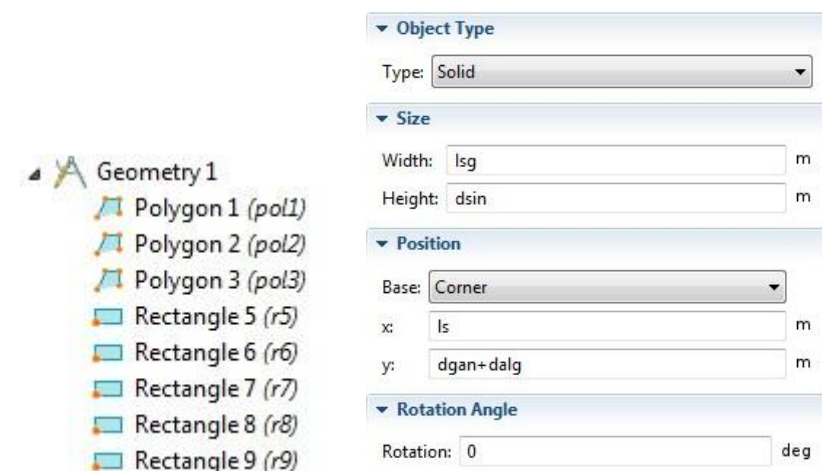
Object Type
Type: Solid

Size
Width: lgd m
Height: dsin m

Position
Base: Corner
x: ls+ls+lg m
y: dgan+dalg m

Rotation Angle
Rotation: 0 deg

3.8. Draw the source-side passivation



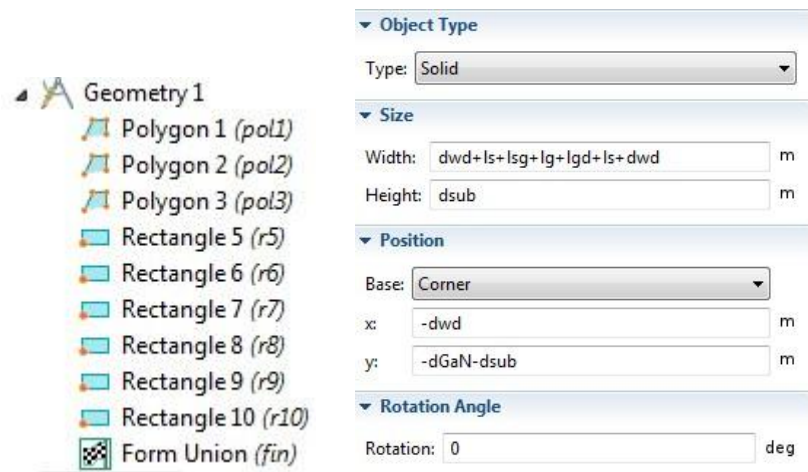
Object Type
Type: Solid

Size
Width: ls m
Height: dsin m

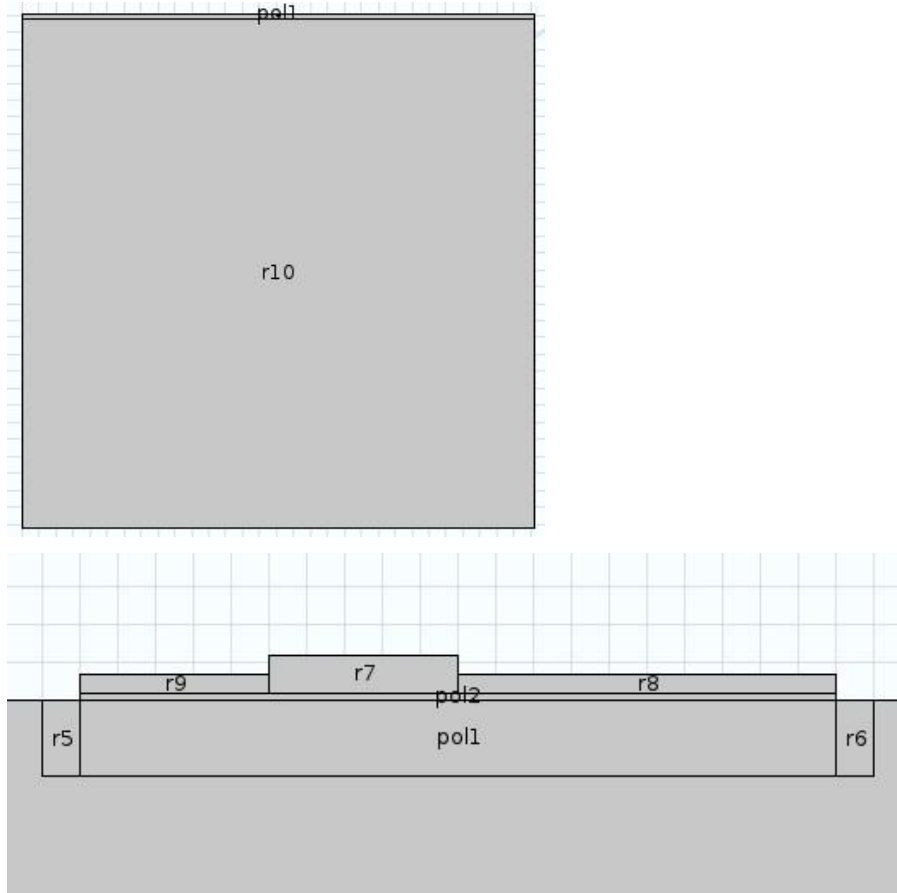
Position
Base: Corner
x: ls m
y: dgan+dalg m

Rotation Angle
Rotation: 0 deg

3.9. Draw the substrate



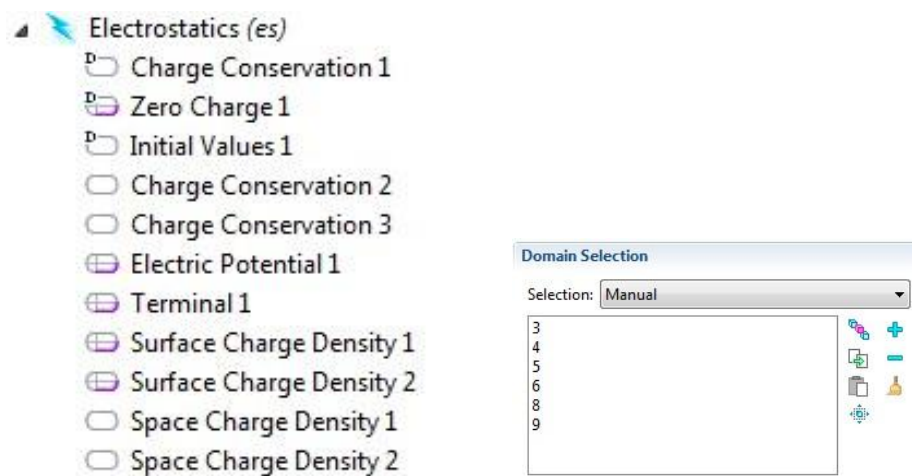
The final geometry (top) and the enlarged view of the device (bottom) are displayed in the following.



4. Add modules for coupling of physical fields

4.1. Device in equilibrium state

For the device in equilibrium state without considering the electron current ($V_{DS}=0$ V), we can solve only the Poisson equation. The solved results of electrical potential distribution can be used as the initial values for the following electron current solving. We can add the module **Electrostatics (es)** in COMSOL describing the Poisson equation as shown in the following graph.



Input the name of the dependent variable as ***faio***. Select the domain (3,4,5,6,8,9) as shown in the right graph for this module.

See the following **SETTINGS** for different **ITEMS**. Note that for the ITEMS not inputted in the following, they were determined by the software automatically.

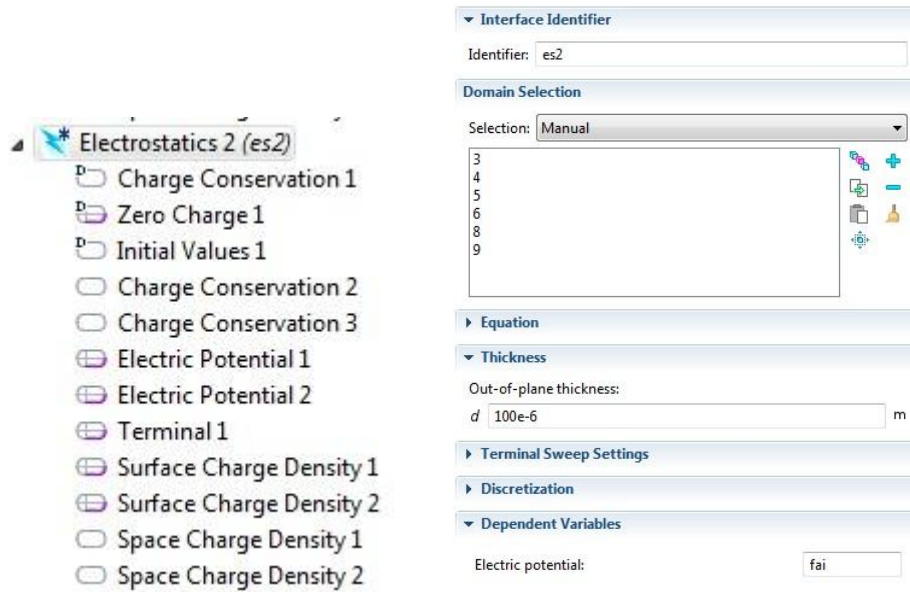
ITEMS	SETTINGS
Charge Conservation 1	Domain: Automatically determined Relative permittivity: eps33_AlGaN
Initial Values 1	0
Charge Conservation 2	Domain: 3,4,9 Relative permittivity: eps33_GaN

Charge Conservation 3	Domain: 6,8 Relative permittivity: eps_SiN
Electric Potential 1 (Description: source and drain contact boundary conditions)	Boundary: 8,31 Electrical potential: faiso1
Terminal 1 (Description: gate contact boundary conditions)	Boundary: 19 Electrical potential: Vgs
Surface Charge Density 1 (Description: AlGaIn/GaN interface charge density)	Boundary: 12,17,23 surface charge density: sur
Surface Charge Density 2 (Description: AlGaIn surface charge density)	Boundary: 14,19,25 Surface charge density: surn
Space charge density 1 (Description: Source and Drain contact space charge density)	domain: 3,9 Space charge density: $q*(-n_{0GaIn}+N_{dom})$
Space charge density 2 (Description: GaN buffer space charge density)	domain: 4 Space charge density: $-q*n_{0GaIn}$

4.2. Electrical simulation

The electrical simulation needs to solve Poisson equation and electron current continuity equation in a full coupling manner. Two modules in COMSOL, i.e., **Electrostatics 2 (es2)** describing the Poisson equation and **PDE (g)** describing the electron current continuity equation, will be used.

The following graphs show the module **Electrostatics 2 (es2)** describing the Poisson equation.



Select the domains: 3,4,5,6,8,9.

Input the dependent variable electric potential: **fai**

See the following SETTINGS for different ITEMS. Most of the SETTINGS are same as that described in 4.1. Here we only list the different ones.

ITEMS	SETTINGS
Initial Values 1 (Description: use the solution <i>fai</i> o as the initial value for this electrical simulation)	fai
Electrical Potential 1 (Description: source contact boundary condition)	Boundary: 8 Electrical potential: faiso1
Electrical Potential 2 (Description: drain contact boundary condition for the device in non-equilibrium state)	Boundary: 31 Electrical potential: faiso1+Vds

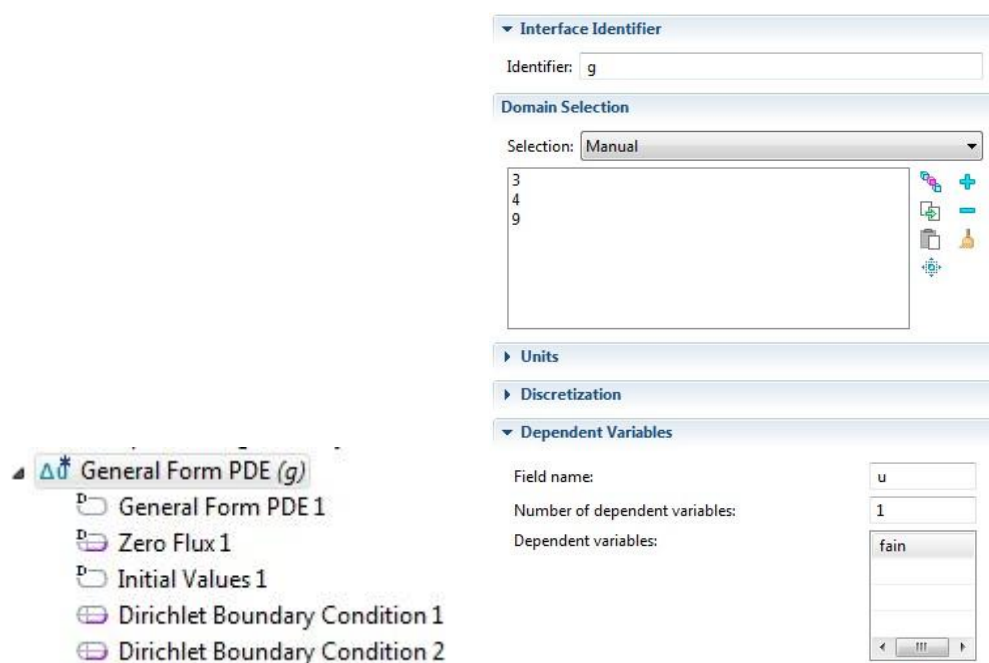
Space charge density 1 (Description: Source and Drain contact space charge density)	domain: 3,9 Space charge density: $q*(-ncGaN+N_{dom})$
Space charge density 2 (Description: GaN buffer space charge density)	domain: 4 Space charge density: $-q*ncGaN$

The following graphs show the module **PDE (g)** describing the electron current continuity equation.

Note that we did not consider the electron transport in the AlGa_N barrier. Not same as the general method, we used the quasi-Fermi level instead of electron concentration as the variable.

Select the domains for the electron transport in GaN buffer: 3,4,9.

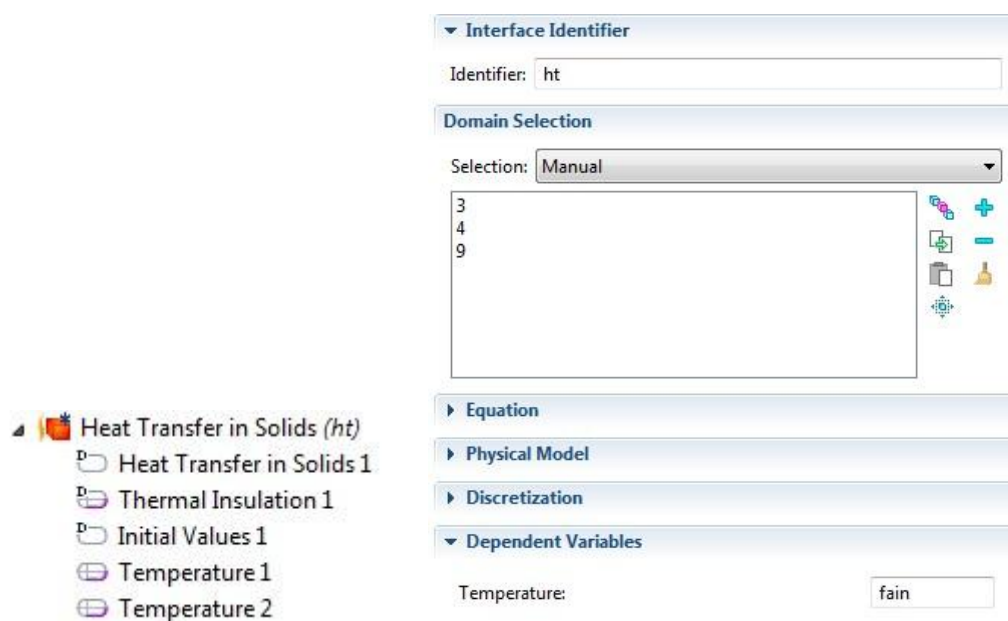
Input the dependent variable: **fain**



See the following **SETTINGS** for different **ITEMS**.

ITEMS	SETTINGS
General Form PDE1 (Description: this setting corresponds to the electron continuity equation. All of other inputs in this item are zero)	Conservative Flux: x: $-\mu_n G_a N_p * n_c G_a N * f_{ainx}$ y: $-\mu_n G_a N_p * n_c G_a N * f_{ainy}$
Initial Values 1	0
Dirichlet Boundary Condition 1 (Description: boundary condition for the quasi-Fermi level at the source contact)	Boundary: 8 Prescribed value: 0
Dirichlet Boundary Condition 2 (Description: boundary condition for the quasi-Fermi level at the drain contact)	Boundary: 31 Prescribed value: Vds

In addition to the module **PDE (g)** described above, we also can use the module **Heat Transfer in Solids (ht)** for describing the electron current continuity equation. See the following graphs. Also input the dependent variable: **fain**. Select the domains: 3,4,9.

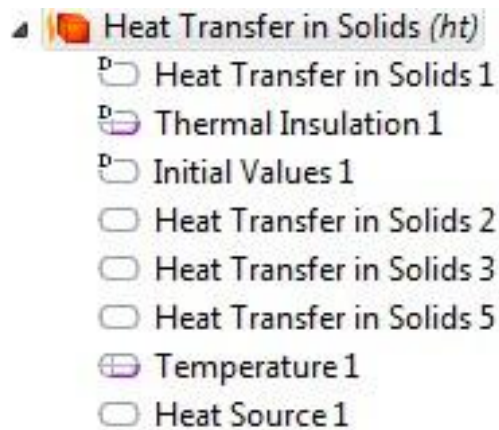


See the following SETTINGS for different ITEMS.

ITEMS	SETTINGS
Heat Transfer in Solids 1 (Description: this setting corresponds to the electron current continuity equation)	Thermal conductivity: -munGaNp*ncGaN
Initial Values 1	0
Temperature 1 (Description: boundary condition for the quasi-Fermi level at the source contact)	Boundary: 8 Temperature: 0
Temperature 2 (Description: boundary condition for the quasi-Fermi level at the drain contact)	Boundary: 31 Temperature: Vds

4.3. Electro-thermal simulation

The electro-thermal simulation needs to solve Poisson equation, electron current continuity equation, and thermal conduction equation in a full coupling manner. Three modules in COMSOL, i.e., **Electrostatics 2 (es2)** describing the Poisson equation, **PDE (g)** describing the electron current continuity equation, and **Heat Transfer in Solids (ht)** describing the thermal conduction equation, will be used. We already have explained the details of the first two modules, in the following the module **Heat Transfer in Solids (ht)** will be explained.



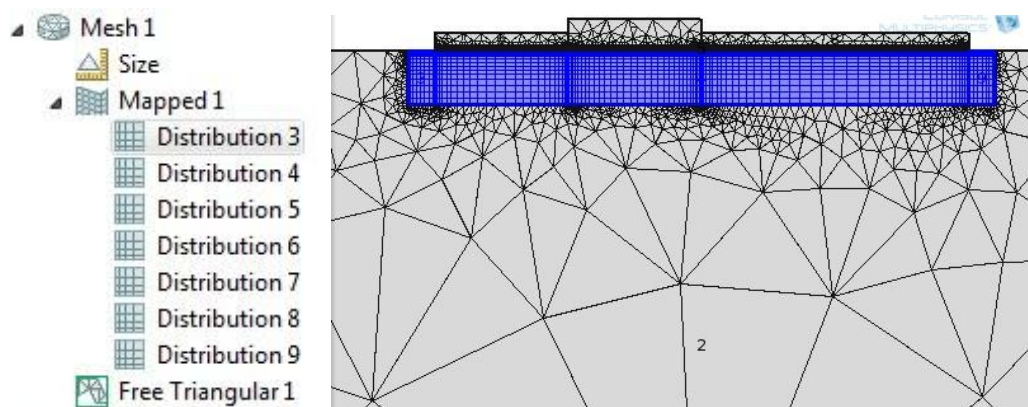
Select all the domains for this module.

See the following **SETTINGS** for different **ITEMS**.

ITEMS	SETTINGS
Heat Transfer in Solids 1 (<i>Description: thermal conduction in GaN</i>)	Domain: 2,3,4,5,9 Thermal conductivity: k _{gan}
Initial Values 1	300
Heat Transfer in Solids 2 (<i>Description: thermal conduction in SiN</i>)	Domain: 6,8 Thermal conductivity: k _{sin}
Heat Transfer in Solids 3 (<i>Description: thermal conduction in gate metal</i>)	Domain: 7 Thermal conductivity: k _{metal}
Heat Transfer in Solids 5 (<i>Description: thermal conduction in Si substrate</i>)	Domain: 1 Thermal conductivity: k _{si}
Temperature 1 (<i>Description: the bottom of the substrate was set as heat sink with temperature 300 K</i>)	Boundary: 2 Temperature: 300

5. Mesh

The mesh for the device geometry is important for the numerical solving. It can influence the numerical convergent behavior. The following graphs show the mesh used in the simulations. For the domains within the AlGaN and GaN, we used the mesh manner **Mapped**, while for other domains the mesh manner is **Free Triangular**.



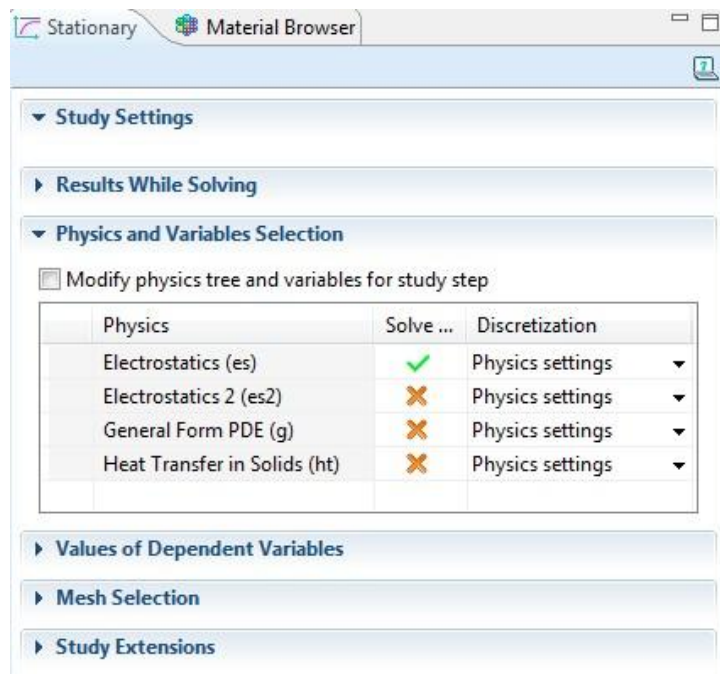
See the following **SETTINGS** for different **ITEMS**.

ITEMS	SETTINGS
Mapped 1	Domain: 3,4,5,9
Distribution 3	Boundary: 11,30 Number of elements: 5
Distribution 4	Boundary: 6,9,28,33 Number of elements: 15
Distribution 5	Boundary: 7,8 Number of elements: 7
Distribution 6	Boundary: 10,12,14 Number of elements: lsg/TrCsg Element ratio: gra (Symmetric distribution)
Distribution 7	Boundary: 16,17,19 Number of elements: lg/TrCg Element ratio: gra (Symmetric distribution)
Distribution 8	Boundary: 22,23,25 Number of elements: lgd/TrCgd Element ratio: gra (Symmetric distribution)

Distribution 9	Boundary: 29,31 Number of elements: 7 Element ratio: 2 (Reverse direction)
Free Triangular 1	Domain: 1,2,6,7,8

6. Solving

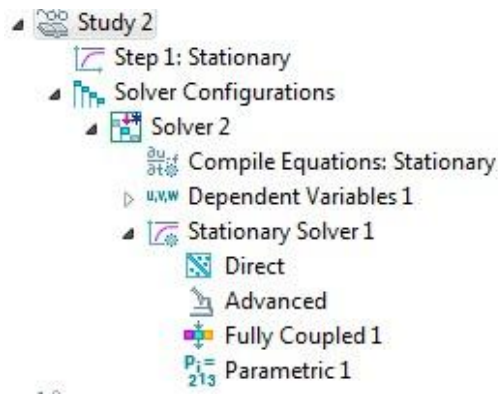
6.1. Solving for the device in equilibrium state



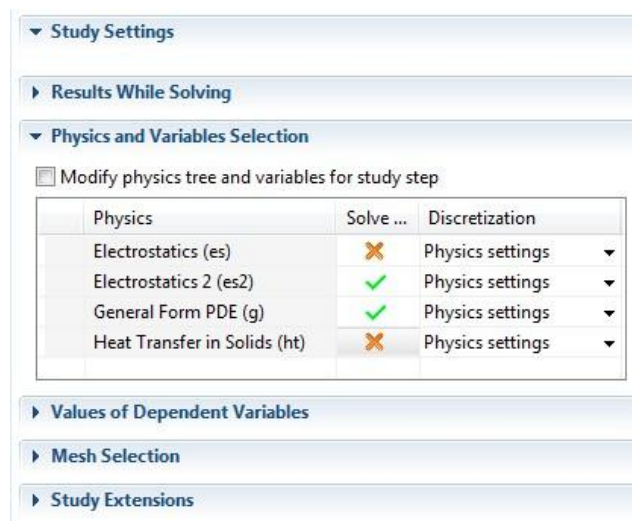
Note that select only **Electrostatics (es)** as shown in the graph for this simulation.

6.2. Electrical solving

Add the item **Parametric 1** as shown in the following graph for scanning the drain voltage.

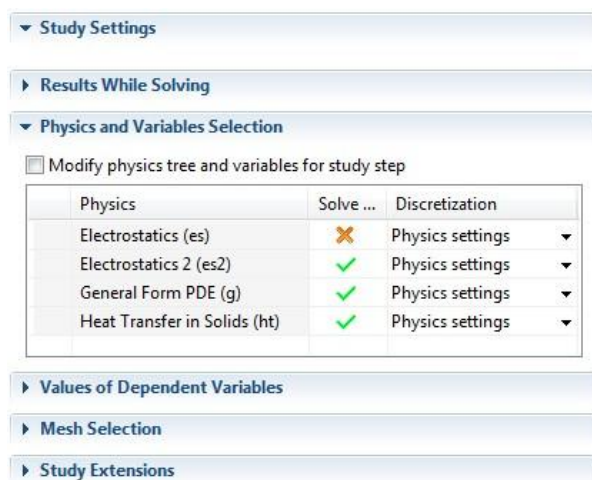


Note that select **Electrostatics 2 (es2)** and **General Form PDE (g)** as shown in the following graph for the electrical simulation.



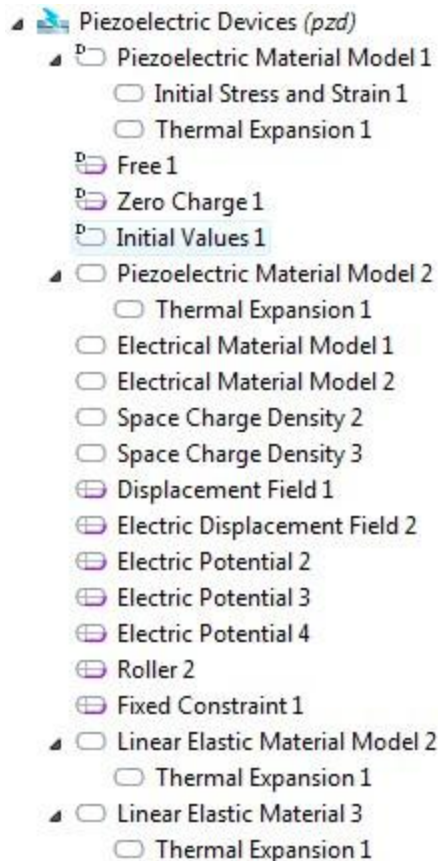
6.3. Electrothermal solving

Note that select **Electrostatics 2 (es2)**, **General Form PDE (g)**, and **Heat Transfer in Solids (ht)** as shown in the following graph for the electro-thermal simulation.



7. Simulations including the device stress effect

We have described the basic steps for the electrical and thermal behavior simulations of GaN HEMTs using COMSOL. In the following, we will show how to build the model using COMSOL for the simulations including the stress effect, i.e., electro-mechanical (EM) coupling and electro-thermo-mechanical (ETM) coupling simulations.



The module **Piezoelectric Devices (pzd)** was used for GaN HEMTs stress effect simulation as shown in the above graph.

See the following SETTINGS for different ITEMS.

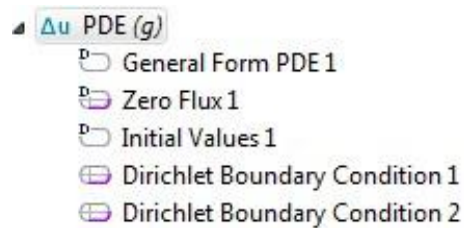
ITEMS	SETTINGS
Piezoelectric Material Model 1	Add domain within the AlGaIn layer
Initial Stress and Strain1	Input the built-in strain in the AlGaIn layer

Thermal Expansion 1	Simulate the thermal expansion in the AlGa _N layer
Piezoelectric Material Model 2	Add domain within the Ga _N layer
Thermal Expansion 1	Simulate the thermal expansion in the Ga _N layer
Electrical Material Model 1	Add the Ohmic contact domain in the Ga _N layer
Electrical Material Model 2	Add the domain within the passivation
Space Charge Density 2	Add the space charge density in the Ohmic contact
Space Charge Density 3	Add the space charge density in the Ga _N buffer
Displacement Field 1	Add the AlGa _N /Ga _N interface spontaneous polarization
Displacement Field 2	Add the AlGa _N surface spontaneous polarization
Electric Potential 2	Add the electrical boundary condition for the source contact
Electric Potential 3	Add the electrical boundary condition for the drain contact
Electric Potential 4	Add the electrical boundary condition for the gate contact
Roller 2	Add the mechanical boundary condition for the sidewall of the device
Fixed Constraint 1	Add the mechanical boundary condition for the bottom of the device
Linear Elastic Material Model 2	Add the domain within the gate metal

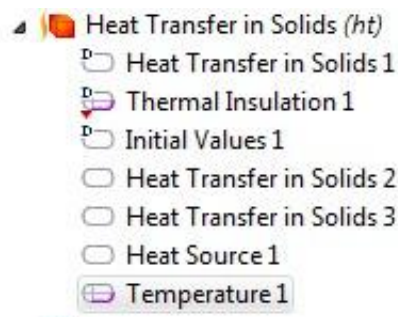
Thermal Expansion 1	Simulate the thermal expansion in the gate metal
Linear Elastic Material 3	Add the domain within the passivation
Thermal Expansion 1	Simulate the thermal expansion in the gate metal

Note that the thermal expansion in different layers was used to simulate the thermal stress.

Add the Module **PDE (g)** for electron current simulation as shown in the following graph. The SETTINGS for different ITEMS are same as that shown in Section 4.2.



Add the Module **Heat Transfer in Solids (ht)** for thermal conduction simulation as shown in the following graph. The SETTINGS for different ITEMS are same as that shown in Section 4.3.



For the EM simulation, the module **Piezoelectric Devices (pzd)** and **PDE (g)** will be full coupled. For ETM simulation, these two modules together with the module **Heat Transfer in Solids (ht)** will be full coupled.

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